

现有论文公式与符号体系文献对照审核稿

内部核验材料

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本稿只回答一个问题：现有 paper_main.pdf 中的公式和符号是否正确、规范、前后一致。审核以已发表论文和学位论文原文为准，不以现有代码反推模型，不在本稿中拼接或创造替代公式。文末附当前稿与各文献的原始模型页，供逐项复核。

目录

1	结论	3
2	审核口径	3
3	逐组公式审核	4
4	符号体系审核	7
4.1	需要补齐的索引与集合	7
4.2	公式中出现但符号表缺失的量	7
4.3	分类或含义需要改正的量	8
5	体系层面的差距	8
6	文献来源与原页索引	8
7	原始模型页	9
7.1	当前论文：被审核的模型页	9
7.2	陈雨蝶等（2025）：同刊模型章表达	17
7.3	陈婉茹等（2023）：混合车队、能耗与碳交易	22
7.4	Goeke-Schneider（2015）：混合车队能耗原式	26
7.5	Zhen 等（2020）：多车场多趟原式	29
7.6	Keskin-Çatay（2016）：线性部分充电原式	32
7.7	Montoya 等（2017）：非线性充电路径模型	35
7.8	Diefenbach 等（2023）：非线性充电排班模型	40
7.9	Froger 等（2022）：容量受限充电站模型	45

7.10 邱莹莹 (2024): 混合车队与动态需求模型	50
7.11 饶卫振等 (2022): 协同配送与成本分摊模型	61
7.12 胡路等 (2025): 充电桩排队与路径模型	67
7.13 Cheng 等 (2022): 分时碳强度充电模型	70
7.14 Lin 等 (2021): 分时电价充放电模型	73

1 结论

现有公式和符号体系不能判为“全部正确、完整且规范”。核心物理关系、成本求和结构、客户流守恒、时间传播、模式覆盖和 Shapley 分摊已有文献依据；但当前模型仍存在会影响数学闭合的实质问题，不能只按排版问题处理。

最重要的结论如下。

1. 车辆能耗核心形式有来源，但车辆和配送趟下标没有闭合。机械功率、电耗与 Goeke-Schneider (2015) 及陈婉茹等 (2023) 的结构一致；现稿中的 P_{ij} 、 u_{ij} 、 e_{ij} 、 f_{ij} 却使用了车辆参数 m_k 、 A_k^f 、 α_k^e 、 β_{0k}^g 和 β_{1k}^g ，公式左端没有 k ，也没有把弧段载重 u_{ij} 与 L_{ir} 建立等式关系。
2. 电耗系数的文字定义不正确。原文中的 $\phi^d \varphi^d$ 是从机械功率到电池功率的乘法换算系数；现稿公式同样采用乘法，却把 α_k^e 称为“效率系数”。若它是通常取值小于 1 的效率，乘法方向错误；若它代表原文两个换算系数的乘积，则名称和取值范围必须按原文说明。
3. 燃油公式可视为 CMEM 的合并系数写法，但现稿没有给出合并关系和单位。陈婉茹等 (2023) 直接写出柴油空气质量比、热值、发动机摩擦项和效率项；现稿写成 $\beta_{0k}^g + \beta_{1k}^g P_{ij}$ 。两者可以代数等价，但当前文本未证明该等价，因而不能把 β_{0k}^g 、 β_{1k}^g 当成已由文献直接给出的符号。
4. 非线性充电链没有闭合。充电时长式采用“按 SOC 分段的恒功率积分”思路，数学上可以解释；但 Montoya 等 (2017) 和 Diefenbach 等 (2023) 原文采用充电函数及其反函数，当前求和式并非所引两篇论文的原式。更关键的是，现稿只说 y_{qt} “由充电曲线积分确定”，没有给出非线性情况下 y_{qt} 与会话、SOC 分段和时段重叠的完整关系。
5. 载重和电量传播只有单边不等式。现稿允许车辆在相邻节点之间无故减少载重或电量；同时 Y_{ir} 没有与实际充电会话的电量严格连接。这会使路线能耗、充电成本与可行性脱节，属于实质模型缺口。
6. 变量域不完整。完整模型只声明了 z_{kp} 为二元变量； x_{ijr} 、 a_{ir} 、 Y_{ir} 、 T_{ir} 、 L_{ir} 、 ε_{ir} 、 a_q^{ch} 、 Δ_q 和 y_{qt} 的域没有完整列出。同时， A_{ikp} 和 O_{skpt} 在模式选择层应是模式系数，现符号表却把 A_{ikp} 称为决策变量。
7. 成本与碳核算存在两处口径冲突。目标文字称“碳结算成本”，式中却使用预测排放 $\hat{E}_{kp}$ ；表中公共站服务费以元/kWh 给出，式 F_5 却用元/min 的占用成本 $c^{\text{occ}} H_{kp}^S$ 。二者至少有一处尚未对齐。
8. Shapley 公式本身正确。当前式与饶卫振等 (2022) 式 (17) 的标准 Shapley 权重一致，且在 $C(\emptyset) = 0$ 时满足效率性。需要核对的是 $C(A)$ 是否始终按同一数据、同一约束和同一成本口径求得，而不是改公式。

2 审核口径

判定	含义	本稿采用的标准
可保留	原式或等价结构成立	文献原式、适用前提、单位和现稿定义能够对应；仅有不影响含义的字母差异。
需校正	核心关系可成立	需要补定义、单位、下标、适用前提或统一命名，校正前不宜作为终稿。
未闭合	当前数学表达不完整	存在自由下标、缺少变量域、传播不闭合、接口未连接或成本单位冲突。
证据不足	暂不能由文献确认	现稿采用项目特有定义，但未给直接来源或完整推导；本稿不替作者创造公式。

3 逐组公式审核

机械功率式	需校正
与原文相同：平路、匀速条件下，空气阻力项与滚动阻力项同 Goeke-Schneider 式 (1)、陈婉茹等式 (1)。 差异及影响：左端缺车辆与趟下标；u_{ij} 未列入符号表，也未与 L_{ir} 连接；公式中使用 k，但量词未给出。	
电耗式	需校正
与原文相同：$P \times d/v$ 得到机械能，再换算为电池电量，与两篇混合车队文献的结构一致；除以 3.6×10^6 可把 J 换成 kWh。 差异及影响：α_k^e 的名称与乘法含义冲突；原文采用 $\phi^d \varphi^d$，并区分放电与能量回收。现稿还缺 k, r 下标。	
燃油式	需校正
与原文相同：在发动机参数固定、机械功率非负时，完整 CMEM 燃油率可合并成常数项与功率项。 差异及影响：文献原式并不直接使用 $\beta_{0k}^g, \beta_{1k}^g$；现稿未给二者由 CMEM 参数合并而来的定义、单位和非负条件。	
线性电耗退化式	需校正
与原文相同：固定速度、平路条件下，按里程电耗对载重呈线性，代数方向成立。 差异及影响：g_v 与 ω^E 实际依赖车辆 k，现符号没有车辆下标；两者也未列入符号表。该式是推导式，不是所引文献的原记号。	
充电时长式	证据不足
与原文相同：SOC 分段功率越低，同一 SOC 增量所需时间越长，物理方向合理；恒功率可作为退化情形。 差异及影响：当前求和式未在所引 Keskin、Montoya 或 Diefenbach 原文中以此形式出现；$(\cdot)_+$ 未定义，L 未入符号表。必须回到一套公开充电函数原式后再定。	
充电窗口式	需校正
与原文相同：“开始不早于窗口下界、结束不晚于上界”是标准连续会话约束。	

差异及影响: $a_q, \bar{a}_q$ 未列入符号表; 窗口如何由站内停留、趟间间隔或首趟前时段生成, 只在文字中说明。

时段重叠式

需校正

与原文相同: 两个时间区间的交集长度写法正确。

差异及影响: g_{qt} 与 $(\cdot)_+$ 均未定义; 它只得到重叠时长, 不能单独给出非线性充电时各时段的电量。

y_{qt} 分时电量

未闭合

与原文相同: 恒功率退化式 $y_{qt} = \pi_{s(q)} g_{qt} / 3600$ 在单位上正确。

差异及影响: 非线性情形没有正式公式, 只有总量守恒; 因此分时电价和分时碳强度尚未与 SOC 充电曲线闭合。

预测与结算碳排放式

需校正

与原文相同: 燃油量乘排放因子、分时充电量乘时变碳强度, 和 Cheng 等式 (2a) 的“碳强度乘分时电量”结构一致, 量纲闭合。

差异及影响: 优化目标使用 $\hat{E}_{kp}$, 正文却称“碳结算成本”; 预测与结算的用途需要统一。

动态触发式

需校正

与原文相同: “累计需求达到阈值”或“固定时间间隔到期”取较早者, 与邱莹莹式 (3-2)–(3-3) 的定时、定量联合触发方向一致。

差异及影响: 原式还含接收期终点; 现稿未定义 $W(\cdot)$ 如何处理新增、取消和改量, 也未把动态符号纳入符号表。

客户集合更新式

证据不足

与原文相同: 新增并入、已服务和取消剔除的集合运算方向清楚。

差异及影响: 该式不是所引邱莹莹模型中的原式; $N_{\text{ser}}^\tau, N_{\text{can}}^\tau, N_{\text{new}}^\tau$ 未定义。

历史成本与排放累计式

未闭合

与原文相同: 分阶段累计的目的合理。

差异及影响: $\bar{Z}^\tau, Z^{\tau-1}, \bar{E}^\tau, \Delta E^{\tau-1}$ 均未在符号表定义, 且 ΔE 与前文 E_{kp} 记号不一致; 无法确认是否重复结算。

碳交易成本 F_1

需校正

与原文相同: “碳价乘排放量减配额”与陈婉茹等式 (5) 的碳交易项一致。

差异及影响: 使用预测排放还是结算排放尚未统一; CE 是全联盟、企业还是阶段配额未说明。

固定、里程和燃油成本 F_2 – F_4

可保留

与原文相同: 以被选模式 z_{kp} 汇总实体车日固定成本、模式里程和模式油耗, 单位与求和层次一致。

差异及影响: 保留的前提是 D_{kp} 、 F_{kp} 均从同一可行模式完整计算, 且每车至多选择一个模式。

充电成本 F_5

未闭合

与原文相同: 分时电价乘分时电量的部分与 Lin 等式 (1) 及 Cheng 等式 (2a) 的结构一致。

差异及影响: y_{qt} 尚未闭合; 公式按元/min 计公共站占用成本, 参数表却给元/kWh 服务费, 单位与含义冲突。

跨场协调成本 F_6

证据不足

与原文相同: 若 A_{ikp} 是模式覆盖系数, 求和可统计跨责任车场服务的客户数, 量纲为元。

差异及影响: 未找到与当前指示函数完全对应的文献原式; A_{ikp} 应是模式系数而非二元决策变量, **1** 也需定义。

总目标函数

需校正

与原文相同: 六类货币成本相加的单一货币目标符合当前研究口径。

差异及影响: 目标本身形式可用, 但继承 F_1 、 F_5 和 F_6 的未决问题。

趟起讫流与客户流守恒

需校正

与原文相同: 单一趟从起点副本出发、到终点副本结束, 客户入度与出度等于访问变量, 属于标准网络流结构。

差异及影响: 必须明确 $d_r^+, d_r^- \in V_r$ 、禁止自环, 并补 x_{ijr} 与 a_{ir} 的二元域。

初始载重与载重上界

未闭合

与原文相同: 起点载重等于本趟全部客户需求的思想正确。

差异及影响: Q_k 中的 k 未与趟 r 或模式 p 绑定; 现式无法判断该趟使用哪辆车的容量。

载重传播

未闭合

与原文相同: 访问客户后载重应减少 q_j 。

差异及影响: 只有上界, 允许在途中无故丢失额外载重; 而能耗依赖载重时会直接影响目标值。

时间窗与时间传播

需校正

与原文相同: 硬时间窗和允许提前等待时, 沿弧给服务开始时刻下界是标准写法。

差异及影响: T_{ir} 的非负域、车场时间界和所有量词未完整列出; 同一个 M 同时用于时间、载重和电量, 单位不统一。

电量边界与传播

未闭合

与原文相同: 电量应位于最低值与容量之间, 并按行驶电耗递减、按充电量增加。

差异及影响: 传播只有单边上界; k 是自由下标; Y_{ir} 未与 E_q^{ch} 、 y_{qt} 或站点会话建立完整等式, 存在免费补电接口。

客户点禁止充电

需校正

与原文相同: 客户点不可充电与当前问题假设一致。

差异及影响: 还需给非客户节点的 Y_{ir} 上界、非负域以及与会话的对应关系。

相邻趟衔接

未闭合

与原文相同: 同一实体车相邻趟需要时间和电量连续, 方向与 Zhen 等多趟衔接思想一致。

差异及影响： $r \rightarrow r'$ 与 $Q_{kp}^{rr'}$ 未正式定义；充后电量没有容量上界，充电时长与两趟间隔只靠段后文字说明。

客户覆盖与每车选模

可保留

与原文相同：这是标准集合划分/集合打包结构；每个客户恰好覆盖一次，每辆实体车至多选择一个日模式。

差异及影响：前提是 A_{ikp} 为已生成模式的固定系数，而不是同时作为另一层决策变量。

站点容量

需校正

与原文相同：以模式在站点时段内的并发占用系数限制桩数，与 Froger 等和胡路等的容量思想一致。

差异及影响：应明确 O_{skpt} 是固定整数系数，并说明核算时段边界上的占用规则。

变量域

未闭合

与原文相同： $z_{kp} \in \{0, 1\}$ 声明正确。

差异及影响：只声明 z_{kp} 不足以构成前面趟层和充电层的完整模型；其余二元变量与连续变量的域缺失。

Shapley 成本分摊

可保留

与原文相同：权重、边际成本和效率性与饶卫振等式 (17) 一致。

差异及影响： $C(A)$ 必须表示联盟 A 在同一输入、约束和成本口径下的最小配送成本；若输入或预算不同，公式正确也不能保证分摊可比。

4 符号体系审核

4.1 需要补齐的索引与集合

现稿没有集中定义 i, j (节点)、 d (车场或企业)、 s (充电地点)、 t (核算时段)、 k (实体车)、 r (配送趟)、 p (日排班模式)、 q (充电会话)、 l (充电曲线分段)、 m (重规划阶段) 以及 A (企业联盟)。 K^g 、 K^e 只出现在并集 $K = K^g \cup K^e$ 中，未分别说明，也未声明二者不相交。

4.2 公式中出现但符号表缺失的量

- $u_{ij}, e_{ij}, f_{ij}, g_v, \omega^E$: 出现在能耗式及其线性退化式，需要车辆、趟或模式下标以及与载重变量的关系。
- $L, (x)_+$: 充电曲线段数和正部算子。
- $\varepsilon_q^{\text{in}}, \varepsilon_q^{\text{out}}, s(q)$: 充电会话进入、离开电量及会话所在地点。
- $\underline{a}_q, \bar{a}_q, g_{qt}$: 充电可行窗口和会话与时段的重叠时长。
- τ_m 和 $W(\tau_{m-1}, u)$: 动态触发时刻与累计需求变动函数。
- $N_{\text{ser}}^\tau, N_{\text{can}}^\tau, N_{\text{new}}^\tau$: 已服务、取消和新增客户集合。

- $o_k^\tau, t_k^\tau, \ell_k^\tau, \varepsilon_k^\tau$: 动态重规划时继承的车辆状态。
- $\bar{Z}^\tau, Z^{\tau-1}, \bar{E}^\tau, \Delta E^{\tau-1}$: 历史成本和排放累计。
- $Q_{kp}^{rr'}$: 相邻两趟之间的充电会话集合。

4.3 分类或含义需要改正的量

- A_{ikp} 和 O_{skpt} 若用于模式选择主模型，应是模式 p 的固定系数；当前把 A_{ikp} 称为决策变量会混淆两层模型。
- D_{kp} 、 F_{kp} 、 H_{kp}^S 、 $\hat{E}_{kp}$ 和 E_{kp} 也是模式属性，不是独立优化变量。
- α_k^e 不能同时被称为“电耗修正系数”和通常意义上的“效率系数”；必须按原能量转换式选定一个含义。
- $p_{s,t}^e$ 实际同时覆盖车场和公共站，地点索引范围应写为 $s \in S \cup D$ ，或另设统一充电地点集合。
- 单个 M 被用于 kg、s 和 kWh 三种约束，不符合量纲清晰的写法；文献中的大数也应按约束量给出可解释上界。

5 体系层面的差距

文献中的模型大多各自闭合：Goeke-Schneider 和陈婉茹等采用弧变量完成混合车队能耗与路径模型；Zhen 等完成多车场多趟排班；Montoya、Diefenbach 和 Froger 等分别在自身问题中闭合非线性充电或站点容量；邱莹莹给出动态触发和静态重求解结构；饶卫振等在路线求解之后完成联盟成本分摊。

现稿把这些层次组合为“趟内弧模型 + 实体车日模式选择 + 分时充电 + 动态继承 + 事后分摊”。这种组合本身并非错误，但目前三处接口尚未完成：

1. 趟 r 没有与车辆 k 的容量、质量和能耗参数严密绑定；
2. 充电会话 q 没有与节点补电量 Y_{ir} 、分时电量 y_{qt} 和模式属性完整绑定；
3. 动态阶段变量没有进入统一符号表，也没有说明每次重规划继承哪一部分模式与成本。

因此，当前最准确的结论不是“公式全错”，也不是“只需统一字母”，而是：若干核心公式有可靠原型，但组合后的完整模型尚未数学闭合。

6 文献来源与原页索引

- 当前论文，PDF 物理页 4–10：被审核的符号表、能耗、充电、动态、成本和完整模型。
- 陈雨蝶等 (2025)，页 5–8：同刊模型章组织、符号表、成本分项和完整模型的呈现方式。
- 陈婉茹等 (2023)，页 4–6：混合车队符号、机械功率、电耗、完整 CMEM 油耗、碳交易目标和约束。

- **Goeke–Schneider (2015)**, 页 4–5: 机械功率、电机与电池换算系数、燃油率、混合车队变量和约束。
- **Zhen 等 (2020)**, 页 4–5: 多车场、多趟、时间衔接的集合、变量与完整约束。
- **Keskin–Çatay (2016)**, 页 4–5: 线性部分充电的变量、目标和约束; 不代表非线性充电。
- **Montoya 等 (2017)**, 页 4–7: 路径问题中的非线性充电函数、分段线性表示和完整模型。
- **Diefenbach 等 (2023)**, 页 6–9: 车辆排班中的充电函数 $\eta(\tau)$ 、反函数和 SOC 传播; 问题类型与路径规划不同。
- **Froger 等 (2022)**, 页 7–10: 容量受限充电站、充电路径和分段充电变量的完整模型。
- **邱莹莹 (2024)**, 页 30–39: 混合车队成本、联合定时/定量动态触发和静态模型约束。
- **饶卫振等 (2022)**, 页 5–9: 协同路径模型、标准 Shapley 公式和服务质量调整公式。
- **胡路等 (2025)**, 页 3–4: 多充电桩排队、符号、充电时间和路径模型。
- **Cheng 等 (2022)**, 页 3–4: 时变电网碳强度、分时充电功率、SOC 和站点功率约束。
- **Lin 等 (2021)**, 页 6–9: 分时电价下路径与充放电模型、符号、目标和变量域。

7 原始模型页

以下页面保持来源原貌。它们用于核对公式、字母、上下标、编号和适用前提, 不表示应把任一套模型整套搬入现稿。

7.1 当前论文: 被审核的模型页

分工,因而所得方案能否被各方接受,须在结算层单独判定。

本文模型具体表述为:多家配送企业各自拥有一个车场及其车队,并各自负责一部分客户;协作配送允许各企业共享客户和车辆资源,联合为全部客户提供服务;车队由燃油车(CV)和电动车(EV)混合组成,每辆车只属于一个企业的车场,但允许服务其他企业的客户;车辆可在一个运营日内连续执行多趟任务,电动车途中可访问共享充电站,也可在首趟出发前和相邻趟之间在车场按需补电;实体车队规模有限且每辆车按日计一次启用成本,因而一辆车通常需承担多趟任务,其日间返场装货的间隔正是车场补能的可用窗口,补能机会由此与配送排班直接绑定;电网碳强度和电价均按48个半小时时段给定,充电发生的时段不同,充电电费和间接碳排放也随之不同;订单新增、取消或改量后,系统只调整尚未执行的任务,已完成或已发车的路径前缀不可撤回。模型的优化目标是在满足载重、时间窗、电量和站点容量等运行约束的前提下,以运营成本和碳结算成本之和最小化确定车辆路径、车型选择和充电安排;合作成员的参与性在方案生成后的结算层判定。

模型假设条件如下:1)在每个规划时刻,已到达订单的客户、车场、车辆信息已知,新订单在运营过程中动态出现。2)每个客户由一辆车服务一次,需求不可拆分。3)车辆在配送全程不超过所属车型的最大载重。4)车辆行驶速度设为常值,能耗按综合模式排放模型(CMEM)系能耗函数^[7,29]计算。5)客户具有硬时间窗,车辆可提前到达后等待,但不得迟于时间窗上界。6)充电功率为电池荷电状态的分段常函数(恒功率为其退化情形),每次充电为一个连续区间,允许部分充电,时段电量由充电曲线和电网时段的重叠确定。7)时段电网碳强度和分时电价为外生参数,企业为价格与碳强度的接受者。8)客户责任按编号顺序均分给各配送企业,该归属为公开披露的构造情景,分法参照饶卫振等^[17];运营成本先按实际执行车辆所属企业归集;协作路线优化完成后,再依据各子联盟的独立经营成本分配协作收益。核仁给出唯一且联盟稳定的结算结果,Shapley值作为成员贡献的比较基准。模型符号说明如表2所示。

表2 符号说明

符号	说明	符号	说明
D	车场集合	c_D	空气阻力系数
N	客户集合 $N = \{1, 2, \dots, n\}$, 其中 n 是客户数目	c_R	滚动阻力系数
S	共享充电站集合	ρ^a	空气密度 (kg/m^3)
T	电网时段集合 $T = \{1, 2, \dots, 48\}$	g_0	重力加速度 (m/s^2)
K	实体车辆集合	α_k^e	车辆 k 的电耗修正系数
K_d	车场 d 的车辆集合	$\beta_{0k}^g, \beta_{1k}^g$	车辆 k 的油耗系数
V_r	趟 r 的节点集合	B	电池容量 (kWh)
V_r^c	趟 r 的客户子集 $V_r^c = V_r \cap N$	$B^{\min}$	电池最低允许电量 (kWh)
Ω_k	车辆 k 的可行排班模式集合	λ^g	燃油排放因子 ($\text{kgCO}_2\text{e}/\text{L}$)
$\mathcal{R}_{kp}$	模式 p 的配送趟集合	π_s	站点 s 的额定充电功率 (kW)
$\mathcal{Q}_{kp}$	模式 p 的充电会话集合	C_s	站点 s 的充电桩数
d_r^+, d_r^-	趟 r 的起讫车场副本	b_l, κ_l	充电曲线第 l 段 SOC 断点与相对功率
$[\underline{t}_t, \overline{t}_t]$	时段 t 的起止时刻 (s)	$\hat{\gamma}_t, \gamma_t$	时段 t 的预测与结算碳强度 ($\text{kgCO}_2\text{e}/\text{kWh}$)
q_i	客户 i 的需求量 (kg)	p^{car}	单位碳价格 (元/kg)
Q_k	车辆 k 的最大载重 (kg)	CE	碳配额 (kg)
$[t_i, \bar{t}_i]$	客户 i 的时间窗	c^{fix}	每趟派遣固定成本 (元/趟)
σ_i	客户 i 的服务时长 (s)	c_k^{km}	车辆 k 的单位里程成本 (元/km)
d_{ij}	节点 i 与节点 j 之间的距离 (m)	p^f	单位燃油价格 (元/L)
t_{ij}	弧 (i, j) 的行驶时间 (s)	$p_{s,t}^e$	站点 s 在时段 t 的电价 (元/kWh)
v	车辆行驶速度 (m/s)	c^{occ}	公共站单位占用时间成本 (元/min)
m_k	车辆 k 的整备质量 (kg)	c^{tr}	单位跨场协调成本 (元/客户)
A_k^f	车辆 k 的迎风面积 (m^2)	ρ	单位货量收入 (元/kg)
$d(k)$	车辆 k 所属车场	$v(A)$	企业联盟 A 的可转移收益 (元)
d_i^0	客户 i 的原责任车场	Π_d^0	车场 d 独立经营的基准收益 (元)
$C(A), R(A)$	联盟 A 的配送成本与收入 (元)	ϕ_d	车场 d 的 Shapley 收益分配 (元)
x_d^v	车场 d 的核仁收益分配 (元)	$e(A, x)$	联盟 A 对分配 x 的不满意额 (元)
n_{kp}	模式 p 的配送趟数	$\overline{W}$	动态触发的累计需求上限 (kg)
D_{kp}	模式 p 的总里程 (km)	T^{int}	动态触发的时间间隔 (s)
F_{kp}	模式 p 的总油耗 (L)	M	足够大的正数
$H_{kp}^S, \hat{E}_{kp}, E_{kp}$	模式 p 的公共站占用时长 (min)、 预测与结算碳排放 (kgCO_2e)	L_{ir}	离开节点 i 时的载重 (kg)
O_{skpt}	模式 p 在站 s 、时段 t 的在充会话数	T_{ir}	节点 i 的服务开始时刻 (s)
A_{ikp}	0-1 变量, 如果模式 p 服务 客户 i , 则为 1, 否则为 0	ε_{ir}	到达节点 i 时的电量 (kWh)
x_{ijr}	0-1 变量, 如果趟 r 使用 弧 (i, j) , 则为 1, 否则为 0	Y_{ir}	节点 i 处的补电量 (kWh)
a_{ir}	0-1 变量, 如果趟 r 服务 客户 i , 则为 1, 否则为 0	$a_q^{\text{ch}}, \Delta_q$	会话 q 的开始时刻与时长 (s)
z_{kp}	0-1 变量, 如果车辆 k 选择 模式 p , 则为 1, 否则为 0	E_q^{ch}, y_{qt}	会话 q 的补电量与其在 时段 t 的补电量 (kWh)

2.2 目标函数

2.2.1 碳排放成本 F_1

碳排放成本由燃油车行驶的直接排放和电动车充电的间接排放构成。燃油油耗由公式 (7) 和 (9) 求得, 规划阶段的充电间接排放按充电发生时段的预测碳强度逐时段核算, 模式 p 的预测碳排放 $\hat{E}_{kp}$ 如公式 (14) 所示。在碳交易机制中, 政府指定碳配额, 企业碳排放量超过配额 CE 时需要购买更多的碳排放权, 相反也可以通过出售碳排放权赚取利润, 此时碳排放成本的计算如公式 (1) 所示。

$$F_1 = p^{\text{car}} \left(\sum_{k \in K} \sum_{p \in \Omega_k} \hat{E}_{kp} z_{kp} - CE \right). \quad (1)$$

2.2.2 固定成本 F_2

固定成本主要包括车辆折旧费、维修费、保险费和驾驶员工资等,与派遣的配送趟数成正比.

$$F_2 = c^{\text{fix}} \sum_{k \in K} \sum_{p \in \Omega_k} n_{kp} z_{kp}. \quad (2)$$

2.2.3 里程成本 F_3

里程成本与行驶距离成正比,反映轮胎磨损、保养等非能源行驶费用.

$$F_3 = \sum_{k \in K} \sum_{p \in \Omega_k} c_k^{\text{km}} D_{kp} z_{kp}. \quad (3)$$

2.2.4 油耗成本 F_4

油耗成本与燃油车运输时产生的油耗成正比,油耗量由能耗模型逐弧累计.

$$F_4 = p^f \sum_{k \in K} \sum_{p \in \Omega_k} F_{kp} z_{kp}. \quad (4)$$

2.2.5 充电成本 F_5

充电成本包括车场充电电费、公共站充电电费和公共站占用成本.车场执行分时目录电价,公共站电价为含服务费的分时总价,公共站另计占用时间费.充电发生的时段不同,电费也随之不同,因此充电成本按会话在各时段的补电量逐时段结算.

$$F_5 = \sum_{k \in K} \sum_{p \in \Omega_k} \left(\sum_{q \in \mathcal{Q}_{kp}} \sum_{t \in T} p_{s(q),t}^e y_{qt} + c^{\text{occ}} H_{kp}^S \right) z_{kp}. \quad (5)$$

其中 $s(q)$ 为会话 q 所在的车场或充电站.

2.2.6 跨场成本 F_6

车辆服务非所属车场的客户时计入跨场协调成本,反映跨场协同产生的信息交换和管理开销.

$$F_6 = c^{\text{tr}} \sum_{k \in K} \sum_{p \in \Omega_k} \sum_{i \in N} A_{ikp} \mathbf{1}_{d(k) \neq d_i^0} z_{kp}. \quad (6)$$

上述公式中, n_{kp} , D_{kp} , F_{kp} , H_{kp}^S , $\hat{E}_{kp}$ 分别为模式 p 的配送趟数、总里程(km)、总油耗(L)、公共站占用时间(min)和预测碳排放(kgCO₂e),由模式 p 所含配送趟和充电会话按公式(7)–(14)逐项累计得到. E_{kp} 为同一模式执行后按结算碳强度复算的碳排放.

2.3 车辆能耗模型

为了准确刻画载重和车型对能耗的影响,本文参照混合车队路径优化研究^[7]采用CMEM系能耗函数^[29,30],车辆在弧 (i, j) 上以给定速度 v 匀速行驶,牵引功率由空气阻力、滚动阻力和载重共同决定:

$$P_{ij} = \frac{1}{2} c_D \rho^a A_k^f v^3 + (m_k + u_{ij}) g_0 c_R v. \quad (7)$$

其中 u_{ij} 为弧段载重(kg),由载重变量 L_{ir} 按弧确定.式(7)中 P_{ij} 的单位为W.为使电量与电池状态变量的单位一致,电动车弧段电耗按 $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ 折算为kWh;燃油车弧段油耗由排放系数直接给出体积单位L.二者分别为

$$e_{ij} = \frac{\alpha_k^e}{3.6 \times 10^6} P_{ij} \frac{d_{ij}}{v}, \quad (8)$$

$$f_{ij} = (\beta_{0k}^g + \beta_{1k}^g P_{ij}) \frac{d_{ij}}{v}. \quad (9)$$

其中 α_k^e 为无量纲的电驱动系统效率系数.在常值速度下,式(8)退化为关于载重线性的按里程电耗式:

$$e_{ij} = (g_v + \omega^E u_{ij}) d_{ij}, \quad g_v = \frac{\alpha_k^e}{3.6 \times 10^6} \left(\frac{1}{2} c_D \rho^a A_k^f v^2 + m_k g_0 c_R \right), \quad \omega^E = \frac{\alpha_k^e g_0 c_R}{3.6 \times 10^6}. \quad (10)$$

其中 d_{ij} 以m计, g_v 与 ω^E 分别为kWh/m和kWh/(kg·m),因而 e_{ij} 与 ε_{ir} , Y_{ir} , B 同为kWh,式(25)的电量传播在量纲上闭合.该线性形式使电量传播约束保持线性,燃油油耗同理退化.

2.4 充电与分时碳排放模型

传统电动车路径研究常将充电近似为恒功率满充, 这与现实中动力电池高荷电状态下功率下降、车辆按需部分补电的运行方式不符. 鉴于此, 本文参照部分充电与非线性充电研究^[10,11], 将站点 s 的充电功率表示为电池荷电状态 (SOC) 的分段常函数. 设 L 个 SOC 分段 $[b_{l-1}, b_l)$ 对应相对功率 κ_l ($b_0 = 0, b_L = 1, \kappa_1 \geq \dots \geq \kappa_L > 0$), 实际充电功率为 $\pi_s \kappa_l$ kW. 充电会话 q 由进入电量 $\varepsilon_q^{\text{in}}$ 充至离开电量 $\varepsilon_q^{\text{out}}$, 补电量 $E_q^{\text{ch}} = \varepsilon_q^{\text{out}} - \varepsilon_q^{\text{in}}$, 充电时长为

$$\Delta_q = \frac{3600B}{\pi_{s(q)}} \sum_{l=1}^L \frac{(\min\{\varepsilon_q^{\text{out}}/B, b_l\} - \max\{\varepsilon_q^{\text{in}}/B, b_{l-1}\})_+}{\kappa_l}. \quad (11)$$

会话须整段落落在其可行窗口 $[a_q, \bar{a}_q]$ 内:

$$a_q \leq a_q^{\text{ch}}, \quad a_q^{\text{ch}} + \Delta_q \leq \bar{a}_q. \quad (12)$$

会话与电网时段 t 的重叠时长为

$$g_{qt} = [\min\{a_q^{\text{ch}} + \Delta_q, \bar{T}_t\} - \max\{a_q^{\text{ch}}, \underline{T}_t\}]_+. \quad (13)$$

各时段补能量 y_{qt} 由时段内充电曲线积分确定, 满足 $\sum_{t \in T} y_{qt} = E_q^{\text{ch}}$; 恒功率 ($L = 1, \kappa_1 = 1$) 退化为 $y_{qt} = \pi_{s(q)} g_{qt} / 3600$.

充电间接排放按补电量所在时段的电网碳强度核算. 决策时使用预测碳强度 $\hat{\gamma}_t$, 执行后按结算碳强度 γ_t 复算, 模式 p 的预测碳排放与结算碳排放分别为

$$\begin{aligned} \hat{E}_{kp} &= \lambda^g \sum_{r \in \mathcal{R}_{kp}} \sum_{(i,j) \in r} f_{ij} + \sum_{q \in \mathcal{Q}_{kp}} \sum_{t \in T} \hat{\gamma}_t y_{qt}, \\ E_{kp} &= \lambda^g \sum_{r \in \mathcal{R}_{kp}} \sum_{(i,j) \in r} f_{ij} + \sum_{q \in \mathcal{Q}_{kp}} \sum_{t \in T} \gamma_t y_{qt}. \end{aligned} \quad (14)$$

第一项是燃油车直接排放, 第二项是电动车充电侧间接排放.

2.5 动态需求处理

参照动态需求批处理研究^[23,25], 本文同时考虑定时和定量两种批事件触发策略. 设第 $m-1$ 次重规划发生于时刻 τ_{m-1} , 其后新到事件的累计需求变动量为 $W(\tau_{m-1}, u)$, 则第 m 次重规划时刻为

$$\tau_m = \min\{\min\{u : W(\tau_{m-1}, u) \geq \bar{W}\}, \tau_{m-1} + T^{\text{int}}\}, \quad (15)$$

即累计需求变动达到上限 $\bar{W}$ 或距上次重规划达到间隔 T^{int} 时, 即触发滚动重规划. 在重规划时刻 τ , 有效客户集合按新增、取消和已完成服务更新:

$$N^\tau = (N^{\tau-1} \setminus N_{\text{ser}}^\tau \setminus N_{\text{can}}^\tau) \cup N_{\text{new}}^\tau. \quad (16)$$

每辆实体车 k 继承当前位置 o_k^τ 、最早可用时刻 t_k^τ 、剩余载重 ℓ_k^τ 和剩余电量 ε_k^τ , 已经完成、已经发车和正在执行的路径前缀不可撤回, 已完成部分的成本、排放和收益按阶段累计:

$$\bar{Z}^\tau = \bar{Z}^{\tau-1} + Z^{\tau-1}, \quad \bar{E}^\tau = \bar{E}^{\tau-1} + \Delta E^{\tau-1}, \quad \bar{\Pi}_d^\tau = \bar{\Pi}_d^{\tau-1} + \Pi_d^{\text{stage}, \tau-1}. \quad (17)$$

状态继承保证滚动重规划不撤回已执行路径、不重复使用实体车、不重复结算历史排放.

2.6 模型建立

模型分两层建立. 第一层为配送趟约束, 对趟 r , d_r^+ , d_r^- 为起讫车场副本; 第二层为模式选择约束, M 为足够大的正数.

$$\min F = F_1 + F_2 + F_3 + F_4 + F_5 + F_6. \quad (18)$$

$$\sum_{j \in V_r} x_{d_r^+ j r} = 1, \quad \sum_{j \in V_r} x_{j d_r^- r} = 1, \quad \sum_{i \in V_r} x_{i d_r^+ r} = 0, \quad \sum_{i \in V_r} x_{d_r^- i r} = 0. \quad (19)$$

$$\sum_{j \in V_r} x_{i j r} = \sum_{j \in V_r} x_{j i r} = a_{ir}, \quad i \in V_r^c. \quad (20)$$

$$L_{d_r^+ r} = \sum_{i \in V_r^c} q_i a_{ir}, \quad 0 \leq L_{ir} \leq Q_k, \quad i \in V_r. \quad (21)$$

$$L_{jr} \leq L_{ir} - q_j + M(1 - x_{ijr}), \quad i, j \in V_r. \quad (22)$$

$$\underline{t}_i a_{ir} \leq T_{ir} \leq \bar{t}_i + M(1 - a_{ir}), \quad i \in V_r^c. \quad (23)$$

$$T_{jr} \geq T_{ir} + \sigma_i + t_{ij} - M(1 - x_{ijr}), \quad i, j \in V_r. \quad (24)$$

$$B^{\min} \leq \varepsilon_{ir} \leq B, \quad \varepsilon_{jr} \leq \varepsilon_{ir} + Y_{ir} - e_{ij} + M(1 - x_{ijr}), \quad i, j \in V_r. \quad (25)$$

$$Y_{ir} = 0, \quad i \in V_r^c. \quad (26)$$

$$T_{d_r^+, r'} \geq T_{d_r^-, r}, \quad \varepsilon_{d_r^+, r'} = \varepsilon_{d_r^-, r} + \sum_{q \in Q_{kp}^{rr'}} E_q^{\text{ch}}, \quad r \rightarrow r' \text{ 相邻}. \quad (27)$$

$$\sum_{k \in K} \sum_{p \in \Omega_k} A_{ikp} z_{kp} = 1, \quad i \in N, \quad \sum_{p \in \Omega_k} z_{kp} \leq 1, \quad k \in K. \quad (28)$$

$$\sum_{k \in K} \sum_{p \in \Omega_k} O_{skpt} z_{kp} \leq C_s, \quad s \in S \cup D, \quad t \in T. \quad (29)$$

规划阶段的车场收益采用预测碳排放 $\hat{E}_{kp}$; 执行后的排放报告采用 E_{kp} .

$$\Pi_d^{\text{stage}} = \sum_{k \in K_d} \sum_{p \in \Omega_k} \left(\rho \sum_{i \in N} q_i A_{ikp} - c^{\text{fix}} n_{kp} - c_k^{\text{km}} D_{kp} - p^f F_{kp} - G_{kp}^e - c^{\text{tr}} n_{kp}^{\text{tr}} - p^{\text{car}} \hat{E}_{kp} \right) z_{kp}. \quad (30)$$

$$z_{kp} \in \{0, 1\}, \quad k \in K, \quad p \in \Omega_k. \quad (31)$$

公式 (18) 表示运营成本与碳结算成本之和最小化. 公式 (19) 和 (20) 定义趟的起讫车场与客户流守恒; 公式 (21) 和 (22) 保证载重不超容量且随服务递减; 公式 (23) 和 (24) 保证服务时刻满足硬时间窗并沿弧递推; 公式 (25) 保证电量在允许范围且逐弧传播, 公式 (26) 规定客户点不可充电. 公式 (27) 保证同一实体车相邻趟的时间衔接与跨趟电量连续, 即趟间按需补电, 其中 $Q_{kp}^{rr'}$ 为安排在两趟之间的车场充电会话集合; 充电会话窗口由车辆实际停留区间界定, 站内为到离时刻, 趟间为 $[T_{d_r^-, r}, T_{d_r^+, r'}]$, 首趟前为运营开始时刻到首趟发车时刻, 所有会话服从公式 (11)–(13) 且互不重叠. 记 Ω_k 为满足公式 (19)–(27) 的车辆 k 全部可行排班模式的集合, 模式 $p \in \Omega_k$ 完整给定其配送趟、节点时刻、车型和充电会话, 第二层在 Ω_k 上进行模式选择. 公式 (28) 保证每个客户由恰好一个模式覆盖且每车至多执行一个模式; 公式 (29) 参照容量受限充电站研究^[31,32], 保证任意时段站点并发充电数不超过桩数, O_{skpt} 由会话开始时刻和时长预先计算. 公式 (30) 按实际执行车辆的所属车场计算转移前运营收益, 其中 G_{kp}^e 为模式 p 的充电总成本 (即公式 (5) 的模式项), $n_{kp}^{\text{tr}} = \sum_{i \in N} A_{ikp} \mathbf{1}_{d(k) \neq d_i^0}$ 为跨场服务客户数. 基准收益 Π_d^0 由同一算法在同等评价预算下, 按车场 d 仅服务自身责任客户、禁用跨场服务的独立经营情形求得. $\bar{\Pi}_d$ 为截至上次重规划的累计收益, 静态情形取 0, 其滚动累计规则见第 2.5 节.

令 D 为参与协作的配送企业集合, 每家企业拥有一个车场及其车队, 并对一部分客户负有配送责任. 对任意非空联盟 $A \subseteq D$, 限定只有联盟成员的车场、车辆和责任客户参与配送, 由同一算法求得联盟成本 $C(A)$ 与收入 $R(A)$, 并定义可转移收益

$$v(A) = R(A) - C(A), \quad v(\emptyset) = 0. \quad (32)$$

$|D|$ 家企业共有 $2^{|D|} - 1$ 个非空联盟. 车场 d 的 Shapley 收益分配为

$$\phi_d = \sum_{A \subseteq D \setminus \{d\}} \frac{|A|!(|D| - |A| - 1)!}{|D|!} [v(A \cup \{d\}) - v(A)]. \quad (33)$$

若分配 x 满足

$$\sum_{d \in D} x_d = v(D), \quad \sum_{d \in A} x_d \geq v(A), \quad \emptyset \neq A \subset D, \quad (34)$$

则没有任何单家或子联盟能通过退出四家合作获得更高收益, 该分配位于核内. 记联盟不满意额为 $e(A, x) = v(A) - \sum_{d \in A} x_d$; 核仁按降序对全部 $e(A, x)$ 作字典序最小化, 得到唯一分配 x^ν . 本文采用核仁作为结算结果, 同时报出 Shapley 值以比较贡献分配与联盟稳定性的差异. 第二层为集合划分结构, 模式

集 Ω_k 随客户规模指数增长, 本模型是标准 VRP 的推广, 属于 NP-难问题, 第 3 节给出其启发式求解算法.

3 算法设计

3.1 算法流程图

上述模型同时包含跨场服务、车型选择、实体车多趟、分时充电排放、成员结算参与和动态订单, 其可行解需同时满足路径层与实体车排班层的耦合约束. 在该结构下, 仅凭路线邻域无法判定一个候选方案是否可行: 可行与否取决于该方案被补全为实体车排班并完成补能之后的结果. 因此本文不另行设计搜索内核, 而是直接采用成熟的混合遗传搜索 (Hybrid Genetic Search, HGS)^[33,34] 作为搜索底座, 并在其上构造面向本问题的特化求解组件: 统一完整评价、实体车多趟补全、碳感知充电排程与动态状态继承. 算法记为 HGS-FRVCP (融合固定路线充电求解的混合遗传搜索), 主流程如图1所示. 算法按代循环推进: 每代由交叉产生子代, 子代依次经过串行改进链的三步问题特化改进, 再由完整评价器裁决能否入群, 如图2所示. 图1给出总流程, 与下文步骤 1–6 逐项对应. 算法全过程坚持同一核算口径: 候选方案能否被接受、能否进入种群、能否存活, 均由同一完整评价器给出的完整成本与完整可行性决定, 不以简化路线代价近似替代完整方案代价. 能源消耗、补能安排、趟间时间与电量衔接、分时电价和时变碳强度进入同一本账; 候选补全失败时保留失败原因, 不以初始解顶替. 第 4 节报告其求解结果.

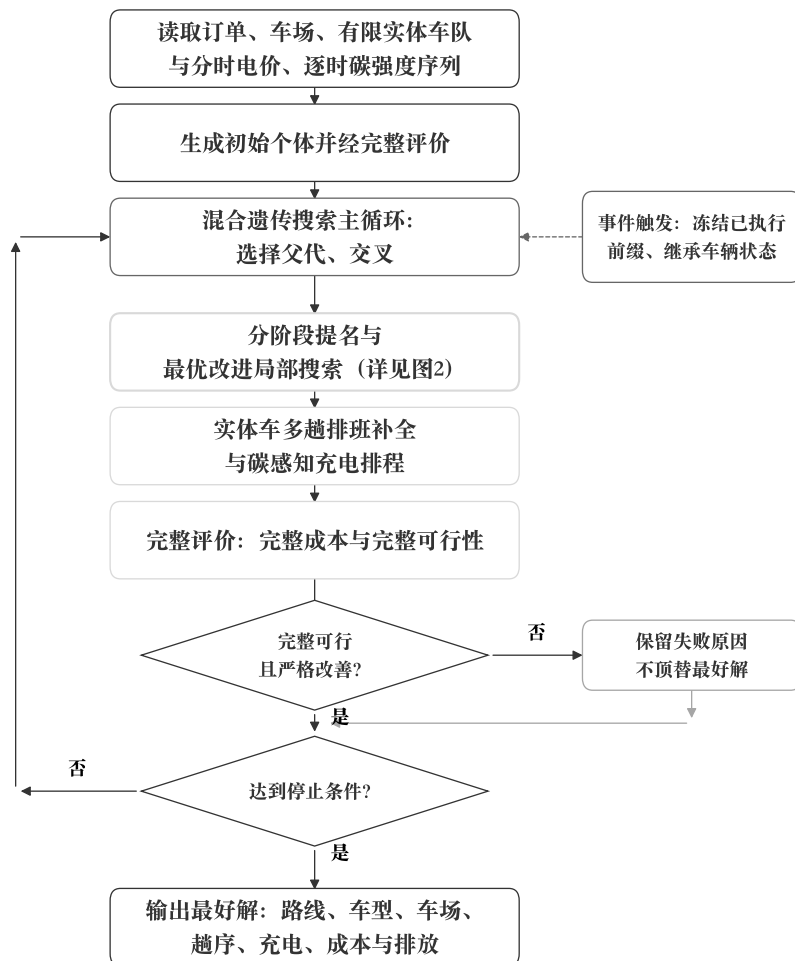


图1 本文算法流程图

3.2 算法步骤

步骤 1: 读取状态并构造可行起点。静态阶段读取订单、车场、有限实体车队与分时电价、逐时电网碳强度序列；动态阶段读取式(16)–(17)定义的剩余客户与车辆状态。初始个体先由完整模型检查，缺客户或未通过完整评价的候选不进入种群。

步骤 2: 执行混合遗传搜索主循环。算法维护可行与不可行两个子种群，按锦标赛选择父代并执行交叉。罚系数依可行解比例自适应调整。停止条件与评价预算由本项目在同一算例、同一机器上的实测收敛记录标定，不直接照搬其他算法的迭代数。

步骤 3: 分阶段提名与最优改进局部搜索。交叉子代进入局部搜索，其候选动作由提名器按阶段顺序供给，如图2所示：先给出路线邻域动作，包括客户重定位、客户交换、路段反转、路段重定位与尾段交换；再给出机制作，包括整车车型交换、充电时刻调整与整趟交换。两个阶段的动作按顺序拼接并去重，汇入同一条提名流。局部搜索每轮扫描该提名流的全部候选，按完整评价结果选出其中最优的一个动作并应用，重复至本轮不再产生改善。提名阶段的先后只决定候选出现的次序，取舍一律由完整评价器裁决，

7.2 陈雨蝶等 (2025): 同刊模型章表达

述为: 多个配送中心共享客户、车辆等资源, 联合为多个客户提供送货和取货服务; 配送中心使用多隔间冷藏车, 每个隔间可以独立控制温度, 以运输储存条件不同或不相容的冷链产品, 例如一辆车可以同时运输需要冷冻的海鲜和需要冷藏的新鲜蔬菜; 此外, 车辆能够在单次行程中完成多个客户点的送货和取货任务, 提升车辆利用率; 所有车辆在同一时刻从配送中心出发, 完成客户服务后就近返回配送中心. 模型的优化目标是在满足载重等约束的前提下, 帮助企业合理制定运输方案, 减少运输成本和碳排放.

模型假设条件如下:1) 客户点、配送中心、车辆等信息已知.2) 每个客户只能由一辆车服务一次.3) 车辆在运输过程中不得超过隔间的最大载重.4) 考虑实际路况等因素, 车辆运输时变速行驶.5) 客户具有软时间窗要求, 车辆早到需等待后才能服务. 模型符号说明如表1所示.

表 1 符号说明

符号	说明	符号	说明
V	点集合 $V = \{0, 1, 2, \dots, m+n-1\}$, 其中 m, n 是客户点和配送中心数目	δ_2^p	车门打开时产品 p 的新鲜度衰减增量系数
V_c	客户点集合 $V_c = \{0, \dots, m-1\}$	Q^e	隔间 e 的最大载重 (t)
V_d	配送中心集合 $V_d = \{m, \dots, m+n-1\}$	Q_i^p	车辆从客户点 i 离开时产品 p 的载重 (t)
K	车辆集合 $K = \{1, \dots, k_s\}$, k_s 是车辆使用数	Q_{ij}	车辆在节点 i 与节点 j 之间的载重 (t)
P	产品集合 $P = \{1, 2, \dots, p, \dots\}$	q	车辆自重 (t)
E	隔间集合 $E = \{1, 2, \dots, e, \dots\}$	v_{ij}	车辆从节点 i 到节点 j 的平均行驶速度 (km/h), $v_{ij} = \frac{d_{ij}}{t_{ij}}$
d_{ij}	节点 i 与节点 j 之间的距离 (km)	t_{ij}	车辆从节点 i 到节点 j 的行驶时间 (h)
s_i	车辆在客户点 i 处的服务时间 (h)	t_i	车辆从客户点 i 返回配送中心所需时间 (h)
$[DET, DLT]$	配送中心时间窗, 其中 DET 是车辆从配送中心出发的时间	ξ	能量单位转化系数
$[ET_i, LT_i]$	客户 i 的时间窗	ψ	车辆引擎效率
t_i	车辆到达节点 i 的时间	g_e	每单位燃油消耗产生的能量 (kW·h/L)
q_i^p	客户点 i 对产品 p 的送货需求 (t)	a_v	车辆行驶过程中的加速度 (m/s ²)
d_i^p	客户点 i 对产品 p 的取货需求 (t)	g	重量加速度 (m/s ²)
c_1	单位碳价格 (元/kg)	θ_{ij}	节点 i 与节点 j 之间的道路坡度
c_2	每辆车的固定成本 (元/辆)	C_r	滚动摩擦系数
c_3	单位燃油价格 (元/L)	C_d	空气阻力系数
c_{41}	运输和等待时单位时间制冷剂成本 (元/h)	A	迎风表面积 (m ²)
c_{42}	服务时单位时间制冷剂成本 (元/h)	ρ_A	空气密度 (kg/m ³)
c^p	产品 p 单位质量价格 (元/t)	η	碳排放系数 (kg/L)
c_{61}	车辆早到单位时间惩罚成本 (元/h)	T_m	碳配额 (kg)
c_{62}	车辆晚到单位时间惩罚成本 (元/h)	x_{ijk}	0-1 变量, 如果车辆 k 从节点 i 到节点 j , 则为 1, 否则为 0
ω_1	运输和等待时单位时间制冷的油耗 (L/h)	y_{ik}	0-1 变量, 如果车辆 k 服务了客户点 i , 则为 1, 否则为 0
ω_2	服务时单位时间制冷的油耗 (L/h)	z_{ep}	0-1 变量, 如果产品 p 储存在隔间 e 中, 则为 1, 否则为 0
δ_1^p	车门关闭时产品 p 的新鲜度衰减系数		

2.2 目标函数

2.2.1 碳排放成本 F_1

碳排放成本根据油耗计算, 采用综合模式排放模型 (CMEM)^[17] 计算车辆行驶时的油耗 f_1 , 其中 a_{ij} 是路况参数, ϕ 是车型参数, 由公式(2)和(3)求得, 计算时应注意单位之间的换算. 公式(4)中 f_2 是车辆制冷时的油耗. 在碳交易机制中, 政府指定碳配额, 企业碳排放量超过限额时需要购买更多的碳排放权, 相反也可以通过出售碳排放权赚取利润, 此时碳排放成本的计算如公式(5)所示.

$$f_1 = \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} x_{ijk} d_{ij} * [a_{ij} * (q + Q_{ij}) + \phi * (v_{ij})^2] / (\xi * \psi * g_e) \quad (1)$$

$$a_{ij} = a_v + g \sin \theta_{ij} + g C_r \cos \theta_{ij} \quad (2)$$

$$\phi = 0.5 C_d A \rho_A \quad (3)$$

$$f_2 = \omega_1 \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} x_{ijk} t_{ij} + \omega_1 \sum_{i \in V_c} \max\{(ET_i - t_i), 0\} + \omega_2 \sum_{i \in V_c} s_i \quad (4)$$

$$F_1 = c_1 * [\eta * (f_1 + f_2) - T_m] \quad (5)$$

2.2.2 固定成本 F_2

固定成本主要是指车辆折旧费、维修费、保险费、制冷设备损耗、驾驶员工资等, 与使用车辆数成正比.

$$F_2 = c_2 * k_s \quad (6)$$

2.2.3 油耗成本 F_3

油耗成本与车辆制冷和运输时产生的油耗成正比.

$$F_3 = c_3 * (f_1 + f_2) \quad (7)$$

2.2.4 制冷成本 F_4

冷藏车利用制冷剂的循环流动实现制冷. 制冷剂成本与时间相关, 制冷时间越长, 成本也就越高. 此外, 冷藏车车门打开会影响车辆的油耗、制冷和损坏成本, 车辆运输和等待时车门紧闭, 车内温度保持稳定; 服务客户时车门打开, 车内冷空气流出, 车外热空气流入, 冷藏车需要消耗更多的燃油和制冷剂来保持车厢内的低温环境, 所以一般情况下 $\omega_1 < \omega_2$ 、 $c_{41} < c_{42}$.

$$F_4 = c_{41} \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} x_{ijk} t_{ij} + c_{41} \sum_{i \in V_c} \max\{(ET_i - t_i), 0\} + c_{42} \sum_{i \in V_c} s_i \quad (8)$$

2.2.5 损坏成本 F_5

服务客户时车门打开, 货物损坏程度与车门关闭时有明显差别, 很多冷链物流相关文献参考货物新鲜度衰减函数把车辆运输和等待时的损坏成本及车辆服务时的损坏成本分开计算并求和. 因为运输过程中车门会多次打开关闭, 时间难以分割, 所以计算车辆运输和等待时的损坏成本会使用车辆开始服务时间进行计算, 这个时间不仅包含了车辆运输和等待的时间, 也包含了已服务客户的服务时间, 所以存在车辆服务时损坏成本重复计算的问题. 为了完善冷链物流模型, 本文重新定义 δ_1^p 和 δ_2^p 两个参数, 前者是车门关闭时产品 p 的新鲜度衰减系数, 后者是车门打开时产品 p 的新鲜度衰减增量系数, 因此公式(9)和(10)中 F_{51} 是车门关闭时的损坏成本, F_{52} 是车门打开时增加的损坏成本, 这样不会产生成本重复计算的问题. 此外, 与独立送货和单隔间模型不同的是, 同时送取货时送货及取货产品均会产生损坏成本, 且多隔间车辆配送多种产品, 每种产品的损坏程度并不相同, 这两点在损坏成本计算时有所体现.

$$F_{51} = \sum_{i \in V_c} \sum_{k \in K} \sum_{p \in P} y_{ik} c^p [q_i^p (1 - e^{-\delta_1^p (\max\{t_i, ET_i\} - DET)}) + d_i^p (1 - e^{-\delta_1^p t_i'})] \quad (9)$$

$$F_{52} = \sum_{i \in V_c} \sum_{k \in K} \sum_{p \in P} y_{ik} c^p Q_i^p (1 - e^{-\delta_2^p * s_i}) \quad (10)$$

$$F_5 = F_{51} + F_{52} \quad (11)$$

2.2.6 惩罚成本 F_6

车辆早到或晚到时客户满意度都会降低, 需要支付惩罚成本; 车辆早到时车辆需等待后服务, 等待时需要制冷、产生油耗、增加碳排放、提高产品损坏风险, 即车辆等待会额外增加成本. 一些文献将车辆早到时的等待成本归类到惩罚成本中, 因此模型仅考虑了车辆晚到时客户满意度降低带来的惩罚成本, 忽略了早到客户满意度降低的潜在风险, 且这样计算时等待成本也只与时间相关, 忽略了载重、速度、距离、客户需求等因素对等待成本的影响. 为了完善冷链物流模型, 本文将车辆早到时的惩罚成本和等待成本独立开, 前者在惩罚成本中计算, 后者参考现实情况和相关文献合理地计算在碳排放、油耗、损坏和制冷成本中^[18].

$$F_6 = \sum_{i \in V_c} [c_{61} * \max\{(ET_i - t_i), 0\}] + \sum_{i \in V_c} [c_{62} * \max\{(t_i - LT_i), 0\}] \quad (12)$$

2.3 时变速度函数

为了简化数学模型, 很多学者将车辆速度假定为常值, 或者采用简单的分段函数或线性函数, 这并不符合现实运输情况. 鉴于此, 本文参考文献^[19]以三角函数表示现实中车辆速度的变化趋势, 速度 $v(t)$ 的具体计算公式如(13)所示, 其中 A 、 B 、 C 、 D 为道路相关参数.

$$v(t) = \begin{cases} A_1 * \sin[(B_1 * t - C_1) * \pi] + D_1 & , 0 \leq t < T_1 \\ A_2 * \sin[(B_2 * t - C_2) * \pi] + D_2 & , T_1 \leq t < T_2 \\ \dots & \\ A_L * \sin[(B_L * t - C_L) * \pi] + D_2 & , T_{L-1} \leq t < 24 \end{cases} \quad (13)$$

速度为常值 v_k 时, $t_{ij} = d_{ij}/v_k$; 速度随时间动态变化时, t_{ij} 可由积分求上限得到. 假设车辆离开节点 i 的时间 t_i'' 在 $[T_l, T_{l+1}]$ 之间, 首先判断车辆是否跨时段.

1) 如果 $d_{ij} \leq \int_{t_i''}^{T_{l+1}} v(t)dt$, 车辆可以在当前时段内从节点 i 到达节点 j , 则车辆到达节点 j 的时间 t_j 可由 $\int_{t_i''}^{t_j} v(t)dt = d_{ij}$ 求出, 进而通过 $(t_j - t_i'')$ 求出 t_{ij} .

2) 如果 $d_{ij} > \int_{t_i''}^{T_{l+1}} v(t)dt$, 车辆需要跨时段才可以从节点 i 到达节点 j . 假设车辆需要跨越 M 个时段, 用 d_{ij} 减去前 $M-1$ 个时段车辆行驶的距离得到车辆在最后一个时段 $[T_{l+M-1}, T_{l+M}]$ 中行驶的距离 d'_{ij} , 则 t_j 可以由 $\int_{T_{l+M-1}}^{t_j} v(t)dt = d'_{ij}$ 求出, 进而通过 $(t_j - t_i'')$ 求出 t_{ij} .

2.4 模型建立

$$\min F = F_1 + F_2 + F_3 + F_4 + F_5 + F_6 \quad (14)$$

$$\sum_{i \in V_d} \sum_{j \in V_c} x_{ijk} = \sum_{i \in V_d} \sum_{j \in V_c} x_{jik} = 1, \forall k \in K \quad (15)$$

$$\sum_{i \in V_d} x_{ijk} = 0, \forall k \in K, \forall j \in V_d, i \neq j \quad (16)$$

$$\sum_{k \in K} \sum_{j \in V} x_{ijk} = 1, \forall i \in V_c, j \neq i \quad (17)$$

$$\sum_{k \in K} \sum_{i \in V} x_{ijk} = 1, \forall j \in V_c, i \neq j \quad (18)$$

$$\sum_{j \in V} Q_{ji} + \sum_{p \in P} (d_i^p - q_i^p) = \sum_{j \in V} Q_{ij}, \forall i \in V_c \quad (19)$$

$$\sum_{i \in V_c} \sum_{p \in P} q_i^p y_{ik} z_{ep} \leq Q^e, \forall k \in K, \forall e \in E \quad (20)$$

$$\sum_{p \in P} z_{ep} Q_i^p \leq Q^e, \forall i \in V_c, \forall e \in E \quad (21)$$

$$t_j = \sum_{k \in K} \sum_{i \in V_c} (\max\{t_i, ET_i\} + s_i + t_{ij}) x_{ijk} + \sum_{k \in K} \sum_{l \in V_d} (DET + t_{lj}) x_{ljk}, \forall j \in V_c \quad (22)$$

$$\sum_{i \in S} \sum_{j \in S} x_{ijk} \leq |S| - 1, \forall S \subseteq V_c, \forall k \in K \quad (23)$$

$$DET \leq t_i \leq DLT, \forall i \in V \quad (24)$$

$$x_{ijk}, y_{ik}, z_{ep} \in \{0, 1\}, \forall i \in V, \forall j \in V, \forall k \in K, \forall e \in E, \forall p \in P \quad (25)$$

公式(14)表示运输总成本最小化; 公式(15)表示车辆可返回任意配送中心; 公式(16)保证车辆不连续经过配送中心; 公式(17)和(18)保证客户只服务一次; 公式(19)保证装卸平衡, 即车辆离开客户点 i 时的载重等于车辆到达 i 时的载重减去客户的送货需求, 加上客户的取货需求; 公式(20)保证车辆从配送中心出发时每个隔间不能超载; 公式(21)保证车辆服务完任意客户点后每个隔间不能超载; 公式(20)和(21)可以表示配送全过程的载重限制; 公式(22)保证车辆到达客户点的时间约束; 公式(23)防止形成子回路; 公式(24)表示时间窗限制条件; 公式(25)表示 0-1 变量。

3 算法设计

3.1 算法流程图

本文建立的 MCTDGVVRPSD 模型综合考虑了联合配送、时间窗、同时送取货、时变路网、多产品、多隔间以及碳排放等因素, 形成了一个复杂的冷链物流模型。这些特点导致模型具有多重约束, 解空间表现出高度的非线性和不确定性。为应对这些挑战, 设计一种具有变邻域搜索操作和动态灾变机制的多种群遗传算法 (Multiple Population Genetic Algorithm with Variable Neighborhood Search and Dynamic Catastrophe Mechanism, MPGA-VNS), 算法流程如图1所示。该算法以遗传算法为主体, 通过改进两点交叉、动态灾变机制和多种群并行进化进行优化, 增强其全局搜索能力。同时, 采用变邻域搜索操作对当前最优解进行优化, 以提升局部搜索能力。这种设计充分发挥了遗传算法的全局搜索优势和变邻域搜索算法的局部优化能力, 从而有效应对模型中的复杂约束, 确保在复杂解空间中实现更高效的探索和求解。

7.3 陈婉茹等 (2023): 混合车队、能耗与碳交易

表 2 模型主要参数

符号	含义
q_i	客户点 i 的货物需求
s_i	客户点 i 的服务 (装卸) 时间
$[e_i, l_i]$	客户点 i 的时间窗
d_{ij}	车辆在弧 (i, j) 上的行驶距离
v_l	车辆在弧 (i, j) 上的行驶速度的下限
v_u	车辆在弧 (i, j) 上的行驶速度的上限
h	车辆一次派遣的最长行程时间
Q_g	燃油车最大载重
Q_e	电动车最大载重
B	电池容量
w_d	配送中心 d 内可用车辆数
$\overline{CE}$	企业的碳排放总量限额
M	一个足够大的正数
x_{ij}^k	二元决策变量, 表示车辆 k 是否从节点 i 直接驶向节点 j , 是为 1, 否为 0
u_{ij}^k	决策变量, 表示车辆 k 由节点 i 到节点 j 的载货量
v_{ij}^k	决策变量, 表示车辆 k 由节点 i 到节点 j 的行驶速度
τ_i^k	决策变量, 表示车辆 k 到达节点 i 处的时间
t_i^k	决策变量, 表示车辆 k 在节点 i 处的等待时间

2.3 车辆能耗模型

采用文献^[16]中的实际能耗模型, 基于机械功率计算车辆在行驶过程中产生的电耗和油耗. 不考虑城市道路的坡度角, 车辆 $k \in K_e^d \cup K_g^d, d \in D$ 在弧 (i, j) 上的恒定机械功率 T_{ij}^k 是关于载重 u_{ij}^k 和速度 v_{ij}^k 的多元函数, 可由下式近似求得:

$$T_{ij}^k(u_{ij}^k, v_{ij}^k) = \frac{1}{2} \cdot c_d \cdot \rho \cdot A \cdot v_{ij}^{k3} + (m_k + u_{ij}^k) \cdot g \cdot c_r \cdot v_{ij}^k, \quad (1)$$

其中 c_d 为空气阻力系数, ρ 为空气密度, A 为迎风面积, g 表示重力常量, c_r 表示滚动阻力系数, m_k 为车辆 k 的空车质量. 电动车 $k \in K_e^d, d \in D$ 在弧 (i, j) 上的电耗 b_{ij}^k 等于机械功率 T_{ij}^k 与电动发动机效率 ϕ^d 、电池放电效率 φ^d 和行驶时间的乘积, 行驶时间由弧 (i, j) 的行驶距离和车辆行驶速度求得:

$$b_{ij}^k(u_{ij}^k, v_{ij}^k) = \phi^d \cdot \varphi^d \cdot T_{ij}^k(u_{ij}^k, v_{ij}^k) \cdot d_{ij}/v_{ij}^k. \quad (2)$$

燃油车 $k \in K_g^d, d \in D$ 在弧 (i, j) 上的单位时间油耗率 G_{ij}^k 与机械功率 T_{ij}^k 满足下式:

$$G_{ij}^k(u_{ij}^k, v_{ij}^k) = \xi \cdot (\varepsilon \cdot R + T_{ij}^k(u_{ij}^k, v_{ij}^k)/\eta)/(\kappa \cdot \psi), \quad (3)$$

其中, ξ 表示柴油空气质量比、 κ 表示柴油热值、 ε 表示摩擦因数、 R 为发动机技术参数、 ψ 为燃油率转换系数、 η 为柴油发动机效率参数. 将单位时间油耗率 G_{ij}^k 乘以燃油车 $k \in K_g^d, d \in D$ 在弧 (i, j) 上的行驶时间即得到相应油耗, 行驶时间由弧 (i, j) 的行驶距离和车辆行驶速度求得:

$$f_{ij}^k(u_{ij}^k, v_{ij}^k) = G_{ij}^k(u_{ij}^k, v_{ij}^k) \cdot d_{ij}/v_{ij}^k. \quad (4)$$

根据式 (2)、(4), 有如下命题成立:

命题 1 b_{ij}^k 为关于 u_{ij}^k, v_{ij}^k 的凸函数, 对于弧 (i, j) , 已知载重 u_{ij}^k , 存在电耗最小路由速度 $v_{ij}^e \in [v_l, v_u]$.

证明 当 $u_{ij}^k > 0, v_{ij}^k > 0$ 时, b_{ij}^k 具有二阶连续偏导数, 参数 $\phi^d, \varphi^d, d_{ij}, c_d, \rho, A$ 均大于 0, 则有: $\frac{d^2 b_{ij}^k}{dv_{ij}^{k2}} > 0, \frac{d^2 b_{ij}^k}{dv_{ij}^{k2}} \cdot \frac{d^2 b_{ij}^k}{du_{ij}^{k2}} - \left(\frac{d^2 b_{ij}^k}{dv_{ij}^k du_{ij}^k}\right)^2 = 0$, 故 b_{ij}^k 为关于 u_{ij}^k, v_{ij}^k 的凸函数. 若已知弧段载重 u_{ij}^k , 有 $\frac{db_{ij}^k}{dv_{ij}^k} > 0$, 故电耗关于速度单调递增, 在给定速度范围内存在唯一速度 v_{ij}^e 使得电耗最小.

命题 2 f_{ij}^k 为关于 u_{ij}^k, v_{ij}^k 的凸函数, 对于弧 (i, j) , 已知载重 u_{ij}^k , 存在油耗最小路由速度 $v_{ij}^g \in [v_l, v_u]$.

证明 同理, 当 $u_{ij}^k > 0, v_{ij}^k > 0$ 时, f_{ij}^k 具有二阶连续偏导数, 且有: $\frac{d^2 f_{ij}^k}{dv_{ij}^{k2}} > 0, \frac{d^2 f_{ij}^k}{dv_{ij}^{k2}} \cdot \frac{d^2 f_{ij}^k}{du_{ij}^{k2}} - (\frac{d^2 f_{ij}^k}{dv_{ij}^k du_{ij}^k})^2 = 0$, 故 f_{ij}^k 为关于 u_{ij}^k, v_{ij}^k 的凸函数. 若已知弧段载重 u_{ij}^k , 令 $\frac{df_{ij}^k}{dv_{ij}^k} = 0$, 得 $v_{ij}^G = \sqrt[3]{\frac{\varepsilon \cdot R \cdot \eta}{c_d \cdot \rho \cdot A}}$: 当 $0 < v_{ij}^k < v_{ij}^G$ 时, $\frac{df_{ij}^k}{dv_{ij}^k} < 0$, 油耗关于速度单调递减; 当 $v_{ij}^k > v_{ij}^G$ 时, $\frac{df_{ij}^k}{dv_{ij}^k} > 0$, 油耗关于速度单调递增, 故在给定速度范围内存在速度 v_{ij}^g 使得油耗最小.

2.4 数学模型

2.4.1 优化目标

模型以固定成本、能耗成本和碳交易成本之和最低作为优化目标, 固定成本与派遣车辆数相关, 能耗成本与车辆配送产生的油耗和电耗相关, 碳交易成本与碳排放量相关, 具体计算方式如式 (5) 所示.

$$\begin{aligned} \text{Min} F = & p_1 \cdot \sum_{k \in K_e^i} \sum_{j \in N} \sum_{i \in D} x_{ij}^k + p_2 \cdot \sum_{k \in K_g^i} \sum_{j \in N} \sum_{i \in D} x_{ij}^k + \\ & p_3 \cdot \sum_{(i,j) \in E} \sum_{k \in K_e^d} \sum_{d \in D} x_{ij}^k \cdot b_{ij}^k(u_{ij}^k, v_{ij}^k) + p_4 \cdot \sum_{(i,j) \in E} \sum_{k \in K_g^d} \sum_{d \in D} x_{ij}^k \cdot f_{ij}^k(u_{ij}^k, v_{ij}^k) + \\ & p_5 \cdot \left(\sum_{(i,j) \in E} \sum_{k \in K_g^d} \sum_{d \in D} \lambda \cdot x_{ij}^k \cdot f_{ij}^k(u_{ij}^k, v_{ij}^k) - \overline{\text{CE}} \right), \end{aligned} \quad (5)$$

其中, p_1 和 p_2 分别为电动车和燃油车的单位固定成本, p_3 和 p_4 分别为单位电耗成本和单位油耗成本, p_5 为单位碳交易价格, λ 为单位燃油二氧化碳排放因子.

2.4.2 模型约束

模型约束如下:

$$\sum_{j \in V \setminus i} \sum_{k \in K_e^d \cup K_g^d} \sum_{d \in D} x_{ij}^k = 1, \quad \forall i \in N, \quad (6)$$

$$\sum_{j \in V \setminus i} x_{ji}^k = \sum_{j \in V \setminus i} x_{ij}^k, \quad \forall i \in N, d \in D, k \in K_e^d \cup K_g^d, \quad (7)$$

$$\sum_{i \in N} x_{ji}^k = \sum_{i \in N} x_{ij}^k, \quad \forall j \in D, k \in K_e^j \cup K_g^j, \quad (8)$$

$$\sum_{j \in N} \sum_{k \in K_e^i \cup K_g^i} x_{ij}^k \leq \omega_i, \quad \forall i \in D, \quad (9)$$

$$\tau_i^k \geq e_i, \quad \forall i \in D, k \in K_e^i \cup K_g^i, \quad (10)$$

$$\tau_i^k + d_{ij}/v_{ij}^k - M \cdot (1 - x_{ij}^k) \leq \tau_j^k, \quad \forall i \in D, j \in N, k \in K_e^i \cup K_g^i, \quad (11)$$

$$\tau_i^k + t_i^k + s_i + d_{ij}/v_{ij}^k - M \cdot (1 - x_{ij}^k) \leq \tau_j^k, \quad \forall i \in N, j \in N \setminus i, d \in D, k \in K_e^d \cup K_g^d, \quad (12)$$

$$e_i \leq \tau_i^k + t_i^k \leq l_i, \quad \forall i \in N, d \in D, k \in K_e^d \cup K_g^d, \quad (13)$$

$$\tau_i^k + t_i^k + s_i + d_{ij}/v_{ij}^k - M \cdot (1 - x_{ij}^k) \leq l_j, \quad \forall i \in N, j \in D, k \in K_e^j \cup K_g^j, \quad (14)$$

$$t_i^k = 0, \quad \forall i \in D, k \in K_e^i \cup K_g^i, \quad (15)$$

$$\sum_{(i,j) \in E} x_{ij}^k \cdot (d_{ij}/v_{ij}^k + t_j^k + s_j) \leq h, \quad \forall d \in D, k \in K_e^d \cup K_g^d, \quad (16)$$

$$\sum_{i \in N} u_{ij}^k = 0, \quad \forall j \in D, k \in K_e^j \cup K_g^j, \quad (17)$$

$$\sum_{i \in V \setminus j} u_{ij}^k - \sum_{i \in V \setminus j} u_{ji}^k + M \left(1 - \sum_{i \in V \setminus j} x_{ij}^k \right) \geq q_j, \quad \forall j \in N, d \in D, k \in K_e^d \cup K_g^d, \quad (18)$$

$$u_{ij}^k \leq Q_e, \quad \forall (i, j) \in E, d \in D, k \in K_e^d, \quad (19)$$

$$u_{ij}^k \leq Q_g, \quad \forall (i, j) \in E, d \in D, k \in K_g^d, \quad (20)$$

$$\sum_{(i,j) \in E} x_{ij}^k \cdot b_{ij}^k(u_{ij}^k, v_{ij}^k) \leq B, \quad \forall d \in D, k \in K_e^d, \quad (21)$$

$$x_{ij}^k = \{0, 1\}, \quad \forall (i, j) \in E, d \in D, k \in K_e^d \cup K_g^d, \quad (22)$$

$$v_l \leq v_{ij}^k \leq v_u, \quad \forall (i, j) \in E, d \in D, k \in K_e^d \cup K_g^d, \quad (23)$$

$$u_{ij}^k \geq 0, \quad \forall (i, j) \in E, d \in D, k \in K_e^d \cup K_g^d, \quad (24)$$

$$\tau_i^k \geq 0, \quad \forall i \in V, d \in D, k \in K_e^d \cup K_g^d, \quad (25)$$

$$t_i^k \geq 0, \quad \forall i \in V, d \in D, k \in K_e^d \cup K_g^d. \quad (26)$$

式 (6) 表示所有客户点的需求都将被唯一一车辆一次满足. 式 (7) 为客户点车辆进出平衡约束. 式 (8) 表示车辆行程起止于同一配送中心. 式 (9) 为车场容量约束. 式 (10) 限制车辆在配送中心的出发时间. 式 (11) 表示配送中心与第一个客户点的时间递推关系. 式 (12) 表示前后相继访问客户点的时间递推关系. 式 (13) 为客户点时间窗约束. 式 (14) 限制车辆返回配送中心的时间. 式 (15) 表示车辆在配送中心的等待时间为 0. 式 (16) 限制车辆一次派遣的最长行驶时间. 式 (17) 表示车辆返回配送中心时的载货量为 0. 式 (18) 为车辆在途的货流量守恒约束. 式 (19) 和 (20) 限制车辆在任意弧段上的载重小于车辆装载能力. 式 (21) 限制电动车配送过程中消耗的电能不超过电池容量. 式 (22)~(26) 限制变量的取值范围.

3 算法设计

3.1 算法框架

多车场带时间窗的车辆路径问题属于 NP-hard 问题^[23], 在此基础上, 本文进一步考虑由电动车和燃油车构成的混合动力车队, 并将与能耗直接相关的速度作为连续决策变量, 解空间更加复杂, 求解难度大. 鉴于变邻域搜索算法在求解多车场带时间窗车辆路径问题和异质车队车辆路径问题中的优异表现^[23-25], 本文设计了考虑速度优化的并行改进变邻域算法 (parallel improved variable neighborhood search algorithm with speed optimization, PIVNS-SOA) 对问题进行求解, 算法流程如图 1 所示.

为兼顾 PIVNS-SOA 算法的搜索深度和广度, 在设计有效扰动策略的同时, 将大邻域搜索算法中的全局破坏-修复搜索嵌入变邻域搜索算法的基本框架; 为提升算法的搜索效率, 采用并行局部搜索策略加速算法的求解; 此外, 根据实际能耗模型中速度与能耗的相关关系, 设计速度优化方法, 并应用于并行局部搜索和全局破坏-修复搜索阶段对解的评价前.

3.2 评价函数

为更好地地区分空间中解的可行性, 在评价函数中除了对解的成本进行评估外, 需要对约束的违反加以惩罚, 以促使搜索朝着可行的方向收敛. 基于此, 本文设计评价函数为:

$$F + \alpha \cdot V_1(x) + \beta \cdot V_2(x) + \gamma \cdot V_3(x) + \chi \cdot V_4(x), \quad (27)$$

其中, F 为由式 (5) 得到的总成本; $V_1(x)$, $V_2(x)$, $V_3(x)$, $V_4(x)$ 分别为当前解 x 的时间窗约束式 (10)、(13)、(14)、行程时间约束式 (16)、载重约束式 (19)、(20) 和电池容量约束式 (21) 的违背量; α , β , γ , χ 为相应惩罚因子.

7.4 Goeke-Schneider (2015): 混合车队能耗原式

force. Let m denote the total vehicle mass, g the gravitational constant, c_r the rolling friction coefficient (which depends on tire pressure, road surface conditions and other factors) and α the gradient angle. Then the rolling resistance F_r can be determined as

$$F_r = c_r \cdot m \cdot g \cdot \cos(\alpha).$$

With v denoting the speed, c_d the aerodynamic drag coefficient, ρ_a the air density and A_f the frontal area of the vehicle, the aerodynamic resistance is:

$$F_a = \frac{1}{2} \cdot \rho_a \cdot A_f \cdot c_d \cdot v^2.$$

Adding the gravitational force $F_g = m \cdot g \cdot \sin(\alpha)$, the total mechanical power P_M is:

$$P_M = \left(m \cdot a + \frac{1}{2} \cdot c_d \cdot \rho \cdot A \cdot v^2 + m \cdot g \cdot \sin(\alpha) + c_r \cdot m \cdot g \cdot \cos(\alpha) \right) \cdot v, \quad (1)$$

where a denotes the acceleration. As mentioned above, a is assumed to be zero in the following as we do not consider acceleration and braking processes.

In the second step, the electric power P_E that is needed to achieve the required mechanical power P_M is calculated, taking into account energy losses that occur in the electric engine. To this end, we use the Quantized State System (QSS) model of an electric engine presented in (Guzzella & Amstutz, 2005). The authors provide values for the engine efficiency at given pairs of torque and rotational velocity. We directly convert torque and rotational velocity to mechanical power. Using homogeneous regression, i.e. a linear regression with a y-axis intercept of zero, on the converted values, we obtain the following relationship between P_M and the discharged electric energy P_E^d (respectively the recuperated electric energy P_E^r) that is described by the regression coefficient ϕ :

$$\begin{aligned} P_E^d &= \phi^d \cdot P_M \text{ for } 0 \text{ kilowatts} \leq P_M \leq 100 \text{ kilowatts} \\ P_E^r &= \phi^r \cdot P_M \text{ for } -100 \text{ kilowatts} \leq P_M < 0 \text{ kilowatts.} \end{aligned}$$

The determined relationship is adequate for a mechanical power demand of up to 100 kilowatts, which we consider a realistic threshold for commonly employed ECV models.

In the third step, we compute the battery power P_B that is necessary to obtain the required electric energy P_E . For this purpose, we use the relationship for converting electric power P_E into battery power P_B for a lithium-ion accumulator presented in van Keulen, de Jager, Serrarens, and Steinbuch (2010). The presented relationship does not account for external effects (e.g., the ambient temperature) or the influence of the current charge level of the battery. We approximate the given relation by means of a homogeneous regression and obtain as final equations for discharging (P_B^d) and recuperating (P_B^r)

$$\begin{aligned} P_B^d &= \phi^d \cdot P_E \text{ for } P_E \geq 0 \text{ kilowatts} \\ P_B^r &= \phi^r \cdot P_E \text{ for } P_E < 0 \text{ kilowatts,} \end{aligned}$$

with the regression coefficient ϕ that describes the battery efficiency.

To compute energy consumption on a graph, vehicle mass is defined as a function of the currently loaded amount of cargo u , i.e., $m(u) = m_c + m_u \cdot u$, where m_c is the curb mass of the vehicle and m_u the mass of one unit of cargo. Let u_j denote the amount of cargo when arriving at customer j (before unloading), i.e., the amount required to fulfill the demand at customer j and all following customers. Further, let $P_{ij}(u_j)$ denote the constant mechanical power demand over the course of an arc (i, j) with a load that is equivalent to the remaining customer demand:

$$P_{ij}(u_j) = \left(\frac{1}{2} \cdot c_d \cdot \rho \cdot A \cdot v^2 + m(u_j) \cdot g \cdot (\sin(\alpha_{ij}) + c_r \cdot \cos(\alpha_{ij})) \right) \cdot v_{ij}$$

Then, the battery power demand on this arc is given by $P_B(P_E(P_{ij}(u_j)))$. Multiplying with the associated travel time t_{ij} of the arc yields the electric energy consumption b_{ij} for traveling this particular arc:

$$b_{ij}(u_j) = \begin{cases} \phi^d \cdot \varphi^d \cdot P_{ij}(u_j) \cdot t_{ij} & \text{if } P_{ij}(u_j) \geq 0 \text{ kilowatts} \\ \phi^r \cdot \varphi^r \cdot P_{ij}(u_j) \cdot t_{ij} & \text{if } P_{ij}(u_j) < 0 \text{ kilowatts.} \end{cases} \quad (2)$$

All variables except the amount of cargo u_j are constant for a given arc. Therefore, it is possible to split $b_{ij}(u_j)$ into a fixed and a load-dependent part and precompute parts of the expression.

3.2. Energy consumption of combustion engines

Mechanical power can be directly translated to diesel fuel consumption as shown in Demir et al. (2012). The fuel rate is given by $FR = \frac{\xi}{\kappa \cdot \psi} \left(kND + \frac{P_M}{\eta \cdot \eta_{tf}} \right)$, where P_M is the mechanical power as defined in Eq. (1), ξ the fuel-to-air mass ratio, κ the heating value of typical diesel fuel, k the engine friction factor, N the engine speed, D the engine displacement, ψ a factor converting the fuel rate from grams per second to liters per second, η the efficiency parameter for diesel engines and η_{tf} the drive train efficiency. We assume that driving downhill in gear can reduce the engine consumption down to 0. The constant fuel rate over the course of an arc is given by:

$$FR_{ij}(u_j) = \max \left(\frac{\xi}{\kappa \cdot \psi} \left(kND + \frac{P_{ij}(u_j)}{\eta \cdot \eta_{tf}} \right), 0 \right).$$

Thus, the fuel consumption of an ICCV traversing an arc (i, j) with cargo u_j can be calculated as:

$$f_{ij}(u_j) = t_{ij} \cdot FR_{ij}(u_j). \quad (3)$$

4. The Electric Vehicle Routing Problem with Time Windows and Mixed Fleet (E-VRPTWMF)

We define E-VRPTWMF on a complete, directed graph $G = (V_{0,N+1}, A)$. $V = \{1, \dots, N\}$ denotes the set of customers, F the set of recharging stations, F' the set of visits to vertices in F and V' denotes the union of V and F' . Vertices 0 and $N+1$ denote instances of the depot, and all routes start at 0 and end at $N+1$. Inclusion of one or both depot instances is indicated by subscripting the respective set, e.g., the set $V'_{0,N+1}$ contains both instances of the depot. The set of arcs is given by $A = \{(i, j) \mid i, j \in V'_{0,N+1}, i \neq j\}$.

With each vertex $i \in V'_{0,N+1}$, we associate a nonnegative demand q_i ($q_i = 0, i \notin V$), a nonnegative service time s_i ($s_i = 0, i \notin V$) and a time window $[e_i, l_i]$, within which service at the customer has to start. Time windows are hard, i.e., starting service late is not allowed but waiting in case of early arrival is possible. The time window $[e_0, l_{N+1}]$ at the depot specifies the scheduling horizon of the problem. Each arc is described by the following properties: distance d_{ij} , travel speed v_{ij} , travel time $t_{ij} = d_{ij}/v_{ij}$ and a factor α_{ij} representing the gradient of the terrain.

A mixed vehicle fleet of fixed size consisting of m_E ECVs and m_{IC} ICCVs is positioned at the depot. All vehicles have the same maximal cargo loading capacity Q . On each visit to a recharging station, ECVs are recharged to their maximum battery capacity B . The recharging time depends on the recharging rate r and the difference between B and charge level on arrival at the station. Electricity and fuel consumption of a vehicle traveling an arc are determined by the models introduced in Section 3.

Thus, E-VRPTWMF can be formulated as a (nonlinear) mixed-integer program. For every arc $(i, j) \in A$, we define the following binary decision variables: Variables x_{ij}^E are equal to 1 if an arc (i, j) is traveled by an ECV, variables x_{ij}^{IC} if the arc is traveled by an ICCV. Otherwise, the decision variables are 0. Variables τ_i define the arrival time, u_i the remaining cargo (before unloading) and y_i the remaining charge level on arrival at vertex $i \in V'_{0,N+1}$. Table 1 summarizes the variables and parameters of our model.

Table 1

Variable and parameter definitions of the E-VRPTWMF model.

$0, N+1$	Depot instances
F'	Set of visits to recharging stations, dummy vertices of the set of recharging stations F
V	Set of customer vertices $V = \{1, \dots, N\}$
V'	Set of customers and recharging visits: $V' = V \cup F'$
d_{ij}	Distance between vertices i and j
t_{ij}	Travel time between vertices i and j
Q	Vehicle capacity
r	Recharging rate
B	Battery capacity
$b_{ij}(u_j)$	Electric energy consumption between between vertices i and j depending on cargo u_j (Eq. (2))
$f_{ij}(u_j)$	Fuel consumption between between vertices i and j depending on cargo u_j (Eq. (3))
q_i	Demand of vertex i , 0 if $i \notin V$
e_i	Earliest start of service at vertex i
l_i	Latest start of service at vertex i
s_i	Service time at vertex i ($s_0, s_{N+1} = 0$)
τ_i	Decision variable specifying the time of arrival at vertex i
u_i	Decision variable specifying the remaining cargo on arrival at vertex i
y_i	Decision variable specifying the remaining battery capacity on arrival at vertex i
x_{ij}^E	Binary decision variable indicating if arc (i, j) is traveled by an ECV
x_{ij}^C	Binary decision variable indicating if arc (i, j) is traveled by an ICCV

In this work, we consider three alternative objective functions for E-VRPTWMF:

1. Minimize traveled distance f_d :

$$f_d = \sum_{i \in V'_0, j \in V'_{N+1}, i \neq j} d_{ij} (x_{ij}^E + x_{ij}^{IC}) \rightarrow \min \quad (4)$$

2. Minimize cost f_c , consisting of the cost for vehicle propulsion and labor cost. Cost factors c^E and c^C denote the cost of electric energy, respective diesel fuel and c^D the driver wage per time unit:

$$f_c = \sum_{i \in V'_0, j \in V'_{N+1}, i \neq j} [c^E \cdot b_{ij}(u_j) \cdot x_{ij}^E + c^C \cdot f_{ij}(u_j) \cdot x_{ij}^{IC}] + \sum_{i \in V'} c^D (x_{iN+1}^E + x_{iN+1}^{IC}) (\tau_i + s_i + t_{iN+1}) \rightarrow \min \quad (5)$$

3. Minimize cost including cost for battery replacement f_{cb} . The distance traveled by ECVs is multiplied by a cost factor c^B that accounts for the cost of battery replacement. It is expressed as cost per kilometer by dividing acquisition cost by the expected total lifetime mileage.

$$f_{cb} = f_c + \sum_{i \in V'_0, j \in V'_{N+1}, i \neq j} c^B \cdot d_{ij} \cdot x_{ij}^E \rightarrow \min \quad (6)$$

The constraints of E-VRPTWMF are as follows:

$$\sum_{j \in V'} (x_{ij}^E + x_{ij}^{IC}) = 1 \quad \forall i \in V \quad (7)$$

$$\sum_{j \in V'_{N+1}, i \neq j} x_{ij}^E \leq 1 \quad \forall i \in F' \quad (8)$$

$$\sum_{j \in V'_{N+1}, i \neq j} x_{ij}^E - \sum_{j \in V'_0, i \neq j} x_{ji}^E = 0 \quad \forall i \in V' \quad (9)$$

$$\sum_{j \in V'_{N+1}, i \neq j} x_{ij}^{IC} - \sum_{j \in V'_0, i \neq j} x_{ji}^{IC} = 0 \quad \forall i \in V' \quad (10)$$

$$\sum_{j \in V'} x_{0j}^E \leq m_E \quad (11)$$

$$\sum_{j \in V} x_{0j}^{IC} \leq m_{IC} \quad (12)$$

$$\tau_i + (s_i + t_{ij}) (x_{ij}^E + x_{ij}^{IC}) - l_0 (1 - x_{ij}^E - x_{ij}^{IC}) \leq \tau_j \quad \forall i \in V_0, \forall j \in V'_{N+1}, i \neq j \quad (13)$$

$$\tau_i + t_{ij} \cdot x_{ij}^E + r (B - y_i) x_{ij}^E - (l_0 + rB) (1 - x_{ij}^E) \leq \tau_j \quad \forall i \in F', \forall j \in V'_{N+1}, i \neq j \quad (14)$$

$$e_i \leq \tau_i \leq l_i \quad \forall i \in V'_{0,N+1} \quad (15)$$

$$u_i - q_i (x_{ij}^E + x_{ij}^{IC}) + Q (1 - x_{ij}^E - x_{ij}^{IC}) \geq u_j \geq 0 \quad \forall i \in V'_0, \forall j \in V'_{N+1}, i \neq j \quad (16)$$

$$0 \leq u_0 \leq Q \quad (17)$$

$$y_i - b_{ij}(u_j) \cdot x_{ij}^E + B (1 - x_{ij}^E) \geq y_j \quad \forall i \in V, \forall j \in V'_{N+1}, i \neq j \quad (18)$$

$$B - b_{ij}(u_j) \cdot x_{ij}^E \geq y_j \quad \forall i \in F'_0, \forall j \in V'_{N+1}, i \neq j \quad (19)$$

$$B \geq y_i \geq 0 \quad \forall i \in V'_{0,N+1} \quad (20)$$

$$x_{ij}^E, x_{ij}^{IC} \in \{0, 1\} \quad \forall i, j \in V'_{0,N+1}, i \neq j \quad (21)$$

Constraints (7) enforce that each customer visit has exactly one successor. Constraints (8) guarantee that each visit to a recharging station is used at most once, i.e., not all recharging visit vertices must be used. Flow conservation constraints (9) and (10) guarantee for each vertex that the number of incoming arcs is equal to the number of outgoing arcs for each vehicle type. Constraints (11) and (12) ensure that the maximum number of employed vehicles adheres to the fleet composition. Constraints (13) and (14) link arrival times at vertices i and j if the arc from i to j is traveled. Constraints (14) covers the case with recharging visits: Here, recharging times that depend on the remaining charge level y_i when arriving at station i have to be taken into account. Constraints (15) guarantee arrival within the time window at each vertex. Constraints (16) enforce the fulfillment of demand at customer vertices. Constraint (17) restricts the initial cargo load level to the maximum capacity of a vehicle.

Constraints (18) set the battery level at a vertex succeeding a customer visit in accordance with the energy consumption for discharging or recuperating. For vertices succeeding a visit to a recharging station or the depot, Constraints (19) set the battery level equal to the maximum battery capacity reduced by the energy required on the respective arc. Here, the battery is either discharged or recuperation takes place. Constraints (20) restricts the charge level to the maximum battery capacity in order to prevent recuperation beyond the maximal capacity. Finally, binary decision variables are defined in constraints (21).

Although our model covers several real-world aspects of ECVs, we still use some simplifications. Gradient and speed are considered fixed over the course of an arc. However, a more fine-grained topology and acceleration patterns can be integrated, e.g., by considering a path $p(i, j)$ instead of the arc (i, j) . A path $p(i, j)$ is defined by a fixed sequence

7.5 Zhen 等 (2020): 多车场多趟原式

- No overlap of time between each trip of the same vehicle.
- Each customer is served exactly once by one vehicle.
- The total demand served on the trip of a vehicle cannot exceed the vehicle's capacity.

3.2. Model formulation

In this section, we formulate the Multi-D&T VRPTW-R as a MIP model.

3.2.1. Notations

Indices and sets:

i, j, l	index of a customer.
N	set of customers.
k	index of a vehicle.
K	set of vehicles.
w	index of a trip.
W	set of trips for any vehicle.

Note: For any vehicle, the number of the real performed trip(s) gradually increases from 1 with a step size of 1. For instance, for two vehicles k and k' with two and three trips, respectively, the real performed trip sets are $\{1,2\}$ and $\{1,2,3\}$ respectively.

Parameters:

e_i	the earliest service time at customer i .
l_i	the latest service time at customer i .
q_i	the demand for customer i .
t_i^R	release date of packages requested by customer i .
t_i^S	service time of customer i .
t_{ij}^N	travel time between customer i and j .
t_k^D	service time of vehicle k 's depot.
$t_{k,i}$	travel time between vehicle k 's depot and customer i .
T^H	time horizon.
Q	the capacity of each vehicle.
M	a sufficiently large positive number.

Decision variables:

$\tau_{k,w}^L$	time point that vehicle k departs from the depot in trip w .
$\tau_{k,w}^B$	time point that vehicle k returns to the depot in trip w .
$\tau_{k,i,w}$	time point that vehicle k arrives at customer i in trip w .
$\delta_{k,w}^L$	cargo load of vehicle k in trip w when departing from the depot.
$\delta_{k,i,w}$	cargo load of vehicle k in trip w after vehicle k visiting customer i .
$\gamma_{k,w}$	binary equals one if vehicle k operates trip w ; otherwise, equals zero.
$\xi_{i,j,k,w}$	binary equals one if vehicle k visits customer j immediately after visiting customer i in trip w ; otherwise, equals zero.
$\lambda_{k,i,w}$	binary equals one if vehicle k serves customer i in trip w ; otherwise, equals zero.
$\mu_{k,i,w}$	binary equals one if customer i is the first customer served by vehicle k in trip w ; otherwise, equals zero.
$\eta_{k,i,w}$	binary equals one if customer i is the last customer served by vehicle k in trip w ; otherwise, equals zero.

3.2.2. Mathematical model

$$\text{Minimize } Z = \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} \sum_{j \in N} t_{ij}^N \xi_{i,j,k,w} + \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} t_{k,i} \mu_{k,i,w} + \sum_{k \in K} \sum_{w \in W} \sum_{i \in N} t_{k,i} \eta_{k,i,w} \quad (1)$$

Subject to:

$$\sum_{k \in K} \sum_{w \in W} \lambda_{k,i,w} = 1 \quad \forall i \in N \quad (2)$$

$$\mu_{k,l,w} + \sum_{i \in N} \xi_{i,l,k,w} = \eta_{k,l,w} + \sum_{j \in N} \xi_{l,j,k,w} = \lambda_{k,l,w} \quad \forall l \in N, k \in K, w \in W \quad (3)$$

$$\sum_{i \in N} \mu_{k,i,w} = \sum_{i \in N} \eta_{k,i,w} = \gamma_{k,w} \quad \forall k \in K, w \in W \quad (4)$$

$$\gamma_{k,w+1} \leq \gamma_{k,w} \quad \forall k \in K, w \in W/|W| \quad (5)$$

$$\sum_{i \in N} \lambda_{k,i,w} \leq \gamma_{k,w} |N| \quad \forall k \in K, w \in W \quad (6)$$

$$\delta_{k,w}^L = \sum_{i \in N} \lambda_{k,i,w} q_i \quad \forall k \in K, w \in W \quad (7)$$

$$\delta_{k,w}^L \leq \gamma_{k,w} Q \quad \forall k \in K, w \in W \quad (8)$$

$$\tau_{k,i,w} \leq \lambda_{k,i,w} l_i \quad \forall i \in N, k \in K, w \in W \quad (9)$$

$$\tau_{k,w}^L \geq \lambda_{k,i,w} (t_i^R + t_k^D) - M(1 - \gamma_{k,w}) \quad \forall i \in N, k \in K, w \in W \quad (10)$$

$$\tau_{k,w+1}^L \geq \tau_{k,w}^B + t_k^D - M(1 - \gamma_{k,w+1}) \quad \forall i \in N, k \in K, w \in W \quad (11)$$

$$\tau_{k,i,w} \geq \tau_{k,w}^L + t_{k,i} - M(1 - \mu_{k,i,w}) \quad \forall i \in N, k \in K, w \in W \quad (12)$$

$$\tau_{k,w}^B \geq \max\{\tau_{k,i,w}, e_i\} + t_i^S + t_{k,i} - M(1 - \eta_{k,i,w}) \quad \forall i \in N, k \in K, w \in W \quad (13)$$

$$\tau_{k,j,w} \geq \max\{\tau_{k,i,w}, e_i\} + t_i^S + t_{i,j}^N - M(1 - \xi_{i,j,k,w}) \quad \forall i, j \in N, k \in K, w \in W \quad (14)$$

$$\tau_{k,w}^B \leq T^H \quad \forall k \in K, w \in W \quad (15)$$

$$\gamma_{k,w} \in \{0, 1\} \quad \forall k \in K, w \in W \quad (16)$$

$$\xi_{i,j,k,w} \in \{0, 1\} \quad \forall i, j \in N, k \in K, w \in W \quad (17)$$

$$\lambda_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (18)$$

$$\mu_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (19)$$

$$\eta_{k,i,w} \in \{0, 1\} \quad \forall i \in N, k \in K, w \in W \quad (20)$$

The objective function (1) minimizes the total traveling time of all the vehicles. Constraints (2) impose that each customer is served exactly once. Constraints (3) show that, for a customer who is served in a trip of a vehicle, if he/she is the first customer in the served customer sequence of this trip, then no customer can be served before him/her in this trip, otherwise, there must exist a customer who is served exactly before him/her. Similarly, if he/she is the last customer in the served customer sequence of this trip, then no customer can be served after him/her in this trip, otherwise, there must be a customer who is served exactly after him/her. Constraints (4) ensure that there must exist only one first customer point and one last customer point in a trip that is performed, on the contrary, if the trip is not performed, then no first customer point exists in this trip, nor the last customer point. Constraints (5) guarantee that trip $w + 1$ of vehicle k can be operated only if vehicle k has operated trip w . Constraints (6) guarantee that a vehicle can serve a customer only if the customer is included in this vehicle's trip. Constraints (7) ensure that the total loads of a vehicle equal the sum of customer demands. Constraints (8) ensure that the total loads of a vehicle cannot exceed its vehicle capacity. Constraints (9) are the service time window constraints. Constraints (10) address the time relationship between the departure time of a vehicle and release dates as well as depot service time in a trip. Constraints (11) state the time relationship between consecutive trips of the same vehicle. Constraints (12)–(13) are travel time conservation constraints. Constraints (14) are service time conservation constraints. Constraints (15) ensure that the time that a vehicle is returned to the depot is no later than the time horizon T^H . Constraints (16)–(20) define the domains of decision variables.

It should be noted that constraints (11), (13), (14) are presented in a nonlinear form with max functions, there is no necessity to linearize them because the current version of the model implementation tools applied in this study (introduced in Section 5) already support the max/min functions.

4. Algorithmic strategies

The model developed in Section 3 can be solved by some off-the-shelf solvers (e.g., CPLEX) for small-scale instances. To deal with large-scale instances, we develop an HPSO algorithm and an HGA algorithm to solve our proposed model and compare their performance through extensive experiments. The classical particle swarm optimization (PSO) algorithm initializes a set of positions of particles (i.e., the swarm), and obtains the optimal position by iteratively updating the positions towards the incumbent best one. In our strategies, HPSO combines the local search-variable neighborhood descent (LS-VND) and the solution translation mechanism to expand search neighborhoods as well as accelerate the algorithm. The classical genetic algorithm (GA) initializes a set of chromosomes (i.e., the population) and the best individual information is found through evolution and iteration mechanisms. In our strategies, a reorder routine is adapted to accelerate our HGA.

In Section 4.1 and 4.2, we separately describe the frameworks for the proposed HPSO and HGA. There are several same procedures in HPSO and HGA, such as LS-VND, labeling procedure. The details of these procedures are all introduced as follows.

4.1. HPSO algorithm

In HPSO, particles in swarm represent solutions. Each particle is identified by a customer id sequence (sorted by increasing order) and a corresponding *random position* sequence (see Fig. 3(a)). Then, a labeling procedure is executed to evaluate the fitness value of

7.6 Keskin-Çatay (2016): 线性部分充电原式

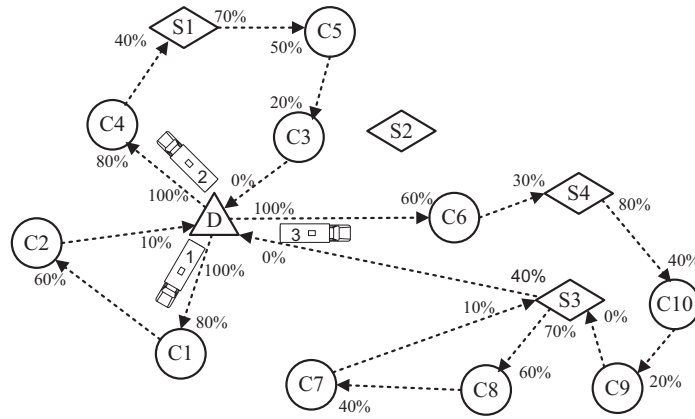


Fig. 1. An illustrative example.

with a distance d_{ij} and travel time t_{ij} . The battery charge is consumed at a rate of h and every traveled arc consumes $h \cdot d_{ij}$ of the remaining battery. Each customer $i \in V'$ has positive demand q_i , service time s_i and time window $[e_i, l_i]$. All EVs have a load capacity of C and a battery capacity of Q . At a recharging station, the battery is charged at a recharging rate of g . The decision variables, τ_i , u_i and y_i keep track of the service starting time, remaining cargo level and battery state of charge on arriving to station $i \in V'_{0,n+1}$, respectively. The binary decision variable x_{ij} takes value 1 if arc (i, j) is traversed and 0 otherwise. In Schneider et al. (2014) the battery is always recharged to full capacity, i.e. the recharge amount is $(Q - y_i)$. EVRPTW-PR allows partial recharges by defining a new decision variable Y_i which represents the battery state of charge on departure from station i .

$$\min \sum_{i \in V'_0, j \in V'_{n+1}, i \neq j} d_{ij} x_{ij} \quad (1)$$

subject to

$$\sum_{j \in V'_{n+1}, i \neq j} x_{ij} = 1 \quad \forall i \in V \quad (2)$$

$$\sum_{j \in V'_{n+1}, i \neq j} x_{ij} \leq 1 \quad \forall i \in F' \quad (3)$$

$$\sum_{i \in V'_0, i \neq j} x_{ij} - \sum_{i \in V'_{n+1}, i \neq j} x_{ji} = 0 \quad \forall j \in V' \quad (4)$$

$$\tau_i + (t_{ij} + s_i)x_{ij} - l_0(1 - x_{ij}) \leq \tau_j \quad \forall i \in V_0, \forall j \in V'_{n+1}, i \neq j \quad (5)$$

$$\tau_i + t_{ij}x_{ij} + g(Y_i - y_i) - (l_0 + gQ)(1 - x_{ij}) \leq \tau_j \quad \forall i \in F', \forall j \in V'_{n+1}, i \neq j \quad (6)$$

$$e_j \leq \tau_j \leq l_j \quad \forall j \in V'_{0,n+1} \quad (7)$$

$$0 \leq u_j \leq u_i - q_i x_{ij} + C(1 - x_{ij}) \quad \forall i \in V'_0, \forall j \in V'_{n+1}, i \neq j \quad (8)$$

$$0 \leq u_0 \leq C \quad (9)$$

$$0 \leq y_j \leq y_i - (h \cdot d_{ij})x_{ij} + Q(1 - x_{ij}) \quad \forall i \in V, \forall j \in V'_{n+1}, i \neq j \quad (10)$$

$$0 \leq y_j \leq Y_i - (h \cdot d_{ij})x_{ij} + Q(1 - x_{ij}) \quad \forall i \in F'_0, \forall j \in V'_{n+1}, i \neq j \quad (11)$$

$$y_i \leq Y_i \leq Q \quad \forall i \in F'_0 \quad (12)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in V'_0, \forall j \in V'_{n+1}, i \neq j \quad (13)$$

The objective function (1) minimizes total distance traveled. Constraints (2) and (3) handle the connectivity of customers and visits to recharging stations, respectively. The flow conservation constraints (4) enforce that the number of outgoing arcs equals to the number of incoming arcs at each vertex. Constraints (5) and (6) ensure the time feasibility of arcs leaving the customers (and the depot), and the stations, respectively. Constraints (7) enforce the time windows of the customers and the depot. In addition, constraints (5)–(7) eliminate the sub-tours. Constraints (8) and (9) guarantee that demand of all customers are satisfied. Constraints (10) and (11) keep track of the battery state of charge and make sure that it is never negative. Constraints (12) determine the battery state of charge after the recharge at a station and make sure that the battery state of charge does not exceed its capacity. Finally, constraints (13) define the binary decision variables.

Proposition 1. *If an optimal solution exists such that an EV leaves the depot with its battery partially charged, i.e. $Y_0^* < 1$, then the same EV departing from the depot fully charged is also optimal, i.e. $Y_0^* = 1$ is also optimal.*

Proof. Let $Y_0^* < 1$ be optimal. Since fully recharging the battery at the depot does not delay the departure time of the EV, $Y_0^* = 1$ must also be optimal. $\square$

Corollary 1. *If $Y_0^* < 1$ is optimal, then the problem has infinite multiple optima.*

Proof. Let $\bar{Y}_0 < 1$ be optimal and $\bar{Y}_0 + \varepsilon \leq 1$ not, where ε is a small positive scalar. Then following Proposition 1, multiple optima exist such that $\bar{Y}_0 \leq Y_0^* \leq 1$. $\square$

Proposition 2. *If an optimal solution exists such that an EV has been recharged at least once and returns to the depot at the end of its route with positive battery state, i.e. $y_{n+1}^* > 0$, then its return to the depot with empty battery is also optimal, i.e. $y_{n+1}^* = 0$ is also optimal.*

Proof. Let $y_{n+1}^* > 0$ be optimal. Since recharging the battery with less energy at the preceding station does not delay the departure time to cause any time window infeasibility, $y_{n+1}^* = 0$ must also be optimal. $\square$

Corollary 2. *If $y_{n+1}^* > 0$ is optimal, then the problem has infinite multiple optima.*

Proof. Let $\bar{y}_{n+1} > 0$ be optimal and $\bar{y}_{n+1} + \varepsilon$ not, where ε is a small positive scalar. Then following Proposition 2, multiple optima exist such that $0 \leq y_{n+1}^* \leq \bar{y}_{n+1}$. $\square$

Without loss of generality, we assume that an EV departs from the depot with a battery charged in full and returns to the depot with its battery fully consumed if it has been recharged at least once along its route.

4. Solution methodology

We propose an ALNS method to solve the EVRPTW-PR. ALNS was introduced by Ropke and Pisinger (2006a) as an extension of the Large Neighborhood Search (LNS) framework put forward by Shaw (1998). Since local search methods can only make small changes to an existing solution their search space is narrow. Hence, they are unable to move from one promising area to another within the feasible region. To overcome this shortcoming, Ropke and Pisinger (2006a) considered large moves that rearrange up to 40% of the vertices instead of using small moves that relocate or exchange only a few arcs or vertices at each iteration. In a subsequent study, Ropke and Pisinger (2006b) developed a unified ALNS heuristic for a large class of VRP with Backhauls. Pisinger and Ropke (2007) improved this heuristic with additional algorithms and showed that the proposed framework gives competitive results in different VRP variants. Since then, ALNS has been successfully implemented to solve various VRPs, e.g. cumulative capacitated VRP (Ribeiro and Laporte, 2012), pollution-routing problem (Demir et al., 2012), two-echelon VRP (Hemmelmayr et al., 2012), pickup and delivery problems with transshipment (Qu and Bard, 2012) and with vehicle transfers (Masson et al., 2013), VRP with multiple routes (Azi et al., 2014), periodic inventory routing problem (Aksen et al., 2014), and production routing problem (Adulyasak et al., 2014).

4.1. Overview of the proposed ALNS approach

4.1.1. Initial solution construction

The initial solution is obtained by iteratively constructing feasible routes. The route construction begins with the nearest customer to the depot. Then, the insertion costs of all unassigned customers to all possible existing positions in the route are determined respecting the time window constraints, i.e. the insertion of customer i between nodes j and k is calculated as

7.7 Montoya 等 (2017): 非线性充电路径模型

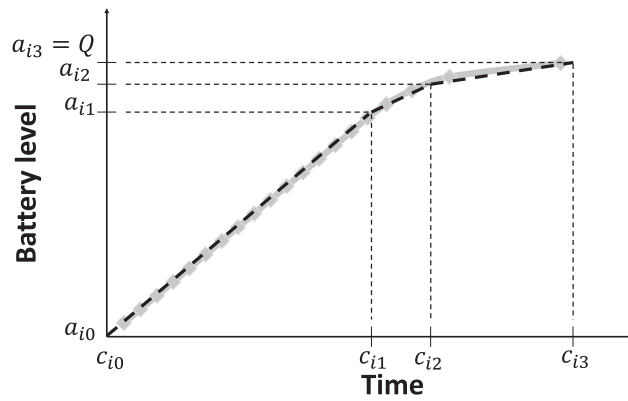


Fig. 3. Real data vs. piecewise linear approximation for a CS of 22 kW charging a battery of 16 kWh.

The remainder of this paper is organized as follows. Section 2 formally introduces the E-VRP-NL. Section 3 describes our hybrid metaheuristic, and Section 4 presents the computational experiments. Finally, Section 5 concludes the paper and discusses future research.

2. Electric vehicle routing problem with nonlinear charging function

2.1. Problem description

Let I be the set of nodes representing the customers, F the set of CSs, and 0 a node representing the depot. Each customer $i \in I$ has a service time p_i . The E-VRP-NL is defined on a directed and complete graph $G = (V, A)$, where $V = \{0\} \cup I \cup F'$ and F' contains the set F and β copies of each CS (i.e., $|F'| = |F| \times (1 + \beta)$). The value of $1 + \beta$ corresponds to the number of times that each CS can be visited. Let $A = \{(i, j) : i, j \in V, i \neq j\}$ be the set of arcs connecting vertices of V . Each arc (i, j) has two associated nonnegative values: a travel time t_{ij} and an energy consumption e_{ij} . The customers are served using an unlimited and homogeneous fleet of EVs. All the EVs have a battery of capacity Q (expressed in kWh) and a maximum tour duration $T_{\max}$. It is assumed that the EVs leave the depot with a fully charged battery, and that all the CSs can handle an unlimited number of EVs simultaneously. Feasible solutions to the E-VRP-NL satisfy the following conditions: each customer is visited exactly once; each route satisfies the maximum-duration limit; each route starts and ends at the depot; and the battery level when an EV arrives at and departs from any vertex is between 0 and Q .

Since the traveled distance is directly related to the energy consumption, most work on E-VRPs with a homogeneous fleet focuses on minimizing the total distance (Schneider et al., 2014; Desaulniers et al., 2016; Hiermann et al., 2016; Keskin and C  tay, 2016). However, this objective function neglects the impact of charging operations. This may lead to solutions that charge the batteries more than needed or charge them even when their level is high. These decisions directly affect the battery's long-term degradation cost (which according to Becker et al., 2009 can be three times the energy cost) and the charging fees at CSs (Bansal, 2015). To better capture the impact of charging operations, in the E-VRP-NL we minimize the total travel and charging time. This objective function was studied by Z  ndorf (2014) and Liao et al. (2016) for related routing problems.

2.2. Modeling of battery charging functions

Each CS $i \in F$ has a charging mode (e.g., slow, moderate, fast) that is associated with a charging function $g_i(q_i, \Delta_i)$. This function maps the charge level when the vehicle arrives at i (q_i) and the time spent charging at i (Δ_i) to the charge level when the vehicle leaves i . To avoid handling a two-dimensional function, we use the transformation proposed by Z  ndorf (2014). Let $\hat{g}_i(l)$ be the charging function when $q_i = 0$ and the battery is charged for l time units; $g_i(q_i, \Delta_i)$ is estimated as $\hat{g}_i(\Delta_i + \hat{g}_i^{-1}(q_i))$. Note that $\Delta_i = \hat{g}_i^{-1}(o_i) - \hat{g}_i^{-1}(q_i)$, where o_i is the charge level when the vehicle leaves i .

The function $\hat{g}_i(l)$ is concave (Bruglieri et al., 2014; H  imoja et al., 2012; Pelletier et al., 2017) with an asymptote at Q . Similarly to Z  ndorf (2014), we argue that $\hat{g}_i(l)$ can be accurately approximated using piecewise linear functions. We support our claim using the data provided by Uhrig et al. (2015). We fit piecewise linear functions to their data and obtain approximations with an average relative absolute error of 0.90%, 1.24%, and 1.90% for CSs of 11, 22, and 44 kW, respectively. Fig. 3 shows the piecewise linear approximation for a CS i of 22 kW charging a vehicle equipped with a battery of 16 kWh. In the plot, c_{ik} and a_{ik} represent the charging time and the charge level for the breakpoint $k \in B$ of the CS $i \in F'$, where $B = \{0, \dots, b\}$ is the set of breakpoints of the piecewise linear approximation.

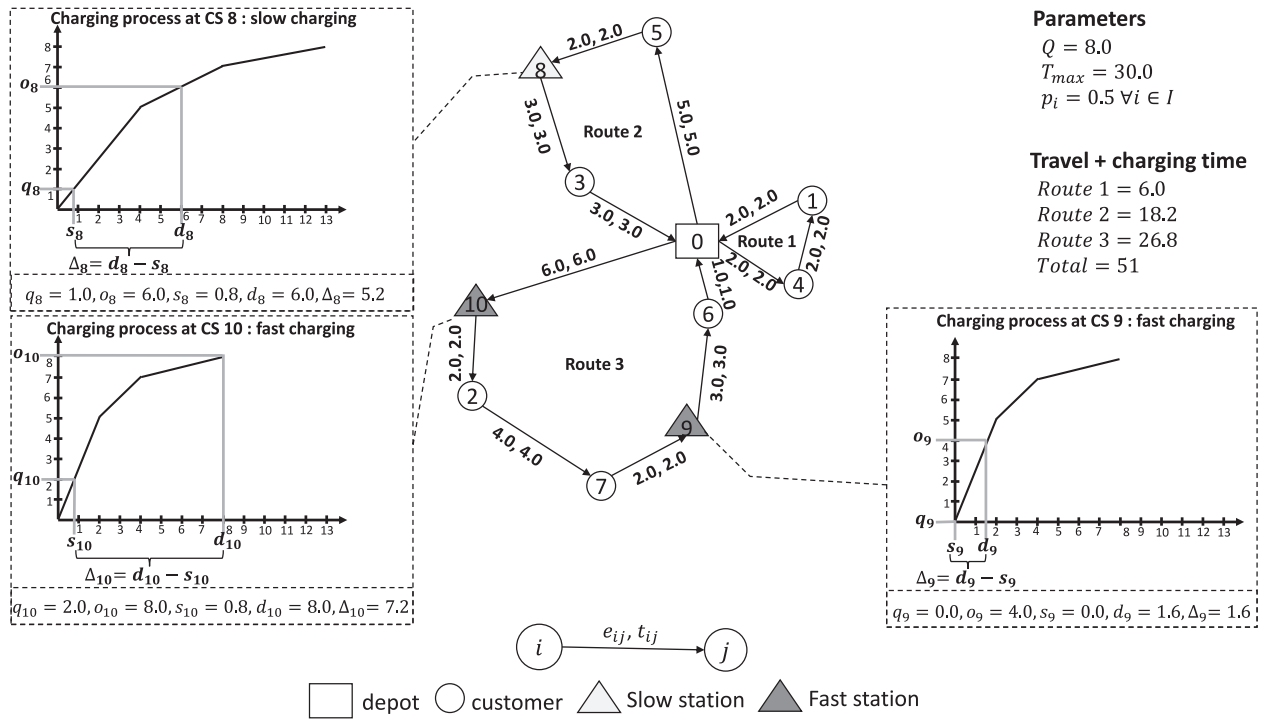


Fig. 4. Example of a feasible E-VRP-NL solution.

2.3. Illustrative example

Fig. 4 presents a numerical example illustrating the E-VRP-NL. The figure depicts a solution to an instance with 7 customers and 3 CSs. The CSs have different technologies (slow and fast), and each technology has a specific piecewise-linear charging function. The charging function maps the battery levels q_i and o_i to the charging times s_i and d_i to estimate the time spent at the CS $i \in F$ (Δ_i). In this example, Route 1 does not visit any CS, because its total energy consumption is less than the battery capacity. Route 2 visits CS 8: the EV arrives at the CS with a battery level $q_8 = 1.0$, and it charges the battery to a level $o_8 = 6.0$. To estimate the time spent at the CS, we use the piecewise-linear charging function: the charging times associated with q_8 and o_8 are $s_8 = 0.8$ and $d_8 = 6.0$, so the time spent at CS 8 is $\Delta_8 = 6.0 - 0.8 = 5.2$. The duration of Route 2 is the sum of the travel time (13.0), the charging time (5.2), and the service time (1.0), i.e., 19.2, which is less than T_{max} . The cost of this route is 18.2 (travel time + charging time). Finally, Route 3 visits CSs 10 and 9, and it spends $\Delta_{10} = 7.2$ and $\Delta_9 = 1.6$ time units charging in these CSs, respectively.

2.4. Mixed-integer linear programming formulation

To help the reader understand the E-VRP-NL, we now provide a mixed integer linear programming (MILP) formulation of the problem. The MILP uses the following decision variables: variable x_{ij} is equal to 1 if an EV travels from vertex i to j , and 0 otherwise. Variables τ_j and y_j track the time and charge level when the EV departs from vertex $j \in V$. Variables q_i and o_i specify the charge levels when an EV arrives at and departs from CS $i \in F'$, and s_i and d_i are the associated charging times. Variable $\Delta_i = d_i - s_i$ represents the time spent at CS $i \in F'$. Variables z_{ik} and w_{ik} are equal to 1 if the charge level is between $a_{i,k-1}$ and $a_{i,k}$, with $k \in B \setminus \{0\}$, when the EV arrives at and departs from CS $i \in F'$ respectively. Finally, variables α_{ik} and λ_{ik} are the coefficients of the breakpoint $k \in B$ in the piecewise linear approximation, when the EV arrives at and departs from CS $i \in F'$ respectively. The MILP formulation follows:

$$\min \sum_{i,j \in V} t_{ij} x_{ij} + \sum_{i \in F'} \Delta_i \quad (1)$$

subject to

$$\sum_{j \in V, i \neq j} x_{ij} = 1, \quad \forall i \in I \quad (2)$$

$$\sum_{j \in V, i \neq j} x_{ij} \leq 1, \quad \forall i \in F' \quad (3)$$

$$\sum_{j \in V, i \neq j} x_{ji} - \sum_{j \in V, i \neq j} x_{ij} = 0, \quad \forall i \in V \quad (4)$$

$$e_{ij}x_{ij} - (1 - x_{ij})Q \leq y_i - y_j \leq e_{ij}x_{ij} + (1 - x_{ij})Q, \quad \forall i \in V, \forall j \in I, i \neq j \quad (5)$$

$$e_{ij}x_{ij} - (1 - x_{ij})Q \leq y_i - q_j \leq e_{ij}x_{ij} + (1 - x_{ij})Q, \quad \forall i \in V, \forall j \in F', i \neq j \quad (6)$$

$$y_i \geq e_{i0}x_{i0}, \quad \forall i \in V \quad (7)$$

$$y_i = o_i, \quad \forall i \in F' \quad (8)$$

$$y_0 = Q \quad (9)$$

$$q_i \leq o_i, \quad \forall i \in F' \quad (10)$$

$$q_i = \sum_{k \in B} \alpha_{ik} a_{ik}, \quad \forall i \in F' \quad (11)$$

$$s_i = \sum_{k \in B} \alpha_{ik} c_{ik}, \quad \forall i \in F' \quad (12)$$

$$\sum_{k \in B} \alpha_{ik} = \sum_{k \in B \setminus \{0\}} z_{ik}, \quad \forall i \in F' \quad (13)$$

$$\sum_{k \in B \setminus \{0\}} z_{ik} = \sum_{j \in V} x_{ij}, \quad \forall i \in F' \quad (14)$$

$$\alpha_{i0} \leq z_{i1}, \quad \forall i \in F', \quad (15)$$

$$\alpha_{ik} \leq z_{ik} + z_{i,k+1}, \quad \forall i \in F', \forall k \in B \setminus \{0, b\} \quad (16)$$

$$\alpha_{ib} \leq z_{ib}, \quad \forall i \in F' \quad (17)$$

$$o_i = \sum_{k \in B} \lambda_{ik} a_{ik}, \quad \forall i \in F' \quad (18)$$

$$d_i = \sum_{k \in B} \lambda_{ik} c_{ik}, \quad \forall i \in F' \quad (19)$$

$$\sum_{k \in B} \lambda_{ik} = \sum_{k \in B \setminus \{0\}} w_{ik}, \quad \forall i \in F' \quad (20)$$

$$\sum_{k \in B \setminus \{0\}} w_{ik} = \sum_{j \in V} x_{ij}, \quad \forall i \in F' \quad (21)$$

$$\lambda_{i0} \leq w_{i1}, \quad \forall i \in F' \quad (22)$$

$$\lambda_{ik} \leq w_{ik} + w_{i,k+1}, \quad \forall i \in F', \forall k \in B \setminus \{0, b\} \quad (23)$$

$$\lambda_{ib} \leq w_{ib}, \quad \forall i \in F' \quad (24)$$

$$\Delta_i = d_i - s_i, \quad \forall i \in F' \quad (25)$$

$$\tau_i + (t_{ij} + p_j)x_{ij} - T_{\max}(1 - x_{ij}) \leq \tau_j, \quad \forall i \in V, \forall j \in I, i \neq j \quad (26)$$

$$\tau_i + \Delta_j + t_{ij}x_{ij} - (S_{\max} + T_{\max})(1 - x_{ij}) \leq \tau_j, \quad \forall i \in V, \forall j \in F', i \neq j \quad (27)$$

$$\tau_j + t_{j0} \leq T_{\max}, \quad \forall j \in V \quad (28)$$

$$\tau_0 \leq T_{\max} \quad (29)$$

$$x_{ij} = 0, \quad \forall i, j \in F' : m_{ij} = 1 \quad (30)$$

$$\tau_i \geq \tau_j, \quad \forall i, j \in F' : m_{ij} = 1, j \leq i \quad (31)$$

$$\tau_j \leq T_{\max} \sum_{i \in V} x_{ij}, \quad \forall j \in F' \quad (32)$$

$$\sum_{i \in V} x_{ih} \geq \sum_{j \in V} x_{jf}, \quad \forall h, f \in F' : m_{hf} = 1, h \leq f \quad (33)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \in V, i \neq j \quad (34)$$

$$\tau_i \geq 0, y_i \geq 0 \quad \forall i \in V \quad (35)$$

$$z_{ik} \in \{0, 1\}, w_{ik} \in \{0, 1\}, \quad \forall i \in F', \forall k \in B \setminus \{0\} \quad (36)$$

$$\alpha_{ik} \geq 0, \lambda_{ik} \geq 0, \quad \forall i \in F', \forall k \in B \quad (37)$$

$$q_i \geq 0, o_i \geq 0, s_i \geq 0, d_i \geq 0, \Delta_i \geq 0, \quad \forall i \in F' \quad (38)$$

The objective function (1) seeks to minimize the total time (travel times plus charging times). Constraints (2) ensure that each customer is visited once. Constraints (3) ensure that each CS copy is visited at most once. Constraints (4) impose the flow conservation. Constraints (5) and (6) track the battery charge level at each vertex. Constraints (7) ensure that if the EV travels between a vertex and the depot, it has sufficient energy to reach its destination. Constraints (8) reset the battery tracking to o_i upon departure from CS $i \in F'$. Constraint (9) ensures that the battery charge level is Q at the depot. Constraints (10) couple the charge levels when an EV arrives at and departs from any CS. Constraints (11)–(17) define the charge level (and its corresponding charging time) when an EV arrives at CS $i \in F'$ (based on the piecewise linear approximation of the charging function). Similarly, constraints (18)–(24) define the charge level (and its corresponding charging time) when an EV departs from CS $i \in F'$. Constraints (25) define the time spent at any CS. Constraints (26) and (27) track the departure time at each vertex, where $S_{\max} = \max_{i \in F'} \{c_{ib}\}$. Constraints (28) and (29) ensure that the EVs return to the depot no later than $T_{\max}$. Constraints (30)–(33) help to avoid the symmetry generated by the copies of the CSs. Parameter m_{ij} is equal to 1 if i and $j \in F'$ represent the same CS. Finally, constraints (34)–(38) define the domain of the decision variables.

3. Solving the E-VRP-NL

Lenstra and Rinnooy Kan (1981) demonstrated that the classical VRP, commonly known as the CVRP, is NP-hard. Since the CVRP is a special case of our E-VRP-NL, the latter is also NP-hard. We therefore propose a metaheuristic approach. Like many metaheuristics for other VRPs, our approach explores new solutions by building new routes or applying changes (moves) to existing ones. In this process, the algorithm makes sequencing and charging decisions. The former fix the order in which the route visits its assigned customers, while the latter determine where and how much to charge the EV serving the route. Sequencing and charging decisions can be made either simultaneously or in two phases (sequencing first, charging second). We use the latter option. To make charging decisions, we solve what we call the *fixed route vehicle charging problem*, or simply FRVCP. In a nutshell, the problem consists in (i) inserting charging stations into a fixed sequence of customers and (ii) deciding how much to charge at each inserted station. Since the FRVCP plays a key role in our approach, we discuss it in the next subsection before introducing our metaheuristic.

7.8 Diefenbach 等 (2023): 非线性充电排班模型

Table 2

Notation.

c	state of charge (SOC)
$\hat{c}$	maximum SOC
D	set of depots; $D = \{1, \dots, d\}$
J	set of trips; $J = \{d+1, \dots, d+n\}$
j	depot, trip or station index
S	set of charging stations; $S = \{d+n+1, \dots, d+n+s\}$
$\tilde{c}_j$	required charge to complete trip j
$\tilde{c}_{j,j'}$	required charge for the transit from depot, station, or (the end of) trip j to depot, station, or (the start of) trip j'
e_j	end time of trip j
$\tilde{m}_j$	maximum vehicle capacity of depot j
m	number of vehicles
r	constant time required to attach a vehicle to a charger (or to exchange the battery, depending on the context)
s_j	start time of trip j
$t_{j,j'}$	transit time from depot, station, or (the end of) trip j to depot, station, or (the start of) trip j'
$\delta_k(l)$	mapping signifying that vehicle k visits station $\delta_k(l) \in S$ or no station at all ($\delta_k(l) = 0$) after the l th trip of its schedule
$\zeta(j, j', j'', c)$	function giving the maximum SOC at the end of trip (or depot) j' for a vehicle coming from j with SOC c visiting station $j'' \in S$ or no station at all ($j'' = 0$) in-between
$\tilde{\zeta}(k, l)$	SOC at the end of the l th trip of vehicle k
$\eta(\tau)$	battery charging function dependent on time τ ; $\eta(\tau)$ is the SOC if an empty battery is charged for time τ
τ	amount of time
$\bar{\tau}(j, j', j'', c)$	time a vehicles spends charging at station j'' while transiting from j with SOC c to j'
ω_k	ordered set denoting which depots are to be visited and trips are to be performed by vehicle k in what order

cle and operator costs tend to dominate the detour costs due to the relatively short driving distances (Emde et al., 2018; Golz et al., 2012).

3.1. Formal problem description

We base the definition of the EVSP-MD-FS on the notation summarized in Table 2. Let $D = \{1, \dots, d\}$ be the set of depots, let $J = \{d+1, \dots, d+n\}$ be the set of trips to be processed during the planning horizon and let $S = \{d+n+1, \dots, d+n+s\}$ be the set of charging stations (called “stations” in the following). Note that we introduce offset indices to ease notation in the following definitions and models. Generally speaking, the locations of stations do not have to coincide with the depots or the start or end points of the trips, but can of course correspond to these locations. In fact, the former is the case for most of the in-plant milk-run logistic systems we observed. Each trip consists of a vehicle visiting multiple locations (i.e., assembly cells) in a given path. The exact path the vehicle takes is immaterial for the EVSP-MD-FS; we assume that this issue has been decided in a previous step. Let s_j be the start time of trip j , i.e., the time when the vehicle arrives at the trip's first location, and let e_j be the end time of trip j , i.e., the time when the vehicle departs from the trip's last location. For notational convenience, we also define $s_j = \infty$ and $e_j = 0$, $\forall j \in D \cup S$, which implies that depots and stations are always available. Without loss of generality, we assume that $s_j \leq s_{j'}$ holds, $\forall j, j' \in J: j < j'$, i.e., the trips are indexed according to their starting time in ascending order. Moreover, let $t_{j,j'}$, $\forall j, j' \in D \cup S \cup J: j \neq j'$, be the transit time of a vehicle from depot, station, or the end of trip j to depot, station, or the starting point of trip j' . For transit times, we assume that the triangle inequality holds in the sense that $t_{j,j'} \leq t_{j,j''} + \max\{e_{j''} - s_{j''}, 0\} + t_{j'',j'}$, $\forall j, j', j'' \in J \cup D \cup S$, where $\max\{e_{j''} - s_{j''}, 0\}$ is the trip's execution time if $j'' \in J$, and zero if $j'' \in D \cup S$. Let the battery capacity, i.e., maximum SOC, of the vehicle be $\hat{c}$, and the charge required to complete trip $j \in J$ be $\tilde{c}_j$, where we assume $\tilde{c}_j \leq \hat{c}$ applies, $\forall j \in J$. Note that the required charge $\tilde{c}_j$ may depend on the total distance covered, the number of stopovers, the carried load, the terrain (e.g., steep slopes), and other factors. We assume that these values have been determined beforehand and are given. To ease notation, we further define $\tilde{c}_j = 0$ and $\tilde{c}_{j,j} = 0$, $\forall j \in D \cup S$. Let $\tilde{c}_{j,j'}$ be the charge required to drive from depot, station or the end

of trip j to depot, station, or the starting point of trip j' . As for travel times, we assume that the triangle inequality also holds for battery requirements in the sense that $\tilde{c}_{j,j'} \leq \tilde{c}_{j,j''} + \tilde{c}_{j'',j'}$, $\forall j, j', j'' \in J \cup D \cup S$. We assume that charging is only possible at stations. If vehicles should be able to charge at depots or at the start or end points of trips, we can add stations that correspond to the respective locations.

Let $\eta(\tau)$ denote the charging function over time τ , i.e., the SOC an empty battery would have after being charged for a time of τ . We assume that $\eta(\tau)$ is either a constant function, a linear function or a non-linear function, which, in the latter two cases, is capped at $\hat{c}$, i.e., when the battery is full. Moreover, we assume that a vehicle arriving at a station requires a time of r before charging can start, where, depending on the assumed charging technology, r is the time required to swap the battery or to connect the battery to the charger. This enables us to account for the two most commonly encountered charging technologies in practice, namely battery swapping and charging via power cable with a realistic non-linear increase of the SOC. Furthermore, we assume the charging function is equal for all vehicles and stations and that vehicles are always charged whenever they spend sufficient time at a charging station. For a vehicle charging for a time of τ beginning from a SOC $c \geq 0$, the resulting SOC is given as $\eta(\eta^{-1}(c) + \tau)$, where $\eta^{-1}(c)$ is the inverse function of $\eta(\tau)$ and states the time τ required to charge an empty battery to SOC c . For the case of $\eta(\tau)$ being a constant function, we define $\eta^{-1}(c) = 0$. To further ease notation, we define $\eta^{-1}(c) = 0$, for $c < 0$, and $\eta^{-1}(c) = \infty$, for $c > \hat{c}$, as well as $\eta(\tau) = 0$, for $\tau < 0$. This makes it easy to accommodate non-linear charging.

A solution for EVSP-MD-FS then consists of

- a collection $\{\omega_1, \dots, \omega_m\}$ of ordered sets ω_k , denoting which depots are to be visited and which trips are to be performed by each vehicle $k = 1, \dots, m$ in what order,
- and m mappings $\delta_k: \{1, \dots, |\omega_k| - 1\} \rightarrow J$, for $k = 1, \dots, m$, where $\delta_k(l) = j$ specifies the deadheading route of vehicle k between the l th trip (or depot for $l = 1$) and the $(l+1)$ th trip (or depot for $l = |\omega_k| - 1$). Two general types of deadheading routes are possible: If $j = 0$, the vehicle drives directly from trip l to $l+1$ without any charging break. Otherwise, if $j \in S$, the vehicle visits station j after the l th trip (or depot) to recharge its battery before heading for the $(l+1)$ th trip (or depot).

Let $\omega_k(l)$ refer to the l th element in the ordered set. For all vehicles $k = 1, \dots, m$, it must hold that $\omega_k(1) \in D$ and $\omega_k(|\omega_k|) \in D$, since each vehicle departs from a depot initially and returns to one at the end. Furthermore, all trips need to be processed exactly once, i.e., $\bigcup_{k=1}^m \bigcup_{l=2}^{|\omega_k|-1} \{\omega_k(l)\} = J$ and $\omega_k(l) \neq \omega_{k'}(l')$, $\forall k, k' = 1, \dots, m; l, l' = 2, \dots, |\omega_k| - 1 : k \neq k' \vee l \neq l'$ must hold.

Note that by these definitions, vehicles are also allowed (but not required) to visit a charging station right after leaving and before returning to a depot. Moreover, our definition of δ_k imposes that vehicles visit at most one station between consecutively executed service trips. This forecloses the possibility of chains of charging, which would be especially useful if distances between service trips are large compared to the vehicles' battery reach, such that multiple charges are required to travel between consecutive service trips. Since distances are comparatively small in intra-logistic transportation systems, visiting multiple charging stations in direct succession is all but unheard of, however.

We say a solution to an instance of EVSP-MD-FS is feasible if it is *feasibly executable*. I.e., for every vehicle, the battery's SOC must never be negative and the executed route must not violate the time constraints. For a formal definition, see Section 3.2.

Moreover, while somewhat uncommon in the in-plant transportation systems that motivated this paper, there are additional constraints that planners may reasonably want to consider. First, due to space constraints, each depot $j \in D$ could be required to host no more than $\bar{m}_j$ vehicles, where $\bar{m}_j = n$ if no limit exists. Second, vehicles could be required to return to their depot of initial departure at the end of their schedule. In the following, we discuss them as optional extensions of the basic EVSP-MD-FS problem. Specifically, $|\{k \in \{1, \dots, m\} : \omega_k(1) = j\}| \leq \bar{m}_j \wedge |\{k \in \{1, \dots, m\} : \omega_k(|\omega_k|) = j\}| \leq \bar{m}_j$, $\forall j \in D$, must hold to account for the limited capacity of the depots. Finally, if vehicles are required to return to their depot of initial departure, it must hold that $\omega_k(1) = \omega_k(|\omega_k|)$, $\forall k \in \{1, \dots, m\}$.

The objective of EVSP-MD-FS is to find a feasible schedule that minimizes the number of vehicles m . Hence, we minimize the number of vehicles m to which trips are assigned.

3.2. Feasibly executable solutions

In order to determine whether a solution is *feasibly executable*, we must determine the recharged battery during charging breaks and, hence, their duration. Each vehicle can charge its battery whenever it visits a station, i.e., whenever $\delta_k(l) \in S$. We can preempt the decision on the charging breaks' duration by taking the following proposition into account.

Proposition 3.1. *Given the trips (and depots) ω_k to be executed (or visited) by vehicle k , the policy of charging the vehicle at every charging station it visits for as long as the trip sequence ω_k permits creates a feasible solution, if any exists. I.e., using this charging policy, the vehicle reaches each trip (and depot) $j \in \omega_k$ with the maximum SOC possible given ω_k .*

Proof. The proof is by an interchange argument. Assume that in some feasible solution, vehicle k reaches trip $j \in \omega_k : j \neq \omega_k(|\omega_k|)$ with SOC c' and the consecutive trip (or depot) j' with c'' , where it executes the deadheading route j'' in-between. We refer to this as case one. Alternatively, assume a second case, where j is reached with SOC $c''' = c' + \Delta c$ and j' with SOC c'''' , where Δc is a positive amount of charge. In the following, we prove that $c'''' \geq c''$ is guaranteed to hold true.

In the second case, the vehicle can execute the exact same deadheading route as in the first case. If $j'' = 0$, it follows that $c'' = c' - \bar{c}_{j,j'}$ and $c'''' = c''' - \bar{c}_{j,j'} = c' + \Delta c - \bar{c}_{j,j'}$, hence $c'''' > c''$ applies. Else, if $j'' \in S$, in case one, the vehicle reaches j'' with SOC $c' - \bar{c}_{j,j''}$. Let τ' denote the time the vehicle charges at station j'' ,

such that it has SOC $\eta(\eta^{-1}(c' - \bar{c}_{j,j''}) + \tau')$ when leaving j'' . Consequently, $c'' = \eta(\eta^{-1}(c' - \bar{c}_{j,j''}) + \tau') - \bar{c}_{j'',j'}$ follows. In the second case, the vehicle reaches station j'' with $c' + \Delta c - \bar{c}_{j,j''}$ and can charge for the exact same duration τ' as in the first case. Consequently, it follows that $c'''' = \eta(\eta^{-1}(c' + \Delta c - \bar{c}_{j,j''}) + \tau') - \bar{c}_{j'',j'}$. Since $\eta^{-1}(c)$ is an increasing function of c and $\eta(\tau)$ is an increasing function of τ , $c'''' \geq c''$ is guaranteed to hold true.

The same reasoning applies for all consecutive trips in ω_k until the final depot $\omega_k(|\omega_k|)$ is reached. Hence, if there exists a feasible solution, there also exists a feasible solution where the remaining charge when reaching each $j \in \omega_k$ is maximized, that is, where charging breaks are never cut short before the vehicle must depart for its next trip. $\square$

Therefore, at each visited station, the vehicle should charge for the maximum possible duration that still allows to timely reach the consecutive trip or depot. Given $\delta_k(l) = j''$ with $j'' \in S$, let $\omega_k(l) = j$ and $\omega_k(l+1) = j'$ be two consecutively executed trips or one of them be a depot. Furthermore, let c be the SOC when the vehicle leaves from trip (or depot) j . The optimal time $\bar{\tau}(j, j', j'', c)$ the vehicle spends charging at station j'' then follows as

$$\bar{\tau}(j, j', j'', c) = s_{j'} - e_j - t_{j,j''} - t_{j'',j'} - r, \quad (1)$$

which is comprised of the available time between the start of trip j' and the end of trip j minus the travel time to and from station j'' minus the plug-in/ battery swap time r .

Given two consecutively executed trips $\omega_k(l) = j$ and $\omega_k(l+1) = j'$ (where either might be a depot instead) and $\delta_k(l) = j''$ (with $j'' \in S \cup \{0\}$) indicating the deadheading route in-between, the maximal SOC at the end of trip (or depot) j' can be calculated as

$$\zeta(j, j', j'', c) = \begin{cases} c - \bar{c}_{j,j'} - \bar{c}_{j'} & \text{if } j'' = 0 \wedge e_j + t_{j,j'} \leq s_{j'} \\ \eta(\eta^{-1}(c - \bar{c}_{j,j''}) + \bar{\tau}(j, j', j'', c)) & \text{if } j'' \in S \wedge \bar{c}_{j,j''} \leq c \\ -\bar{c}_{j'',j'} - \bar{c}_{j'} & \wedge \bar{\tau}(j, j', j'', c) \geq 0 \\ -1 & \text{otherwise} \end{cases} \quad (2)$$

If no station is visited at the deadheading route, the first case applies. Here, the SOC at the end of trip j' is the SOC at the end of trip j minus the battery charge consumed for the transit from j to j' and for the execution of j' . If a station is visited in-between j and j' , the second case applies. Here, the vehicle reaches station j'' with SOC $c - \bar{c}_{j,j''}$ and charges for a duration of $\bar{\tau}(j, j', j'', c)$, which results in a SOC of $\eta(\eta^{-1}(c - \bar{c}_{j,j''}) + \bar{\tau}(j, j', j'', c))$ at the end of the charging procedure. Afterwards, trip j' must be reached and executed, which reduces the SOC at the end of trip j' by $\bar{c}_{j'',j'} + \bar{c}_{j'}$. If either the charge c is not sufficient to reach the station j'' or there is not enough time, $\zeta(j, j', j'', c)$ assumes a value < 0 to indicate infeasibility.

We define $\bar{\zeta}(k, l)$ as the SOC of vehicle k at the end of the l th trip or depot, where $\bar{\zeta}(k, |\omega_k|)$ denotes the SOC when vehicle k reaches the final depot. For $l \geq 2$, it can be recursively calculated as

$$\bar{\zeta}(k, l) = \zeta(\omega_k(l-1), \omega_k(l), \delta_k(l-1), \bar{\zeta}(k, l-1)),$$

with $\bar{\zeta}(k, 1) = \hat{c}$ for $l = 1$.

Based on these definitions, a solution to EVSP-MD-FS is *feasibly executable* if and only if $\zeta(k, l) \geq 0$, $\forall k \in \{1, \dots, m\}, l \in \{1, \dots, |\omega_k|\}$. I.e., for every vehicle, the battery's SOC must never be negative and the executed route must not violate the time constraints (since, otherwise, $\zeta(j, j', j'', c)$ and, hence, $\bar{\zeta}(k, l)$, would assume the value -1).

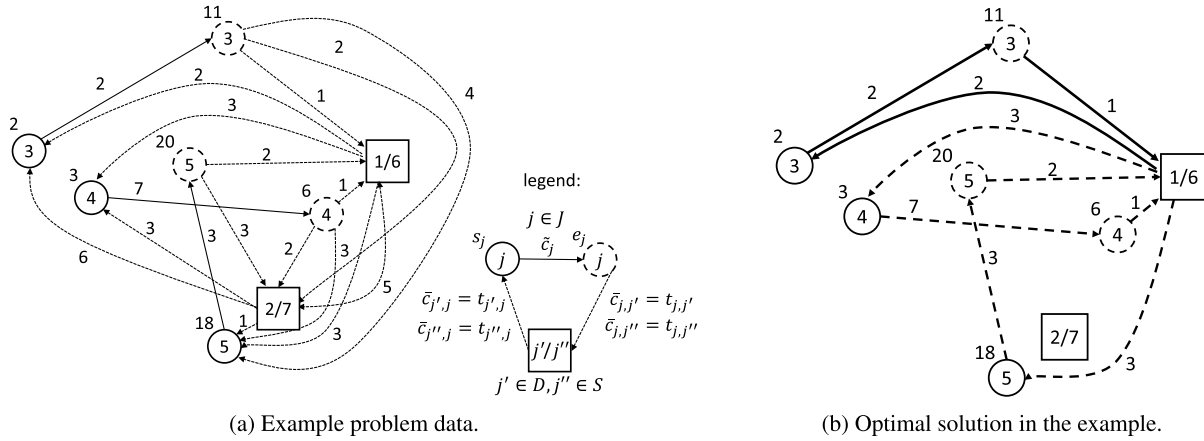


Fig. 2. Example data and solution.

Example. Consider the example problem depicted in Fig. 2a, consisting of $n=3$ trips and $d=s=2$ depots that are also charging stations. Let the battery capacity of the vehicles be $\hat{c}=11$, the charge function be linear $\eta(\tau)=\tau$ (with the cap $\eta(\tau)=11$, for $\tau \geq 11$) and the constant charge time be $r=0$. Furthermore, let $\bar{m}_j=n$, $\forall j \in D$, and let the vehicles be not required to return to their initial depot. Note that in this example, we assume that all driving times ($t_{j,j'}$) are equal to the energy expenditures ($\bar{c}_{j,j'}$), which generally does not need to be the case. The optimal solution is depicted in Fig. 2b, corresponding to $m=2$ vehicles being used. Solid arrows denote the route of vehicle 1, dotted arrows the route of vehicle 2. Vehicle 1 processes only trip 3 (i.e., $\omega_1=\{1, 3, 1\}$, $\delta_1(1)=0$, $\delta_1(2)=0$, $\bar{\zeta}(1,1)=11$, $\bar{\zeta}(1,2)=7$, $\bar{\zeta}(1,3)=6$). Vehicle 2 processes trips 4 and 5 (in that order), returning to, first, station 6 – where it charges its battery for a time of $\bar{\tau}(4,5,6,1)=18-6-1-3-0=8$ – and, second, depot 1 after the trips (i.e., $\omega_2=\{1, 4, 5, 1\}$, $\delta_2(1)=0$, $\delta_2(2)=6$, $\delta_2(3)=0$, $\bar{\zeta}(2,1)=11$, $\bar{\zeta}(2,2)=1$, $\bar{\zeta}(2,3)=2$, $\bar{\zeta}(2,4)=0$).

3.3. Complexity

Concerning the time complexity of EVSP-MD-FS, classic (single-depot) non-electric vehicle scheduling is well-known to be tractable (Saha, 1970), as it can be reduced to finding a minimum cost perfect matching in a bipartite graph (Bertossi, Carraraesi, & Gallo, 1987). This result also holds for the special single-depot case of EVSP-MD-FS where the battery capacity is not a limiting factor ($\hat{c}=\infty$). In this case, charging breaks need not be considered and the driving time t to/from the depot is immaterial. The problem then is to assign the given trips to as few vehicles as possible such that no trips overlap, which is equivalent to classic vehicle scheduling with the objective of minimizing the fleet size.

However, as soon as the battery capacity becomes a limiting factor, single-depot electric vehicle scheduling with finite battery capacity and the objective of minimizing the fleet size becomes intractable, as is proven by Emde et al. (2018, Proposition 3) by a reduction from bin packing. Obviously, this result extends to the multi-depot case as considered in EVSP-MD-FS. Finally, the result also holds if additional constraints are considered, since they merely generalize the basic EVSP-MD-FS problem.

3.4. Feasible deadheading routes

By the definitions in Section 3.1, a solution to EVSP-MD-FS consists of two distinctive parts. The ordered subsets $\omega_1, \dots, \omega_m$ denote which depots are to be visited and which trips to be executed

by which vehicle in which sequence. The mappings $\delta_1, \dots, \delta_m$ indicate the deadheading route in-between the trips and depots for all vehicles $1, \dots, m$. Given only ω_k , the respective feasible deadheading routes δ_k can be determined efficiently as follows.

Let $\omega_k(l)=j$ and $\omega_k(l+1)=j'$ be two consecutively executed trips or either be a depot. Furthermore, let c be the SOC when the vehicle leaves from trip (or depot) j . By Proposition 3.1, provided that any feasible solution exists at all for a given ω_k , there is always a feasible solution that reaches j' with the highest possible SOC and, consequently, completes j' with the maximum possible SOC (since $\bar{c}_{j,j'}$ is constant). The latter SOC is given by $\bar{\zeta}(j, j', j'', c)$ (cf., Eq. (2)), which beyond j, j' and c is solely dependent on $\delta_k(l)=j''$, the deadheading route between j and j' . Hence, it follows that

$$\delta_k^*(l) = \arg \max_{j'' \in S \cup \{0\}} \{ \bar{\zeta}(j, j', j'', c) \}$$

is guaranteed to be feasible (if feasibility is attainable at all). Consequently, the maximum and therefore feasible charge at the end of trip j' is given as

$$\bar{\zeta}^*(j, j', c) = \max_{j'' \in S \cup \{0\}} \{ \bar{\zeta}(j, j', j'', c) \}.$$

4. IP model

To enable the use of default solvers, we propose an IP model based on the notation given in Table 3. We incorporate non-linear battery charging by discretizing the SOC. I.e., the SOC can only assume discrete values $c \in C$. We note that the discretization is only required for the IP model; the solution procedure presented in Section 5 does not rely on discretization. For linear and constant-time charging, the discrete SOC is exact, as long as we assume that all problem parameters, i.e., travel times and required battery charges, are integer. For non-linear charging, using discrete SOC is an approximation. By scaling the problem parameters accordingly, we can, however, approximate the actual SOC to any desired precision.

We introduce binary variables $\beta_{j,c}$ to track the discrete SOC $c \in C$ at the end e_j of each trip $j \in J$ and when a vehicle reaches the final depot $j \in D$. Note that the number of variables $\beta_{j,c}$ grows pseudo-polynomially with $O(\hat{c} \cdot n)$. Moreover, we define set $N = \{(j, c) \in (J \cup D) \times C \mid c \neq \lfloor \bar{\zeta}^*(j', j, c') \rfloor, \forall j' \in J \cup D, c' \in C\}$ as the set of pairs (j, c) indicating that trip j is never completed with SOC c in the optimal solution. Note that set N can be determined in $O((n+d)^2 \cdot s \cdot \hat{c})$, by simply finding all combinations $j, j' \in J \cup D$: $j < j'$, $c \in C$, where $\max_{j'' \in S \cup \{0\}} \{ \bar{\zeta}(j, j', j'', c) \} = \bar{\zeta}^*(j, j', c) < 0$.

Table 3

Notation for the IP model.

C	set of discrete SOC values; index c ; $C = \{0, \dots, \hat{c}\}$; a value of c' refers to $\frac{c'}{\hat{c}}$ of the maximum SOC
$v_{j,j'}$	binary variable: 1, if trip j is associated with depot j' , else 0
$x_{j,j'}$	binary variable: 1, if a vehicle transits from trip or depot j to trip or depot j' (with possible detours for charging in-between), else 0
$\beta_{j,c}$	binary variable: 1, if trip j is completed (or depot j is reached) with SOC c

Using these definitions, Eqs. (3) to (13) define an IP model for the case that vehicles are not required to return to their initial depot at the end of their schedules.

$$[\text{EVSP-MD-FS}] \text{ Minimize } F(\mathbf{x}, \boldsymbol{\beta}) = \sum_{j \in D} \sum_{j' \in J} x_{j,j'} \quad (3)$$

subject to

$$\sum_{\substack{j' \in J \cup D: \\ j' > j \vee j' \in D}} x_{j,j'} = 1 \quad \forall j \in J \quad (4)$$

$$\sum_{\substack{j \in J \cup D: \\ j < j'}} x_{j,j'} = 1 \quad \forall j' \in J \quad (5)$$

$$\sum_{\substack{c \in C: \\ (j,c) \notin N}} \beta_{j,c} = 1 \quad \forall j \in J \quad (6)$$

$$0 \geq x_{j,j'} + \beta_{j,c} - 1 \quad \forall j \in J, j' \in J \cup D: j < j' \vee j' \in D, \\ c \in C: (j, c) \notin N \wedge \lfloor \zeta^*(j, j', c) \rfloor < 0 \quad (7)$$

$$\beta_{j', \lfloor \zeta^*(j, j', c) \rfloor} \geq x_{j,j'} + \beta_{j,c} - 1 \quad \forall j \in J, j' \in J \cup D: j < j' \vee j' \in D, \\ c \in C: (j, c) \notin N \wedge \lfloor \zeta^*(j, j', c) \rfloor \geq 0 \quad (8)$$

$$\beta_{j', \lfloor \zeta^*(j, j', \hat{c}) \rfloor} \geq x_{j,j'} \quad \forall j \in D, j' \in J: \lfloor \zeta^*(j, j', \hat{c}) \rfloor \geq 0 \quad (9)$$

$$\sum_{j \in J} x_{j',j} \leq \bar{m}_{j'} \quad \forall j' \in D \quad (10)$$

$$\sum_{j \in J'} x_{j,j'} \leq \bar{m}_{j'} \quad \forall j' \in D \quad (11)$$

$$x_{j,j'} \in \{0, 1\} \quad \forall j, j' \in J \cup D: (j, j' \in J \wedge j < j') \vee (j \in J \wedge \\ j' \in D) \vee (j \in D \wedge j' \in J \wedge \lfloor \zeta^*(j, j', \hat{c}) \rfloor \geq 0) \quad (12)$$

$$\beta_{j,c} \in \{0, 1\} \quad \forall j \in J \cup D, c \in C: (j, c) \notin N \quad (13)$$

Objective function (3) minimizes the number of required vehicles. Constraints (4) and (5) enforce that every trip has exactly one preceding and one succeeding trip or depot. Constraints (6) make sure that each trip is finished with exactly one non-negative SOC.

Constraints (7) to (9) ensure consistent SOC values. Inequalities (7) prohibit the execution of trip j' after trip j has been completed with SOC c if the SOC would be insufficient for completing the trip or time constraints would be violated (i.e., $\lfloor \zeta^*(j, j', c) \rfloor < 0$). Likewise, if the SOC is sufficient and no time constraints are violated (i.e., $\lfloor \zeta^*(j, j', c) \rfloor \geq 0$), Inequalities (8) set the SOC at the end of trip j' to the appropriate value. Constraints (9) determine the SOC of the trips that are initially executed after the vehicles leave the depots.

Constraints (10) and (11) enforce limits on the number of vehicles per depot. Finally, Constraints (12) and (13) define the domains of the decision variables. Note that – in order to reduce the search space – we omit $\beta_{j,c}$ for $(j, c) \in N$, since the states can never be feasibly reached.

Moreover, to formulate the IP model, we note that determining $\zeta^*(j, j', c)$ requires a pseudo-polynomial amount $O((n+d)^2 \cdot s \cdot \hat{c})$ of pre-calculations. Furthermore, the optimal deadheading routes need to be determined from the IP's solution in a $O((n+d) \cdot s)$ post-processing procedure (cf., Section 3.4).

To enforce the optional constraint that each vehicle return to the depot of its initial departure, we introduce the binary variables $v_{j,j'}$ that are 1 if trip j is associated with depot j' , and 0 otherwise. Using these variables, we extend the IP with the following equations:

$$\sum_{j' \in D} v_{j,j'} = 1 \quad \forall j \in J \quad (14)$$

$$v_{j',j''} \geq x_{j,j'} + v_{j,j''} - 1 \quad \forall j, j' \in J: j < j', j'' \in D \quad (15)$$

$$v_{j,j'} \geq x_{j',j} \quad \forall j \in J, j' \in D \quad (16)$$

$$v_{j,j'} \geq x_{j,j'} \quad \forall j \in J, j' \in D \quad (17)$$

$$v_{j,j'} \in \{0, 1\} \quad \forall j \in J, j' \in D \quad (18)$$

Constraints (14) make sure that every trip $j \in J$ is associated with exactly one depot. Inequalities (15) force each pair of consecutively executed trips, j and j' , to be associated with the same depot. Constraints (16) force the trip executed directly after leaving a depot to be associated with the respective depot. Likewise, Constraints (17) associate each final trip before returning to a depot with the respective depot. In conjunction, these constraints enforce each tour to start from and end at the same depot.

5. Branch-and-check

Our computational tests reveal that a default solver using the IP model from Section 4 performs subpar for instances with $n \geq 100$ (see Section 6.2). We therefore propose a decomposition scheme based on branch-and-check (BCH, Beck, 2010; Thorsteinsson, 2001).

The general idea of BCH is to split a complicated problem into two parts, which are individually easier to solve: a mixed-integer programming master model, which is a relaxation of the original problem, and a subproblem, which encodes the remaining variables and/or constraints, albeit not necessarily in the form of a MIP. The master model is solved using classic branch-and-bound methods. Periodically, when it is advantageous, the subproblem is solved at a node of the branch-and-bound tree to check the feasibility of the current master solution and/ or determine its exact objective value. This information is then communicated to the overarching branch-and-bound process, usually via cuts. The search terminates when there are no more unfathomed nodes in the enumeration tree left to explore. The best found solution is optimal.

7.9 Froger 等 (2022): 容量受限充电站模型

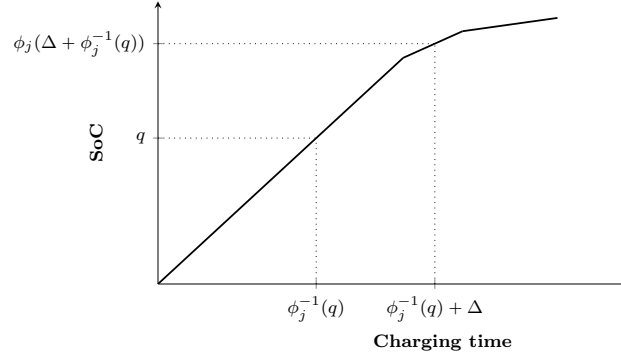


Figure 1: Piecewise linear charging function of a CS $j \in F$.

sorted in non-decreasing order of time.

Each CS $j \in F$ also has a *capacity*, given by the number of available chargers C_j . Due to the limited capacity of CSs, vehicles may incur waiting times while they queue for a charger. We therefore note that optimal solutions are not necessarily *left-shifted schedules*, as is the case for the E-VRP-NL and other E-VRP variants assuming uncapacitated CSs.

Feasible solutions to the E-VRP-NL-C must satisfy the following conditions: 1) each customer is visited exactly once by a single vehicle, 2) each route starts at the depot not before time 0 and ends at the depot not later than time T_{max} , 3) each route is energy-feasible, i.e., the SoC of an EV when it enters and leaves from any location lies between 0 and Q , and 4) no more than C_j EVs simultaneously charge at each CS $j \in F$. The objective of the E-VRP-NL-C is to minimize the total time needed to serve all customers, including driving, service, charging, and waiting times. Note that to avoid congestion at CSs, the starting times of the routes can be delayed at no cost. Moreover, an E-VRP-NL-C instance may have no feasible solution.

3 A mixed-integer linear programming formulation

We can classify explicit the mixed integer linear programming formulations for E-VRPs into two categories: CS replication-based formulations and path-based formulations. CS replication-based formulations are the compact MILP formulations typically used in the E-VRP literature to mathematically define the problem and solve toy instances. As their name suggests, in these formulations the nodes representing CSs are replicated and the number of stops at each CS replication is constrained to one. Note that in this case the number of stops at a CS is limited to its number of replications. This modeling artifice allows for tracking of the routes' travel time, distance, and SoC whenever there are several stops at the CS. To ensure that no optimal solution is cut off, the number of replications of each CS must be very large. For instance, Froger et al. (2019) provided an example where this value is equal to $4|I|$, when the congestion at each CS is neglected. CS replication-based formulations yield intractable MILPs and introduce symmetry issues.

Path-based formulations are based on a more intricate modeling strategy by which the problem is defined on a multigraph. In a nutshell, in these formulations, the nodes represent locations (the depot and the CSs or the depot and the customers) and the arcs represent possible paths between every pair of locations. A path may be defined as a sequence (possibly empty) of visits to customers between two CSs or the depot (Bruglieri et al., 2019a,b) or as a sequences of stops at CSs between two customers or the depot (Roberti and Wen, 2016; Andelmin and Bartolini, 2017; Froger et al., 2019). We refer to the former case as *customer path* and to the latter as a *CS path*. In both cases, dominance rules are defined to limit the number of paths taken into consideration. These formulations no longer require CS vertices replication.

There also exist set partitioning-based formulations (see for instance Desaulniers et al. (2016)), but they are out of the scope of this paper. Table 1 provides details on the formulations used in the E-VRP research.

We introduce a path-based formulation for the E-VRP-NL-C. We decided to use CS paths rather than customer

paths for two reasons. First, the number of customer paths grows much faster than the number of CS paths. Second, customer path-based formulations require introducing a sufficient number of clones of paths that do not visit any customer between two CSs, whereas a CS path can at most be traveled by a single EV. We point out that the modeling of the CS capacity constraints we introduce below can easily be adapted to customer path-based formulations.

The concept of CS paths leads to a redefinition of the problem on a directed multigraph $G = (V, P)$, where $V = \{0\} \cup I$, and P is the set of arcs connecting the vertices of V . An arc in P represents a CS path p , starting in vertex $\text{org}(p)$ visiting a sequence of CSs or none and ending in vertex $\text{dest}(p)$. We denote $\mathbf{n}(p)$ the number of CSs in p . If $\mathbf{n}(p)$ is equal to 0, then p does not stop at any CS between $\text{org}(p)$ and $\text{dest}(p)$. Otherwise, we denote by $\text{cs}(p, l)$ the l -th CS visited in p (with $l \in \{1, \dots, \mathbf{n}(p)\}$). We define inputs $\mathbf{e}(p)$ and $\mathbf{t}(p)$ as the energy consumption (energy used to travel the arcs of the path) and the driving time (time to travel the arcs of the path) initially associated with CS path p . Given two vertices $i, j \in V$, we define P_{ij} as the set of CS paths connecting i to j .

Our path-based formulation of the E-VRP-NL-C involves the following decisions variables. The binary variable x_p is 1 if and only if an EV travels CS path $p \in P$. The continuous variables τ_p and y_p track the time and SoC of an EV when it departs from vertex $\text{org}(p)$ to $\text{dest}(p)$ using CS path p . To model the charging process at $\text{cs}(p, l)$, we introduce a group of variables. The continuous variables q_{pl} and $\bar{q}_{pl}$ specify the SoC of an EV when it enters and leaves $\text{cs}(p, l)$. For $k \in B_i \setminus \{0\}$, the binary variables $\underline{w}_{plk}$ and $\bar{w}_{plk}$ are equal to 1 if and only if the SoC is between $q_{\text{cs}(p, l), k-1}$ and $q_{\text{cs}(p, l), k}$ when the EV enters and leaves $\text{cs}(p, l)$. The variables $\underline{a}_{pl}$ and $\bar{a}_{pl}$ are the scaled arrival and departure times, according to the charging function of $\text{cs}(p, l)$. The continuous variables $\underline{\Delta}_{plk}$ and $\bar{\Delta}_{plk}$ represent the coefficients associated with the breakpoints $(a_{\text{cs}(p, l), k}, q_{\text{cs}(p, l), k})$ of the piecewise linear charging function, when the EV enters and leaves $\text{cs}(p, l)$. The continuous variable Δ_{pl} and ∇_{pl} represents the duration of the charging operation performed at $\text{cs}(p, l)$ and the waiting time incurred before charging at this CS. Lastly, the continuous variables $\underline{s}_{pl}$ and $\bar{s}_{pl}$ represent the starting and completion time of the charging operation performed at $\text{cs}(p, l)$.

To model the CS capacity constraints, we propose a flow-based formulation inspired from the one introduced by Artigues et al. (2003) for the Resource Constrained Scheduling Problem. For every CS $j \in F$, we consider C_j resources (represented by the chargers). Each resource can execute at most one operation at any given time. For convenience, we introduce the set O_j of potential charging operations at CS j . Every operation $o \in O_j$ represents a stop at j at a specific position $l(o)$ in a CS path going from $\text{org}(o)$ to $\text{dest}(o)$. Although there may exist several CS paths between two specific vertices that stop at j at the same position, at most one these CS paths can be selected in a solution. We also introduce two dummy operations $\mathbf{o}_j^+$ and $\mathbf{o}_j^-$, which act as the source and the sink of the flow for each CS j . Specifically, the resources flow from $\mathbf{o}_j^+$ to $\mathbf{o}_j^-$ through the charging operations. Let $(o, o') \in (O_j \cup \{\mathbf{o}_j^+\}) \times (O_j \cup \{\mathbf{o}_j^-\})$ be a couple of charging operations; then the continuous flow variable $f_{oo'}$ denotes the number of chargers that are transferred from operation o to operation o' . When both these operations are not dummy, $f_{oo'}$ is equal to one if and only if these operations are scheduled on the same charger, o' is scheduled after o , and no other operation is scheduled on the charger between the completion of o and the beginning of o' . Binary variable $u_{oo'}$ is equal to 1 if and only if operation o' starts after the completion of operation $o \neq o'$. For notational convenience, we define $\mathbf{C}(o) := 1$ for all $o \in O_j$ and $\mathbf{C}(\mathbf{o}_j^+) := \mathbf{C}(\mathbf{o}_j^-) := C_j$.

A path-based formulation for the E-VRP-NL-C, denoted as $[P^{\text{path}}]$, is as follows:

$$[P^{\text{path}}] \quad \text{minimize} \quad \sum_{p \in P} \left(\mathbf{t}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \right) \quad (1)$$

$$\text{subject to} \quad \sum_{j \in V, i \neq j} \sum_{p \in P_{ij}} x_p = 1 \quad i \in I \quad (2)$$

$$\sum_{j \in V, i \neq j} \sum_{p \in P_{ji}} x_p - \sum_{j \in V, i \neq j} \sum_{p \in P_{ij}} x_p = 0 \quad i \in V \quad (3)$$

$$\sum_{j \in V, j \neq i} \sum_{p \in P_{ji}} \left(y_p - \mathbf{e}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\bar{q}_{pl} - q_{pl}) \right) = \sum_{j \in V, j \neq i} \sum_{p \in P_{ij}} y_p \quad i \in I \quad (4)$$

$$y_p - e_{\text{org}(p), \text{cs}(p, 1)} x_p = \underline{q}_{p1} \quad p \in P \quad (5)$$

$$\bar{q}_{p, l-1} - e_{\text{cs}(p, l-1), \text{cs}(p, l)} x_p = \underline{q}_{pl} \quad p \in P, l \in \{2, \dots, \mathbf{n}(p)\} \quad (6)$$

$$y_p - \mathbf{e}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\bar{q}_{pl} - \underline{q}_{pl}) \geq 0 \quad i \in I, p \in P_{i0} \quad (7)$$

$$y_p = Qx_p \quad i \in V \setminus \{0\}, p \in P_{0i} \quad (8)$$

$$y_p \leq Qx_p \quad p \in P \quad (9)$$

$$\underline{q}_{pl} = \sum_{k \in B_{cs(p,l)}} \lambda_{plk} q_{cs(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (10)$$

$$\underline{a}_{pl} = \sum_{k \in B_{cs(p,l)}} \lambda_{plk} a_{cs(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (11)$$

$$\sum_{k \in B_{cs(p,l)}} \lambda_{plk} = \sum_{k \in B_{cs(p,l)} \setminus \{0\}} \underline{w}_{plk} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (12)$$

$$\sum_{k \in B_{cs(p,l)} \setminus \{0\}} \underline{w}_{plk} = x_p \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (13)$$

$$\lambda_{pl0} \leq \underline{w}_{pl1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (14)$$

$$\lambda_{plk} \leq \underline{w}_{plk} + \underline{w}_{pl,k+1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0, b_{cs(p,l)}\} \quad (15)$$

$$\lambda_{plb_{cs(p,l)}} \leq \underline{w}_{plb_{cs(p,l)}} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (16)$$

$$\bar{q}_{pl} = \sum_{k \in B_{cs(p,l)}} \bar{\lambda}_{plk} q_{cs(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (17)$$

$$\bar{a}_{pl} = \sum_{k \in B_{cs(p,l)}} \bar{\lambda}_{plk} a_{cs(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (18)$$

$$\sum_{k \in B_{cs(p,l)}} \bar{\lambda}_{plk} = \sum_{k \in B_{cs(p,l)} \setminus \{0\}} \bar{w}_{plk} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (19)$$

$$\sum_{k \in B_{cs(p,l)} \setminus \{0\}} \bar{w}_{plk} = x_p \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (20)$$

$$\bar{\lambda}_{i0} \leq \bar{w}_{pl1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (21)$$

$$\bar{\lambda}_{plk} \leq \bar{w}_{plk} + \bar{w}_{pl,k+1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0, b_{cs(p,l)}\} \quad (22)$$

$$\bar{\lambda}_{plb_{cs(p,l)}} \leq \bar{w}_{plb_{cs(p,l)}} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (23)$$

$$\Delta_{pl} = \bar{a}_{pl} - \underline{a}_{pl} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (24)$$

$$\sum_{j \in V, j \neq i} \sum_{p \in P_{ji}} \left(\tau_p + \mathbf{t}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \right) + g_i = \sum_{j \in V, j \neq i} \sum_{p \in P_{ij}} \tau_p \quad i \in I \quad (25)$$

$$\tau_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \leq (T_{max} - \mathbf{t}(p) - g_{\text{dest}(p)} - t_{\text{dest}(p),0}) x_p \quad p \in P \quad (26)$$

$$\tau_p + t_{\text{org}(p),cs(p,1)} x_p + \nabla_{p1} = \underline{s}_{p1} \quad p \in P \quad (27)$$

$$\bar{s}_{p,l-1} + t_{cs(p,l-1),cs(p,l)} x_p + \nabla_{pl} = \underline{s}_{pl} \quad p \in P, l \in \{2, \dots, \mathbf{n}(p)\} \quad (28)$$

$$\bar{s}_{pl} = \underline{s}_{pl} + \Delta_{pl} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (29)$$

$$\sum_{o' \in O_j \cup \{o_j^+\}} f_{o'o} = \sum_{\substack{p \in P_{\text{org}(o),\text{dest}(o)} \\ :cs(p,l(o))=j(o)}} x_p \quad j \in F, o \in O_j \quad (30)$$

$$\sum_{o' \in O_j \cup \{o_j^+\}} f_{o'o} - \sum_{o' \in O_j \cup \{o_j^-\}} f_{oo'} = 0 \quad j \in F, o \in O_j \quad (31)$$

$$\sum_{o \in O_j \cup \{o_j^-\}} f_{o_j^+,o} = C_j \quad j \in F \quad (32)$$

$$\sum_{o \in O_j \cup \{o_j^+\}} f_{o,o_j^-} = C_j \quad j \in F \quad (33)$$

$$\sum_{\substack{p \in P_{\text{org}(o),\text{dest}(o)} \\ :cs(p,l(o))=j(o)}} \underline{s}_{p,l(o)} - \sum_{\substack{p \in P_{\text{org}(o'),\text{dest}(o')} \\ :cs(p,l(o'))=j(o')}} \bar{s}_{p,l(o')} \geq T_{max} (u_{o'o} - 1) \quad j \in F, o, o' \in O_j \quad (34)$$

$$f_{oo'} \leq \min(\mathcal{C}(o), \mathcal{C}(o')) u_{oo'} \quad j \in F, (o, o') \in (O_j \cup \{o_j^+\}) \times (O_j \cup \{o_j^-\}) \quad (35)$$

$$x_p \in \{0, 1\} \quad p \in P \quad (36)$$

$$\tau_p \geq 0, y_p \geq 0 \quad p \in P \quad (37)$$

$$\underline{q}_{pl}, \bar{q}_{pl}, \underline{a}_{pl}, \bar{a}_{pl}, \underline{s}_{pl}, \bar{s}_{pl}, \Delta_{pl}, \nabla_{pl} \geq 0 \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (38)$$

$$\underline{\lambda}_{plk} \geq 0, \bar{\lambda}_{plk} \geq 0 \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0\} \quad (39)$$

$$\underline{w}_{plk} \in \{0, 1\}, \bar{w}_{plk} \in \{0, 1\} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0\} \quad (40)$$

$$u_{oo'} \in \{0, 1\} \quad j \in F, o, o' \in O_j \quad (41)$$

$$f_{oo'} \geq 0 \quad j \in F, (o, o') \in (O_j \cup \{\mathbf{o}_j^+\}) \times (O_j \cup \{\mathbf{o}_j^-\}) \quad (42)$$

The objective function (1) minimizes the total driving, charging, and waiting time. Constraints (2) ensure that each customer is visited exactly once. Constraints (3) impose flow conservation: at each vertex the number of incoming EVs is equal to the number of outgoing EVs. Constraints (4) track the SoC of EVs at each customer location. Constraints (5) track the SoC at the arrival at the first CS of each CS path. Constraints (6) couple the SoC of an EV that leaves a CS with its SoC upon arrival to the next CS of the same path. Constraints (7) ensure that if the EV travels from a customer to the depot, it has sufficient energy to reach its destination. Constraints (8) state that every EV leaves the depot with a fully charged battery. Constraints (9) couple the SoC tracking variables with the path travel variables. Constraints (10)–(24) model the relationship between the SoCs of an EV upon entering and leaving a CS and the charging time. Constraints (25) track the departure time at each customer. Constraints (26) couple the time tracking variable with the path travel variables, and impose the route duration limit. Constraints (27) and (28) define the starting and completion times of every charging operation, as well as the potential waiting time before the start of the operation. Constraints (29) define the completion time of the charging operations based on their starting time and duration. Constraints (30) state that a charger has to be allocated to every charging operation of each selected CS path (at most one CS path in the sum can be selected in a solution). Constraints (31) ensure flow conservation of the chargers of each CS. Constraints (32) and (33) compute the flow value at the beginning and at the end of the time horizon, which is equal to the number of available chargers at each CS. Constraints (34) couple the sequencing variables with the starting time of the corresponding charging operations. Constraints (35) couple the flow variables with the operation sequence variables. Specifically, a charger can be sent from a charging operation o' to another charging operation o if o starts after the completion of o' . Finally, constraints (36)–(42) set the domains of the decision variables.

Without preprocessing, the number of CS paths explodes with the number of CSs and the number of customers. However, a large number of these arcs cannot be part of an optimal solution. Froger et al. (2019) presented a filtering procedure to reduce the number of CS paths based on the definition of a dominance rule between two CS paths having the same origin and destination. Due to the potential waiting times that can occur at CSs, we cannot apply the dominance rule described in the work of Froger et al. (2019) between CS paths with the same origin and destination if they both contain CSs. This is due to the fact that waiting times depend on all the CS paths selected to build a complete solution. In our computational experiments, we apply the dominance rule between the unique CS path with no CS between i and j (if it exists) and every other CS path of P_{ij} , with i and j in V .

4 Solution method

In this section, we introduce a matheuristic to solve the E-VRP-NL-C. The method relies on an algorithmic framework based on two interacting components: a route generator and a solution assembler. The first component builds a pool of high-quality and diverse routes exploring the solution space of the problem without CS capacity constraints, while the second component recombines routes from the pool trying to build a solution satisfying the CS capacity constraints.

Algorithm 1 outlines the general structure of the method. The algorithm starts by generating an initial solution without taking the CS capacity constraints into account (line 1). In our implementation, this step is carried out using a modified version of the classical Clarke and Wright heuristic. Next, the algorithm enters the main loop (lines 3–27). During n_{max} iterations the algorithm alternates between the *route generation* and *solution assembly* phases. In particular, the algorithm uses procedure **generateRoutes**($\cdot$) to retrieve a triplet $(\Phi_O, \Phi_R, \Omega^{ST})$, where Φ_O, Φ_R are sets of high-quality solutions for the *original* and *relaxed* problems (i.e., with and without CS capacity constraints), and Ω^{ST}

7.10 邱莹莹 (2024): 混合车队与动态需求模型

第三章 模型构建

3.1 问题描述

在实际物流配送中，有两个关键问题需要综合考虑。首先，传统燃油车辆的广泛使用导致了大量碳排放，这与环保和可持续发展的目标相矛盾。其次，电动汽车在物流配送中的应用受到一系列限制，包括电池容量、充电时间等。这些问题迫使物流运输企业需要寻找创新解决方案，以减少碳排放同时控制运营成本。

为了应对这一挑战，物流运输企业组建了一支由传统燃油车和电动汽车组成的混合配送车队。假设该车队有 K^C 辆传统燃油车， K^E 辆电动汽车， m 个顾客点， n 个充电站和一个配送中心，其中，客户点的需求和各点的位置已知，车辆需要在客户时间窗内到达，不许早到、迟到。离开配送中心的车辆必须在配送服务完成后返回配送中心，每辆车只能为一条路径服务，过程中客户的需求不能分割，只由一辆车为其提供服务。由于电动汽车续航里程的限制，允许电动汽车在中途寻找合适的充电站充电，充电方式为快充，每次充电都充满。

混合车队物流配送问题的关键目标是在规定的时间内满足客户需求，同时规划最经济、低碳的配送路径，使混合车队的总配送成本最低。这一问题要求综合考虑车型、路径、电池电量管理、充电站选择等多个因素，从而优化整个物流配送过程，提高运营效率，降低碳排放，实现可持续发展目标。

3.2 混合车队车辆路径规划问题模型构建

3.2.1 参数定义

根据以上问题描述及假设，相关参数及变量的符号定义如表 3.1 所示，其中 C 表示传统燃油车， E 表示电动汽车， $A = N \cup \{0\} \cup R$ ， $O = N \cup \{0\}$ ：

表 3.1 符号及定义

符号	含义	符号	含义
N	顾客点集合, $N = \{1, 2, \dots, n\}$	c^C / c^E	车辆启动成本(元/辆)
0	配送中心	m^C / m^E	单位距离行驶成本(元/km)

第三章 模型构建

续表 3.1 符号及定义

B	电动汽车电池容量(kwh)	τ_i	车辆到达顾客 i 时间(min)
R	充电站集合	$[e_i, l_i]$	节点 i 的服务时间窗
Q^C	燃油车最大负载(kg)	h_{ik}^1	电车 k 到达顾客 i 时剩余电量
Q^E	电车最大负载(kg)	h_{ik}^2	电车 k 离开顾客 i 时剩余电量
K^C	燃油车集合	w_r	单位电量成本(元/kwh)
K^E	电车集合	s_i	顾客 i 的服务时间(min)
p_i^C / p_i^E	车辆到达 i 点时的载重(kg)	r_i	电车在充电站 i 充电量(kwh)
q_i	顾客 i 的需求量(kg)	T_i	电车在充电站充电时间(min)
t_{ij}	从顾客 i 到 j 的行驶时间 (min)	w_o	单位燃油成本(元/L)
d_{ij}	顾客 i 到 j 的距离(km)	w_c	单位碳排放成本(元/kgCO2)
G_{ij}	电车的电量消耗率(kwh/min)	v	车辆行驶速度(km/h)
L_{ij}	燃油车从 i 到 j 的油耗量(L)	x_{ijk}^C / x_{ijk}^E	0-1 变量, 车辆 k 从 i 驶向 j 时为 1, 否则为 0
E_{ij}	燃油车从 i 到 j 的碳排放量		
g_r	充电站充电速率(kwh/h)	y_i	0-1 变量, 电车在 i 处充电为 1, 否则为 0.

3.2.2 目标函数

本文同时考虑燃油车的碳排放和车队的能源消耗, 目标是使车辆的使用成本、行驶成本、能耗成本等运营成本的最小化

(1) 车辆启动成本

车辆的固定启动成本不受行驶里程、载重量和行驶时间约束，是车辆被安排从配送中心出发所产生的费用，一般包含司机的工资、货物装卸搬运的费用、车辆的折损费等。在一些情况下，物流公司更倾向于选择租赁车辆而不是购买车辆。这时，租赁费用可能被视为固定启动成本的一部分。具体来说，每当物流公司启动一辆租赁货车，都需要支付一辆车的固定启动费用。这种做法允许公司更加灵活地应对需求波动，降低养护整个车队的成本，从而能够更灵活地控制物流成本，此时的物流配送产生的启动成本为：

$$f_1 = c^C \sum_{k \in K^C} \sum_{j \in O} x_{ojk}^C + c^E \sum_{k \in K^E} \sum_{j \in A} x_{ojk}^E$$

(2) 行驶成本

通常情况下，随着车辆行驶距离的增加，车辆的运营费用也会逐渐上升。随着行驶里程的增加，车辆的机械部件和系统会更频繁地暴露于磨损和损坏的风险之下，这包括定期保养、零部件更换以及车辆维护以及维修所需的费用。此外，行驶里程的增加还涉及到支付通行费用，例如使用特定道路或桥梁，这些费用也会对车辆的运营成本产生直接的影响。则物流配送中产生的行驶费用为：

$$f_2 = m^C \sum_{k \in K^C} \sum_{(i,j) \in O} x_{ijk}^C d_{ij} + m^E \sum_{k \in K^E} \sum_{(i,j) \in A} x_{ijk}^E d_{ij}$$

(3) 充电成本

电动汽车的耗电量受多重因素的影响，本研究中将电动汽车配送中的耗电成本直接计算为电动汽车在充电站的充电成本，不考虑电车的行驶速度、车辆的载重、地形等因素对耗电率的影响。则电动汽车在配送中产生的充电成本为：

$$f_3 = w_r \sum_{k \in K^E} \sum_{(i,j) \in A} x_{ijk}^E r_i$$

(4) 油耗成本

在不同的油耗估算模型中，车辆的耗油量受载重、车速、路况等因素的影响。本文采用综合模式排放模型(comprehensive modal emission model, CMEM)进行油耗量估算。该模型可以计算出不同驾驶条件下每秒的尾气管排放值和油耗量^[60]，相较于其他油耗估算模型，CMEM 模型简单易于执行且估算结果准确率较高^[61]。推导过程如下：

$$FR = \frac{\varphi(\lambda HV + \frac{P}{\eta})}{\mu}$$

第三章 模型构建

FR : 油耗率(g/s)

μ : 柴油热量值

φ : 油气质量比

λ : 发动机摩擦系数

H : 发动机转速

V : 发动机排量

η : 发动机效率参数

P : 发动机功率, 计算公式如下:

$$P = \frac{\left[(M^C + N_{ij}) g C_r + \frac{1}{2} \rho C_a A v^2 \right] v}{\varepsilon}$$

ε : 传动效率

当燃油车在路段 (i, j) 上匀速行驶时, 则油耗量为:

$$\begin{aligned} L_{ij} &= \frac{FR}{\zeta} t_{ij} \\ &= \frac{\varphi(\lambda HV + \frac{P}{\eta})}{\zeta \mu} \frac{d_{ij}}{v} \end{aligned}$$

ζ : 单位转换系数, g/s 转换为 L/h

则燃油车在物流配送中产生的油耗成本为:

$$f_4 = w_o \sum_{k \in K^C} \sum_{(i,j) \in O} x_{ijk}^C L_{ij}$$

(5) 碳排放成本

碳排放量与油耗量呈线性相关, 则燃油车从 i 行驶到 j 的碳排放量为:

$$E_{ij} = L_{ij} e$$

e : 燃油排放因子

在物流配送过程中, 电动汽车是“零碳排放”运载工具, 故本文只考虑传统燃油车的碳排放, 则物流企业配送中产生的碳排放费用为:

$$f_5 = w_c \sum_{k \in K^C} \sum_{(i,j) \in O} x_{ijk}^C E_{ij}$$

本研究中, 混合车队物流配送车辆路径研究的目标是总成本最小化, 则该模

型的目标函数为：

$$\min F = f_1 + f_2 + f_3 + f_4 + f_5 \quad (3-1)$$

3.3 考虑动态需求的混合车队车辆路径规划

3.3.1 目标函数

处理动态需求时的一个关键点是新需求的及时处理，灵活的调整以满足新需求的，可以应对不断变化的市场和客户需求，同时不会对物流配送的效率和成本产生重大影响。即使总成本有轻微增加，但需要维持在一个可控制的范围内，尤其是考虑到动态需求的不可预测性。假设该车队沿用混合车队的配置，车辆仍需要在客户时间窗内到达，不许早到、迟到。同样规定离开配送中心的车辆必须在配送服务完成后返回配送中心，每辆车只能为一条路径服务，过程中客户的需求不能分割，只由一辆车为其提供服务。允许电动汽车在中途寻找合适的充电站充电，充电方式为快充，每次充电都充满。

动态规划相关的参数设置如表 3.2 所示。

表 3.2 动态规划符号和含义

符号	含义	符号	含义
N'	第 n 次更新后待配送顾客	k'	重新派遣的车辆
K^C	第 n 次更新未出发的燃油车集合	z^C	第 n 次更新发出燃油车
K^E	第 n 次更新未出发的电车集合	z^E	第 n 次更新发出电动汽车

（1）车辆启动成本

第 n 次更新后，车辆将继续执行配送任务，包括为未服务的顾客提供服务以及处理新增的顾客需求。然而，在这个阶段，可能会出现一些挑战，如车辆的容量不足以满足某些顾客的需求或者时间窗约束的问题。当已经出发的车辆在满足这些限制方面遇到问题时，配送中心需要采取一些调整措施，以确保高效的物流配送。在这种情况下，重新派车配送是一种常见的解决方法，需要综合考虑启动成本，以寻求最佳的解决方案，确保顾客的需求得到满足，并保持成本在可控范围内。此时，车辆的启动成本为：

$$f_1' = f_1 + c^C \sum_{k \in z^C} \sum_{j \in N'} x_{ojk}^C + c^E \sum_{k \in z^E} \sum_{j \in N'} x_{ojk}^E$$

第三章 模型构建

(2) 车辆行驶成本

第 n 次更新后, 车辆的行驶成本包括已服务的运输成本和规划后的待服务节点运输成本, 此时, 车辆行驶成本为:

$$f_2' = m^C \sum_{k' \in k \cup K^{C'}} \sum_{(i,j) \in N \cup N'} x_{ijk'}^C d_{ij} + m^E \sum_{k' \in k \cup K^{E'}} \sum_{(i,j) \in A \cup N'} x_{ijk'}^E d_{ij}$$

(3) 耗电成本

第 n 次更新后, 派遣的车辆若为电动汽车, 则此时耗电成本为:

$$f_3' = w_r \sum_{k' \in k \cup K^{E'}} \sum_{(i,j) \in N' \cup A} x_{ijk'}^E r_i$$

(4) 油耗成本

第 n 次更新后, 派遣的车辆若为传统燃油车, 则此时油耗成本为:

$$f_4' = w_o \sum_{k' \in k \cup K^{C'}} \sum_{(i,j) \in O \cup N'} x_{ijk'}^C L_{ij}$$

(5) 碳排放成本

当第 n 次更新后, 派遣的车辆为传统燃油车, 此时将涉及到的碳排放成本, 计算公式为:

$$f_5' = w_c \sum_{k' \in k \cup K^{C'}} \sum_{(i,j) \in O \cup N'} x_{ijk'}^C E_{ij}$$

此时, 考虑动态需求的混合车队车辆路径规划的目标函数为:

$$\min F' = f_1' + f_2' + f_3' + f_4' + f_5'$$

3.3.2 动态信息处理策略

在面对复杂的动态车辆路径规划问题(DVRP)时, 通常将其转化为多阶段的静态车辆路径问题(SVRP)。目前常见的信息处理策略包括单事件处理策略和批事件处理策略^[62]。

单事件处理策略是指, 每当系统收到一条动态信息时, 立刻处理该信息, 更新当前的顾客数据, 对瞬时静态车辆路径进行规划, 该策略处理静态信息的方案仅与有无动态信息出现有关^[65]。该策略的优点是能够及时处理动态信息, 满足顾客的需求, 显著提高消费者满意度。但是当处理大批量动态信息同时出现时, 该方法处理信息较繁琐, 频繁地更新、优化配送路径可能会使效率低下, 最终可能导致配送系统的瘫痪。批事件处理策略是指当特定时间点的动态需求满足物流系

统中预先定义的条件时，批量处理将接收到的动态信息。该策略包含定时优化和定量优化。定时优化将配送中心的配送时间窗分成 n 等份，每一时间段的动态需求都在该时间段末集中处理，如此，动态车辆路径问题就分为 n 个静态路径问题。定量优化处理则是提前设置处理需求顾客个数等的上限，达到该上限就立刻对累积的动态需求进行处理。批事件处理策略的优点在于减少了优化的次数，更有利于配送系统的信息管理。

本研究考虑混合车队物流配送过程中顾客需求的新增、静态顾客需求的增减和取消三种动态信息出现的情况。其中需求的增减处理方法为先取消该订单，后将其作为一个新增的动态顾客需求，所有新增的需求都不可拆分，必须由一辆车对其服务。

本文的动态需求处理策略同时考虑了定时和定量两种批事件处理策略，进一步增强物流配送的灵活性。配送中心的服务时间窗为 $[e_0, l_0]$ ，设定在 $[t_0, t_n]$ 时刻内接受动态需求信息，将该时段均分为 n 个时间区间 $[u(0), u(1), u(2), \dots, u(n-1), u(n)]$ 。设置动态需求和上限为 $\bar{q}$ ，当顾客需求总量达到该上限时，将顾客加入待配送行列。同时，设置动态需求处理时间间隔 T ，当需求总量未达到上限，经过一个时间间隔也能够处理动态顾客，动态信息处理流程如图 3.1 所示。当动态顾客的需求量和达到 $\bar{q}$ 或动态需求的处理时间间隔达到 T ，该时刻 $u(t)$ 称为批事件处理时刻，则 $u(t)$ 取值如式子(3-2)所示：

$$u(t) = \min\{u_q, u(t-1) + T, t_n\} \quad (3-2)$$

$$u_q = \min\{u_q \mid q(u(t-1), u_q) \geq \bar{q}, u(t-1) > u_q\} \quad (3-3)$$

其中， $u(0) = t_0$ ， $u(n) = t_n$ 。

第三章 模型构建

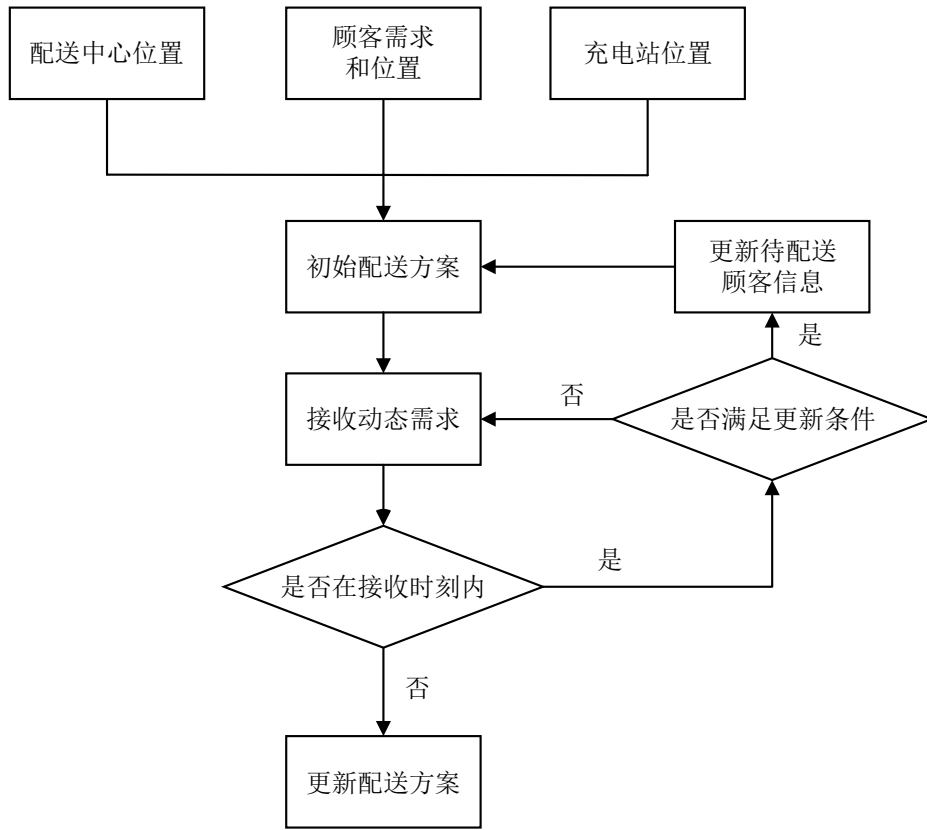


图 3.1 动态信息处理流程

3.4 约束条件

本文对混合车队的物流配送路径研究主要包含四种约束，即为节点访问、载重约束、时间窗约束、电动汽车的电量约束。

(1) 节点访问约束

$$\sum_{i \in O, i \neq j} \sum_{k \in K} (x_{ij}^C + x_{ij}^E) = 1, \forall j \in O \quad (3-4)$$

$$\sum_{k=1}^{K^C} \sum_{(i,j) \in O} x_{ijk}^C = \sum_{k=1}^{K^C} \sum_{(j,i) \in O} x_{jik}^C \quad (3-5)$$

$$\sum_{k=1}^{K^E} \sum_{(i,j) \in A} x_{ijk}^E = \sum_{k=1}^{K^E} \sum_{(j,i) \in A} x_{jik}^E \quad (3-6)$$

其中，约束式(3-4)表示每个客户只被服务一次；约束式(3-5)、(3-6)表示流量的平衡，即每辆车进入节点的次数等于其离开的次数，且使用的车辆数不超过车

队车辆的总数。

(2) 载重约束

$$\sum_{i \neq j} \sum_{k \in K^C} q_i x_{ijk}^C \leq Q^C, \forall i \in N, j \in O \quad (3-7)$$

$$\sum_{i \neq j} \sum_{k \in K^E} q_i x_{ijk}^E \leq Q^E, \forall i \in N, j \in A \quad (3-8)$$

$$p_j^C \leq p_i^C - q_i x_{ijk}^C + Q^C (1 - x_{ijk}^C), \forall i, j \in O, k \in K^C \quad (3-9)$$

$$p_j^E \leq p_i^E - q_i x_{ijk}^E + Q^E (1 - x_{ijk}^E), \forall i, j \in A, k \in K^E \quad (3-10)$$

其中，约束式(3-7)和(3-8)表示燃油车和电动汽车的车辆载重不超过车辆最大负载；约束式(3-9)和(3-10)表示车辆在访问两相邻顾客之间的载重量约束。

(3) 时间窗约束

$$e_i \leq \tau_i \leq l_i, \forall i \in A \quad (3-11)$$

$$T_i = r_i y_i / g_r, \forall i \in A \quad (3-12)$$

$$\tau_j \geq \tau_i + (t_{ij} + s_i) x_{ijk}^C - l_0 (1 - x_{ijk}^C), \forall i, j \in O, k \in K^C \quad (3-13)$$

$$\tau_j \geq \tau_i + (t_{ij} + s_i) x_{ijk}^E - l_0 (1 - x_{ijk}^E), \forall i, j \in A, k \in K^E \quad (3-14)$$

$$\tau_j \geq \tau_i + t_{ij} x_{ijk}^E + T_i y_i - l_0 (1 - x_{ijk}^E), \forall i \in R, j \in O, k \in K^E \quad (3-15)$$

其中，约束式(3-11)表示服务时间必须满足顾客时间窗约束；约束式(3-12)，表示电动汽车在充电站的充电时长，(3-13)、(3-14)表示两种车辆分别访问相邻的两个顾客节点的时间窗约束；约束式(3-15)表示电动汽车在访问充电站后的时间窗约束。

(4) 电量约束

$$h_{jk}^1 = \sum_{i \in A, i \neq j} (h_{ik}^2 - d_{ij} g_v) x_{ijk}^E, \forall j \in A, k \in K^E \quad (3-16)$$

$$h_{ik}^2 = B y_i, \forall i \in R, k \in K^E \quad (3-17)$$

$$r_i = (B - h_{ik}^1) y_i, \forall i \in R, k \in K^E \quad (3-18)$$

$$h_{ik}^1 \geq G_{ij} d_{ij} / v_{ij} x_{ijk}^E, \forall i \in N, j \in A, k \in K^E \quad (3-19)$$

$$h_{ik}^1 = h_{ik}^2, \forall i \in N, k \in K^E \quad (3-20)$$

其中, 约束式(3-16)表示电动汽车在服务两相邻顾客的剩余电量; 约束式(3-17)表示电动汽车在离开充电桩时是充满电的状态; 约束式(3-18)表示电动汽车到达充电站时充电量的计算式; 约束式(3-19)表示电动汽车在节点处的剩余电量可以到达最近的充电站; 约束式(3-20)表示电动汽车在服务顾客时不消耗电量。

3.5 本章小结

首先, 本章着重描述了混合车队物流配送中的车辆路径问题, 将问题的背景和范围明确定义, 并对研究中的情景进行了合理的假设, 以确保问题的可行性和实际应用性。这一部分的重点在于确定问题的核心要素和特征。

接下来, 本节对问题中涉及的变量和参数进行了详细的定义, 以建立清晰的数学模型。这些变量和参数包括了车辆的数量、货物的数量、车辆的路径、货物的配送时间窗口等。

本章还分析了研究模型的目标函数, 将焦点放在了多个方面, 包括车辆的行驶速度、载重量、燃油车的油耗、碳排放量和电动汽车的电能。综合考虑这些因素, 本章节构建了一个目标函数模型, 旨在使总成本最小化, 其中包括启动成本、行驶成本、能源消耗成本以及碳排放成本等。

最后, 本文详细列举了在解决问题的过程中必须满足的约束条件。这些包括了载重约束, 时间窗约束, 节点访问约束以及电动汽车电池的能量约束。考虑这些约束条件能够确保问题的解决方案是可行且符合实际情况的。

7.11 饶卫振等 (2022): 协同配送与成本分摊模型

表 4 服务质量评估指标、含义及量化方法

一级指标	二级指标	量化方法	指标方向	指标名称
关系绩效	受理需求所等待的时间	自客户下单至订单开始配送的时间间隔	负指标	A1
	响应时间便利程度	配送中心营业时长	正指标	A2
	配送人员知识经验	配送业务培训场次与受训人数比值	正指标	A3
	企业社会声誉	企业商誉资产价值	正指标	A4
	操作流程及客户观察透明度 ^[28]	订单配送信息实时推送覆盖率	正指标	A5
运营绩效	配送周期	平均配送时长	负指标	A6
	配送网络覆盖率	企业县级网点覆盖面积占总区域面积比重	正指标	A7
	准时送达率	72 小时内订单送达的比率	正指标	A8
	运输能力与需求匹配程度 ^[29]	运输车辆数量与客户订单量比值	正指标	A9
成本绩效	运价水平	企业配送首重价格	正指标	A10
协作绩效	成本信息共享程度	不共享或全部共享	正指标	A11
	车辆投入量	投入协作联盟中的自有车辆数量	正指标	A12
	响应近距离程度	配送中心与客户间的平均距离	负指标	A13
	协作前后价差补偿	协作前后配送费用差值累积量	负指标	A14

表 5 服务质量指标数据说明

指标大类	指标细类	数据来源	数据属性
基本指标	A6、A8	国家邮政局快递服务测试通告	历史数据
	A4、A7、A9	企业年度报告	历史数据
	A3	企业年度社会责任报告	历史数据
	A1、A2、A10	企业官方运费时效公告	历史数据
协作指标	A5、A11、A12	企业上传信息	实时数据
	A13、A14	协作优化数据	实时数据

弈论综合赋权法的求解模型.

博弈论综合赋权法的核心思想是, 对初始赋权进行组合, 以 Nash 均衡为协调目标, 利用博弈论均衡思想在不同赋权中寻求一致, 得到最优权重, 使得最优权重与初始赋权之间的权重离差和最小.

求解模型如下: 假设采用 L 种方法对 n 个指标分别赋权, 任意权重向量为 $w_t = \{w_{t1}, w_{t2}, \cdots, w_{tn}\}$, 其中 $(t = 1, 2, \cdots, L)$, 记 α_t 为线性组合系数, 则 L 组权重向量的任意线性组合表示为式 (1).

$$\begin{bmatrix} w_{11} & w_{21} & \cdots & w_{L1} \\ w_{12} & w_{22} & \cdots & w_{L2} \\ \cdots & \cdots & \cdots & \cdots \\ w_{1n} & w_{2n} & \cdots & w_{Ln} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_L \end{bmatrix} = \sum_{t=1}^L \alpha_t w_t. \tag{1}$$

依据博弈论均衡思想, 求解最优综合向量 w_t^* , 即转化为求解 L 组向量的最优线性组合系数 α_t^* , 目标为最优综合向量与各初始向量间的离差和最小. 目标函数及约束条件如下:

$$\begin{cases} \min \sum_{t=1}^L \|w^* - w_t\|_2 = \min \sum_{t=1}^L \left\| \sum_{l=1}^L \alpha_l^* w_l - w_t \right\|_2 \\ \text{s.t.} \quad \sum_{t=1}^L \alpha_t^* = 1, \alpha_t^* \geq 0. \end{cases} \tag{2}$$

3 MOCVRPTWC 模型构建

本文提出考虑客户时间窗和配送能力的多方协作车辆路径问题 (multi-owner collaborative vehicle routing problem with time window and capability, MOCVRPTWC). 其思路是在传统 MOCVRP 模型的基础上, 刻画客户时间窗要求及企业配送资源对配送成本的影响, 得到准确反映客户服务需求和企业协作资源投入的子联盟配送成本.

3.1 问题描述

假设共有 n 个企业, 每个企业有多个配送中心及有限数量的自有车辆, 车辆容量相同, 各点之间的距离已知, 每位客户的需求量已知且为随机数, 客户具有时间窗要求. 已知 n 个企业组建联盟进行协作, 交换距离自身较远但距离其他企业较近的订单, 规划配送路线. 若某一企业车辆短缺, 可立即向其他企业租赁同类型车辆.

MOCVRPTWC 问题的核心是: 在各企业及配送中心资源既定的情况下, 规划每辆车的行驶路径, 以满足客户要求及时间窗限制, 并使联盟总配送成本最低.

3.2 符号定义

$G: \{g | g = 1, 2, \dots, |G|\}$ 为客户集合;

$D: \{d | d = 1, 2, \dots, |D|\}$ 为配送中心集合;

k_{ij} : 第 i ($1 \leq i \leq n$) 个企业的第 j ($1 \leq j \leq m$) 个配送中心的自有车辆数, $\sum_{i=1}^n \sum_{j=1}^m k_{ij}$ 表示所有企业的全部自有车辆;

$K: \{k | k = 1, 2, \dots, |K|\}$ 为企业自有车辆集合;

Q : 车辆额定载重;

V : 车辆行驶速度, 为恒定值;

u_c : 单位距离运输成本;

f_c : 自有车辆固定成本;

β : 租赁车辆固定成本系数, $\beta > 1$, 则租赁车辆固定成本为 βf_c ;

q_g : 客户的需求量;

$[e_g, l_g]$: 客户 g 的服务时间窗, 车辆必须早于客户时间窗之前到达;

s_g : 服务客户所需的时间;

d_{ij} : 节点与节点之间的距离, $\forall i, j \in G \cup D$;

$E: \{e | e = 1, 2, \dots, |E|\}$ 为配送企业租赁车辆集合;

e_i : 配送企业 i 需要租赁的车辆数量, $e_i = q_d / Q$, q_d 为企业未服务的客户需求量;

t_g : 车辆到达客户 g 的时间;

x_{ijk} : 0-1 变量, 当车辆经过弧 (i, j) , $i, j \in G \cup D$ 时, $x_{ijk} = 1$, 否则为 0;

y_{swk} : 0-1 变量, 当车辆经过弧 (s, w) , $s, w \in D$ 时, $y_{swk} = 1$, 否则为 0.

对应任意企业组建的第 j 个子联盟, 定义 G^j 为子联盟 S^j 中包含的客户集合, D^j 为配送中心集合, K^j 为子联盟 S^j 中包含的自有车辆集合, E^j 为子联盟租赁车辆集合.

3.3 数学模型

根据以上假设和定义, 考虑客户时间窗和企业配送资源的多方协作车辆路径问题混合整数规划模型如下:
目标函数:

$$\min TC(S^j) = \sum_{k \in K^j \cup E^j} u_c \left(\sum_{i, j \in G^j \cup D^j} d_{ij} x_{ijk} + \sum_{s, w \in D^j} d_{sw} y_{swk} \right) + f_c \sum_{d \in D^j} k_d + \alpha f_c \sum_{e \in E^j} k_e, \quad (3)$$

其中, $\sum_{e \in E^j} k_e$ 表示子联盟 S 的租赁车辆数, 即子联盟 S 应有车辆数减去目前自有车辆数, 计算公式为:

$$\sum_{e \in E^j} k_e = TK(S^j) - \sum_{d \in D^j} k_d. \quad (4)$$

约束条件:

$$\sum_{i \in G^j \cup D^j} \sum_{k \in K^j \cup E^j} x_{ijk} = 1, \quad j \in G^j, \quad (5)$$

$$\sum_{j \in G^j \cup D^j} \sum_{k \in K^j \cup E^j} x_{ijk} = 1, \quad i \in G^j, \quad (6)$$

$$\sum_{i \in D^j} x_{ijk} - \sum_{i \in D^j} x_{jik} = 0, \quad i, j \in G^j \cup D^j, k \in K^j \cup E^j, \quad (7)$$

$$\sum_{i \in G^j \cup D^j} x_{ifv} - \sum_{i \in G^j \cup D^j} x_{fjv} = 0, \quad f \in G^j \cup D^j, k \in K^j \cup E^j, \quad (8)$$

$$\sum_{i \in D^j} y_{ihk} - \sum_{i \in D^j} y_{hjk} = 0, \quad h \in D^j, k \in K^j \cup E^j, \quad (9)$$

$$\sum_{i \in G^j \cup D^j} \sum_{j \in G^j \cup D^j} q_i x_{ijk} \leq Q, \quad i, j \in G^j \cup D^j, k \in K^j \cup E^j, \quad (10)$$

$$e_j \leq \sum_{k \in K^j \cup E^j} \sum_{i \in G^j} \left(t_i + \frac{d_{ij}}{V} + s_i \right) x_{ijk} = t_j \leq l_j, \quad (11)$$

$$\sum_{k \in K^j \cup E^j} \sum_{n \in D^j} \sum_{i \in G^j} y_{nik} \leq \sum_{n \in D^j} (k_n + e_n), \quad (12)$$

$$\sum_{k \in K^j \cup E^j} \sum_{n \in D^j} \sum_{i \in G^j} y_{nik} \leq k_n + e_n, \quad (13)$$

$$y_i - y_j + (|G^j| + |D^j|) x_{ijk} \leq G^j + D^j - 1, \quad i, j \in G^j, k \in K^j \cup E^j, \quad (14)$$

$$x_{ijk} \in \{0, 1\}, \quad i, j \in G^j \cup D^j, k \in K^j \cup E^j, \quad (15)$$

$$y_{swk} \in \{0, 1\}, \quad s, w \in D^j, k \in K^j \cup E^j, \quad (16)$$

其中, 式 (3) 表示运输成本 (包括货物调运和配送两阶段) 及车辆固定成本加和最小. 式 (5) 和式 (6) 保证每个客户均被服务一次, 且为同一辆车进行配送. 式 (7) 表示配送车辆的起终点一致, 行驶路径为闭环. 式 (8) 和式 (9) 确保节点流量均衡. 式 (10) 为车辆装载上限约束. 式 (11) 为车辆到达客户的时间满足客户时间窗需求. 式 (12) 和式 (13) 表示企业及配送中心的车辆数量限制. 式 (14) 表示每辆车的行驶路径为一个简单圈, 消除子回路. 式 (15) 和式 (16) 为 0-1 变量的取值范围.

3.4 求解 MOCVRPTWC 问题的 ALNS 算法

为实现一次性求解 $2^n - 1$ 个 MOCVRPTWC 问题, 本文采用 Luo^[31] 提出的自适应大邻域搜索算法 (adaptive large neighborhood search heuristic algorithm, ALNS) 进行运算, 该算法已被证明能够有效求解协作优化问题^[32]. 其具体运算流程如下:

Step1: 读取参与协作的企业信息, 包括企业及所属配送中心数量、位置, 配送中心拥有的车辆数量、车载量, 客户位置、需求量及时间窗要求;

Step2: 确定 $2^n - 1$ 个子联盟所包含的配送企业、客户需求及车辆信息, 利用 k -means 聚类方法, 按照最近原则分配客户订单, 形成 CVRP 问题;

Step3: 采用贪婪插入启发式求解 CVRP 问题, 得到初始解 InitialSolution;

Step4: 确定移除和插入操作权重, 借助轮盘赌方法, 依照概率 $\omega_i / \sum_{j=1}^h \omega_j$ 选择下一步操作, 操作权重根据规则不断更新;

Step5: 基于 InitialSolution 改变路径方案, 采用随机、最大后悔值、邻域图移除初始解中的 w 个客户;

Step6: 采用深度贪婪启发式, 将移除的 w 个客户节点插入到路径 r 的最优位置, 得到重构解 ModifiedSolution;

Step7: 采用模拟退火判断准则, 比较初始解与重构解, 若 ModifiedSolution 为全局最优解, 转入 Step9, 否则, 转入 Step8 继续求解;

Step8: 判断重构解是否为改进解, 若是, 则更新 InitialSolution; 若为非改进但被接受的解, 转入 Step4 更新操作权重, 继续运算流程;

Step9: 程序求解得到最终解, 结束运算, 返回最优方案 BestSolution.

4 基于 Shapley 值法的协作配送成本分摊调整思路

4.1 Shapley 值法成本分摊原理

协作配送中 n 个成员的成本分摊问题属于合作博弈问题, 现有分摊方法的核心是计算成员的边际贡献, 即成员加入子联盟前后协作成本的差值, 在此基础上设计运算规则得到分摊结果. 本文采用合作博弈理论中

经典的 Shapley 值法作为调整成本分摊结果的理论基础. Shapley 值法是由 Shapley^[33] 提出的一种求解 n 人合作的成本分摊方法, 其求解思想如下:

假设 n 个参与者组建联盟, 会产生 $2^n - 1$ 个非空子联盟 S , $C(S)$ 表示子联盟协作成本, 其中大联盟 $N = (1, 2, \dots, n)$ 的总成本为 $C(N)$, 大联盟中每个成员的成本分摊值记为 $\varphi_i (1 \leq i \leq n)$. 计算公式如下所示:

$$\varphi_i = \sum_{i \in S^j, S^j \in N} \frac{(|S^j| - 1)!(n - |S^j|)!}{n!} [C(S^j) - C(S^j \setminus \{i\})]. \quad (17)$$

上式中 $C(S^j) - C(S^j \setminus \{i\})$ 表示成员 i 加入联盟 S^j 对 $C(S^j)$ 的影响, 反映其对联盟的边际贡献. 根据 Shapley 值法的性质, 由上式计算得到的分摊方案必然使得: 1) n 个企业成本分摊值加和等于大联盟总成本 $C(N)$; 2) 每个企业的分摊成本 φ_i 必定小于该企业独立配送的成本 C_i , 联盟形成实质合作.

然而, 由于 Shapley 值法的对称性假设, 合作成员的个体差异被忽略. 根据第 2 节的分析, 联盟成员服务质量影响整体效益以及客户满意度, 因此有必要对分摊值进行调整, 使得企业服务质量与分摊值相对应, 提升分摊公平性.

4.2 考虑协作企业服务质量差异的 Shapley 值调整模型

由表 4 可得, 评价指标根据与服务质量之间的关系可分为两类: 协作前后价差补偿、配送周期、受理需求等待时间、配送中心响应距离为负指标, 该指标值越大, 意味着企业配送服务与客户期望之间的差距越大, 与理想配送时间的差值越大, 因此其服务质量评价越低. 而正指标值越大, 反映企业为联盟付出的贡献越多, 满足客户需求的程度越高, 因此服务质量评价越高, 成本分摊值相应减少.

4.2.1 确定企业服务质量评定值

依据表 4 并结合实际数据, 得到 n 个企业在关系绩效、运营绩效、成本绩效以及协作绩效下的 14 个指标数据, 据此构造数据矩阵 A , 如下所示:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & \dots & a_{114} \\ a_{21} & a_{22} & a_{23} & a_{24} & \dots & a_{214} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & a_{n4} & \dots & a_{n14} \end{bmatrix}. \quad (18)$$

每一列代表 n 个企业在同一指标下的数据值, 现将矩阵 A 中的元素做如下变化:

$$\alpha_{ij} = a_{ij} - \frac{1}{n} \sum_{i=1}^n a_{ij}. \quad (19)$$

即元素 a_{ij} 减去其所在列元素之和的平均值, 转变为 α_{ij} , 矩阵 A 转变为矩阵 A' . 与二级指标对应的最优权重值相乘, 得到 n 个企业在所有指标下的综合权重值 $(W_1, W_2, \dots, W_n)^T$.

$$(W_1, W_2, \dots, W_n)^T = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} & \dots & \alpha_{114} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} & \alpha_{24} & \dots & \alpha_{214} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \alpha_{n3} & \alpha_{n4} & \dots & \alpha_{n14} \end{bmatrix} \begin{bmatrix} w_1^* \\ w_2^* \\ \vdots \\ w_{14}^* \end{bmatrix}. \quad (20)$$

根据服务质量指标分类, 提取企业 i 在四个负指标下的权重值, 将其加和记为 XW_i , 十个正指标权重值加和记为 YW_i , 记 QW_i 为协作企业服务质量的综合评定值, 计算公式如下:

$$QW_i = YW_i - XW_i = \sum_{j=1}^k W_{ij} - \sum_{j=k+1}^p W_{ij}, \quad (21)$$

其中, p 表示指标总数, k 表示正指标数量, 则 $p - k$ 表示与服务质量呈反向关系的指标数量.

上述计算过程的核心思想是: 联盟企业服务质量由两类指标共同反映. 在正向指标中, 若企业 i 的表现高于联盟平均水平, 则证明企业 i 在协作中投入足量资源保证任务完成, 并在配送过程中保持客户满意. 反之, 在反向指标中, 若企业 i 的数值高于联盟平均值, 说明其配送质量与客户理想情况相距较大, 一定程度上造成客户满意度和企业利益损失. 将两类指标相减, 得到服务质量综合评定值 QW_i , 若 $QW_i \geq 0$, 表明企业 i 服务质量高于联盟平均水平, 反之, 则表明企业 i 服务质量与其他成员之间差距较大.

4.2.2 确定最终分摊成本

基于 4.2.1 节提出的思路, 求得 n 个企业服务质量评定值 QW_i , $QW_i = YW_i - XW_i$, 并对其从高到低进行排序, 记排序结果为 $\{1, 2, \dots, i, j, \dots, n\}$. 根据大联盟协作路线, 得到企业 j 配送前 $j-1$ 个企业客户的订单数量 U . 根据 Shapley 值求解公式, 结合企业间服务质量差异和交换订单数量, 对子联盟成本进行调整, 过程如下:

假设共有 n 家物流企业组建协作联盟, 共生成 $2^n - 1$ 个子联盟, $C(S)$ 表示子联盟成本, 企业分摊成本记为 φ_i . 对除大联盟之外的子联盟成本进行调整, 记 $AC(S)$ 为调整后的子联盟成本, $A\varphi_i$ 为调整后的分摊成本, 具体公式如下所示:

$$AC(S) = C(S) + \sum_{i,j \in S, i \neq j} \max(QW_i - QW_j, 0)U_j^i \mu, \quad (22)$$

$$A\varphi_i = \frac{(n-|S|)! (|S|-1)!}{n!} [AC(S) - AC(S \setminus \{i\})], \quad (23)$$

其中, $\max(QW_i - QW_j, 0)U_j^i$ 表示, 若联盟 S 中企业 i 服务质量评定值高于企业 j , 则需计算企业 j 配送企业 i 的订单数量, 将其与 $QW_i - QW_j$ 相乘, 得到企业 j 由于服务质量较低对企业 i 客户造成的影响. 反之, 若企业 j 服务质量高于企业 i , 则无需调整成本. μ 为惩罚成本系数, 将服务质量差距转化为成本, 代入 Shapley 值求解公式计算最终分摊值.

由上述过程可知, 在联盟成本 $C(N)$ 不变的前提下, 包含服务质量较低成员的子联盟 $AC(S) > C(S)$, 而包含服务质量较高成员的 $C(S)$ 保持不变. 根据 Shapley 值求解规则, 服务质量较低成员的 $A\varphi_i > \varphi_i$, 服务质量较高成员的 $A\varphi_i < \varphi_i$, 且 $\sum A\varphi_i = \sum \varphi_i = C(N)$.

4.3 考虑企业服务质量差异的成本分摊调整流程

基于现实中各配送企业的服务质量具有差异性, 本文提出基于 Shapley 值法的成本分摊调整流程, 依据服务质量差异对初始分摊方案进行调整, 使得成员服务质量与成本节约量相匹配, 保证分摊方案公平性.

成本分摊调整流程分为两个阶段, 第一阶段为计算初始成本分摊结果, 第二阶段基于企业服务质量评价结果进行调整, 具体步骤如图 2 所示.

第一阶段具体流程:

Step1: 参与协作的 n 个企业向平台提交其客户位置、需求及配送车辆数量等信息.

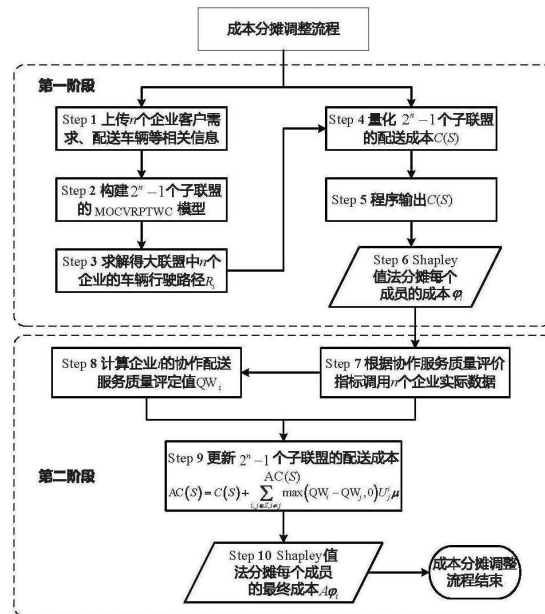


图 2 成本分摊调整流程图

7.12 胡路等 (2025): 充电桩排队与路径模型

车辆运输成本、时间窗惩罚成本、车辆固定成本和充电桩成本在内的总成本最低。

2 数学模型

2.1 模型假设

本文基于 Keskin 等^[20]的研究,作出以下基本假设:

假设 1 电动货车从配送中心出发,完成任务后需回到配送中心,每辆电动货车只行驶一条路线。

假设 2 每个客户只能由一辆车提供服务。

假设 3 每条路线上每个客户的需求总量不超过车辆的承载能力。

假设 4 车辆在充电站完成充电后方可继续行驶。

2.2 符号说明

本文的电动货车路径规划定义在一个完全有向图中,有向图的顶点 i 包括客户点、充电站及车辆中心出口和入口,其相关集合定义为: $V = \{1, 2, \dots, n\}$,

为客户集合,共有 n 个客户; F 为充电站的集合,顶点 0 和 $n+1$ 分别表示车辆中心出口和入口; $V_F = V \cup F$, 为客户和充电站的合集; $V_0 = \{V, \{0\}\}$, 为客户和顶点 0 的合集; $V_{n+1} = \{V, \{n+1\}\}$, 为客户和顶点 $n+1$ 的合集; $V_{0,n+1} = \{V, \{0\}, \{n+1\}\}$, 为客户点与车辆中心出口和入口的合集; $V_{F,0} = \{V_F, \{0\}\}$ 和 $V_{F,n+1} = \{V_F, \{n+1\}\}$ 为客户、充电站分别与顶点 0 和 $n+1$ 的合集; $V_{F,0,n+1} = \{V_F, \{0\}, \{n+1\}\}$, 为有向图中所有顶点的合集; $\tilde{V} = \{(i, j) | i \in F, j \in V_{n+1} \text{ 或 } i \in V_0, j \in V_{F,n+1}\}$ 为充电站与客户点、顶点 $n+1$ 或客户点、顶点 0 与所有顶点组成的边合集。EFVRP 的完全有向图定义为 $G = (V_{F,0,n+1}, E)$, 其中, $E = \{(i, j) | i, j \in V_{F,0,n+1}, i \neq j\}$, 表示有向图中所有顶点之间的边合集。

本文模型构建需要的符号及其说明如表 1 所示。表中, $x_{i,j}$ 、 y_i^a 、 y_i^d 、 v_i 、 a_i 、 f_i 、 u_i 、 $w_{i,k}$ 、 $z_{i,k}$ 、 $a_{i,k}$ 为决策变量。

表 1 符号说明
Tab. 1 Symbol descriptions

符号	说明	符号	说明
$d_{i,j}$	顶点 i 到顶点 j 的行驶距离	c_s	所有充电站安装充电桩的总费用
$t_{i,j}$	顶点 i 到顶点 j 的行驶时间	$x_{i,j}$	当顶点 i 与顶点 j 在有向图中的路径被访问时为 1, 否则为 0
q_i	客户点 i 的需求	y_i^a	到达顶点 i 时剩余电量
s_i	客户点 i 的服务时间	y_i^d	离开顶点 i 时剩余电量
e_i	客户点 i 的最早服务时间	v_i	到达顶点 i 时的迟到时长
l_i	客户点 i 的最晚服务时间	a_i	到达顶点 i ($i \in V_{0,n+1}$) 的时间
G_0	车辆装载容量	f_i	在返回仓库之前,以顶点 i 为最后访问节点的路线所花费的总时间
Q	车辆电池容量	u_i	抵达顶点 i 时的剩余货物容量
g	电池充电速率	$w_{i,k}$	顶点 i ($i \in F$) 进行第 k 次所服务车辆需要的等待时间
h	电池单位距离消耗速率	$z_{i,k}$	顶点 i ($i \in F$) 进行第 k 次所服务车辆需要的充电时间
c_e	每单位距离消耗的电量成本	$a_{i,k}$	顶点 i ($i \in F$) 第 k 次服务的车辆到达时间
c_p	每单位时间惩罚成本	$y_i^a(a_{i,k})$	当车辆到达顶点 i 时间为 $a_{i,k}$ 时的剩余电量
c_f	每个顶点的运行成本,包括维护、购置成本以及在客户点工作的成本	$w_{i,k}(a_{i,k})$	当车辆到达顶点 i 时间为 $a_{i,k}$ 时的等待时间
c_d	每单位时间司机工资		

2.3 模型建立

2.3.1 充电站排队模型

假设每个充电站都配备了相同数量 c 的充电桩。每个充电站构成一个排队系统。其中,充电桩为服务设施,电动货车为排队顾客。电动货车在充电站根据先到先服务原则接受服务。当电动货车到达充电站时,如果有充电桩是空闲的,车辆就立即接受服

务;否则,到达的电动货车被分配到具有最短等待时间的充电桩(如果有多个充电桩可供选择,则选择编号最低的设备)。每个充电站均可描述为一个独立的 $G(t)/G(t)/c$ 排队系统。

以顶点 i 为例($i \in F$)描述其排队模型。一旦充电站 i 有充电桩可以使用,第 k 次服务的车辆使用充电桩的时间为

$$z_{i,k} = (Q - y_i^a(a_{i,k})) / g, \quad i \in F, k \in \mathbf{N}. \quad (1)$$

为计算与排队结果有关的统计数据, 排队输出和评价指标采用递归方法, 并引入一些中间变量来跟踪排队系统在一段时间内的状态. 充电站 i 中第 m 个充电桩提供服务前的排队时间矩阵 $D_{i,k,m} = (D_{i,k,m})$, 从 $a_{i,k}$ 开始计算, 其中 $i \in F, k \in \mathbf{N}, m = 1, 2, \dots, c$. 设置临时排队时间矩阵 $\hat{D}_{i,k,m} = (\hat{D}_{i,k,m})$, 表示充电站 i 第 k 次服务在每个充电桩需要排队的时间, 其计算如式(2)所示.

$$\hat{D}_{i,k,m} = \begin{cases} 0, & k = 0, \\ \max\{D_{i,k-1,m} - (a_{i,k} - a_{i,k-1}), 0\}, & k \in \mathbf{N}. \end{cases} \quad (2)$$

利用该模型可以计算出 2 个重要的数量指标: 充电站 i 进行第 k 次所服务车辆需要的等待时间 $w_{i,k}$ (式(3))、充电站 i 第 k 次所服务车辆使用的充电桩序号 m (式(4)). m 是通过一个反函数计算临时排队矩阵 $\hat{D}_{i,k,m}$ 中排队时间最小时的变量值, 即所使用的充电桩序号. 结合原来的输入和输出参数, 可以得到车辆的到达时间、服务开始时间、服务结束时间, 离开时间.

$$w_{i,k} = \min \hat{D}_{i,k,m}, \quad i \in F, k \in \mathbf{N}, \quad (3)$$

$$m = \arg \min \hat{D}_{i,k,m}, \quad i \in F, k \in \mathbf{N}. \quad (4)$$

当车辆被分配到一个充电桩时, 排队矩阵更新为下一个时间段, 并将该车辆所花费的时间添加到所分配的充电桩的排队中. 更新排队矩阵如式(5)所示.

$$D_{i,k,m} = \hat{D}_{i,k,m} + z_{i,k} + a_{i,k}, \quad i \in F, k \in \mathbf{N}. \quad (5)$$

2.3.2 电动货车路径优化数学规划模型

本文的电动货车路径优化模型如下:

$$\min c_e \sum_{i \in V_{F,0}} \sum_{j \in V_{F,n+1}} t_{i,j} x_{i,j} + c_p \sum_{i \in V} v_i + c_f \sum_{i \in V_{F,0}} \sum_{j \in V_{F,n+1}} x_{i,j} + c_d \sum_{i \in V_F} f_i + c_s, \quad (6)$$

$$\sum_{j \in V_{F,n+1}} x_{i,j} = 1, \quad i \in V, i \neq j, \quad (7)$$

$$\sum_{i \in V_{F,0}} x_{i,j} = \sum_{i \in V_{F,n+1}} x_{j,i}, \quad j \in V_F, i \neq j, \quad (8)$$

$$a_i + (t_{i,j} + s_i) x_{i,j} - M(1 - x_{i,j}) \leq a_j, \quad i \in V_0, j \in V_{F,n+1}, \quad (9)$$

$$a_{i,k} + (t_{i,j} + z_{i,k} + w_{i,k}(a_{i,k})) x_{i,j} - M(1 - x_{i,j}) \leq a_j, \quad i \in F, j \in V_{n+1}, k \in \mathbf{N}, \quad (10)$$

$$0 \leq u_j \leq u_i - q_i x_{i,j} + (1 - x_{i,j}) G_0, \quad i \in V_{F,0}, j \in V_{F,n+1}, i \neq j, \quad (11)$$

$$0 \leq u_0 \leq G_0, \quad (12)$$

$$y_j^a \leq y_i^d - h d_{i,j} x_{i,j} + (1 - x_{i,j}) Q, \quad (i, j) \in \tilde{V}, \quad (13)$$

$$y_j^a \geq y_i^d - h d_{i,j} x_{i,j} - (1 - x_{i,j}) Q, \quad (i, j) \in \tilde{V}, \quad (14)$$

$$0.1 Q \sum_{j \in V_{F,0}} x_{j,i} \leq y_i^a, \quad i \in V_{F,n+1}, i \neq j, \quad (15)$$

$$y_i^a \leq y_i^d \leq Q, \quad i \in V_{F,0}, \quad (16)$$

$$e_i \leq a_i, \quad i \in V, \quad (17)$$

$$v_i \leq a_i - l_i, \quad i \in V, \quad (18)$$

$$x_{i,j} \in \{0, 1\}, \quad i \in V_{F,0}, j \in V_{F,n+1}, i \neq j, \quad (19)$$

式中: M 为一个足够大的数字.

式(6)为目标函数, 使得总成本最小, 由 5 个部分组成: 能源成本、与违反时间窗有关的惩罚成本、车辆的固定成本、司机工资、设立充电桩的总成本. 式(7)表示每个客户只能被服务 1 次; 式(8)为车流量守恒约束; 式(9)、(10)为跟踪车辆分别从客户和充电站出发后到达下一个顶点的时间; 式(11)、式(12)确保每个客户的需求得到满足; 式(13)、(14)为跟踪车辆分别离开充电站和客户时的电池状态; 式(15)、(16)将电池的充电状态限制在 10%~100%; 式(17)是确保顶点 i 在时间窗的最早服务时间开始之后才服务; 式(18)确保当车辆晚到时间晚到时间是非负数. 式(1)~(19)构成本文考虑多充电桩排队和时间窗的电动货车路径优化数学规划模型.

3 算 法

EVRPTW 是一个经典的 NP-Hard 问题, 且本文建立的模型为非线性模型. 为此, 设计一个混合启发式算法求解路径规划问题, 充电站排队模型采用递归算法计算数量指标.

3.1 两阶段启发式算法

在路径规划求解算法中, 节约里程法(C-W)简单易行, 能够生成较好的可行解, 但不一定是最优解. Shaw^[21] 首次提出大邻域搜索(LNS)算法, 通过对初始解进行优化, 包含移除部分客户并重新插入客户的过程, 最终得到优化解. 刘小兰等^[22]提出的改进 LNS 算法具有更强的寻优能力, 其结果非常接近

7.13 Cheng 等 (2022): 分时碳强度充电模型

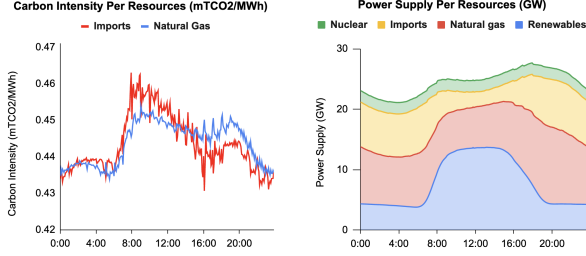


Fig. 2. (Left) The carbon intensity of each resources. We exclude biogas, biomass, geothermal, and coal since it has relatively low carbon contribution in California. (Right) Daily average power supply per resource. We exclude coal and hydro since it is less significant.

Carbon intensity depends on the power grid's operational plan and carbon intensity of each energy resource. In this paper, we specifically examined the carbon intensity for the California grid with data provided by CAISO (California Independent System Operator)². To calculate the carbon intensity, we consulted the carbon emission of the grid resources as well as the supply trend of the grid. Figure 2 describes the feature of the two data sets. In Fig 2 (left) there are two main contributors to the state's grid emission: imports and natural gas.

$$C(t) = \sum_{i=1}^N C_i(t) \cdot \frac{P_i(t)}{P_{total}(t)}, \quad \forall i \in \text{grid mix}; \quad (1)$$

where $C_i(t)$ is a time-series data of the carbon intensity of each energy source, $P_i(t)$ is the power supplied by the energy source and $P_{total}(t)$ is the total power generated on the specific day at time t . From the formula and the data, we can expect the carbon intensity to be high during the night and the early morning when solar energy is less available. On the other hand, the carbon intensity of the grid tends to be low during the day as solar energy and other renewables become a major contributor to the state's power supply.

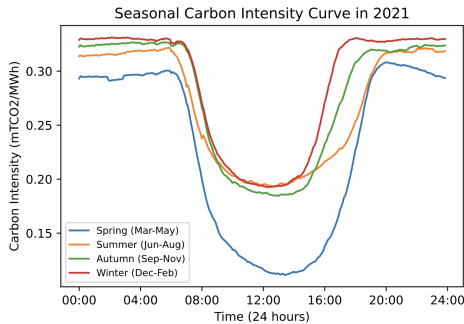


Fig. 3. Average carbon intensity during different seasons.

Carbon intensity is not only grid-mix dependent and time sensitive, but is also seasonally sensitive. Fig 3 shows an

²<http://www.caiso.com/todaysoutlook/pages/emissions.html>

example of carbon intensity calculated using the above formula throughout the year. We notice that carbon intensity is lower especially during the spring. On top of that, we also notice a shift in time for the carbon intensity curve in autumn as well as winter. We can attribute this to the daylight saving hours policy in California.

B. Charging Station Operation Model

As mentioned in Section II, our charging framework takes into account of the user constraints as well as the infrastructure constraints. These constraints are discussed below. We use the discrete time model, where the time step is indexed in $\mathcal{T} := \{1, 2, 3, \dots, T\}$. $\mathcal{V}$ is the set of all EVs. For each vehicle $i \in \mathcal{V}$, we denote the charging demand information is $(x_{i,arrival}, x_{i,depart}, t_{i,arrival}, t_{i,depart})$, where $x_{i,arrival}$ is the state-of-charge SoC of EV i upon its arrival of the charging station, $x_{i,depart}$ is the desired state-of-charge SOC of EV i upon its departure. and $t_{i,arrival}, t_{i,depart}$ is the time index when the vehicle i arrive in / depart from the charging station, $t_{i,arrival}, t_{i,depart} \in \mathcal{T}$. The above four factors are all externally determined by the user. Arrival time $t_{i,arrival}$ and initial energy level $x_{i,arrival}$ are determined when user plugged in the EV. Departure time $t_{i,depart}$ and charge $x_{i,depart}$ are input by the user when they initiate the charging session.

The model aims at deciding the charging power $u_i(t)$ for vehicle i at time t , and the state-of-charge (SoC) $x_i(t)$ for EV i at time t given the specific objective. We are interested in solving the following optimization problem:

$$\min_{\mathbf{u}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{V}} C(t) u_i(t) \Delta T + \lambda \sum_{i \in \mathcal{V}} |x_i(T) - x_{i,depart}| \quad (2a)$$

$$s.t. \quad x_i(0) = x_{i,arrival}, \quad (2b)$$

$$x_i(t) = x_i(t-1) + \frac{\delta u_i(t)}{E_i}, \quad t \geq 1; \quad (2c)$$

$$0 \leq x_i(t) \leq \bar{x}_i, \quad i \in \mathcal{V}; \quad (2d)$$

$$0 \leq u_i(t) \leq \bar{u}_i, \quad i \in \mathcal{V}; \quad (2e)$$

$$\sum_{i \in \mathcal{V}} u_i(t) \leq \bar{P}, \quad t \in \mathcal{T}; \quad (2f)$$

$$u_i(t) = 0, \quad t \notin [t_{i,arrival}, t_{i,depart}), \quad i \in \mathcal{V}. \quad (2g)$$

The above optimal charging scheduling problem minimizes the total carbon emissions while satisfying charging sessions' time and energy constraints, defined by the objective function (2a). We firstly consider the case when $C(t)$ and EV charging demand information for next T steps are available and communicated to the charging station operator. We denote ΔT to be the discretization interval, and in our case, ΔT is 5 minutes aligned with the carbon intensity measurement interval published by CAISO. λ is a balanced factor to trade off between carbon reduction and energy delivery. The operational constraints considered in this framework are described as follows:

EV charging constraints: Constraint (2b) defines the SoC level of EV upon arrival; and (2c) depicts the dynamics between charging power and battery SoC, where E_i is the battery

energy capacity (kWh). δ is an EV-specific charging parameter, which is known before the operation and is computed based on the charging efficiency and discretization, e.g., if charging efficiency is 90%, and discretization interval is 15 minutes, $\delta = 0.9 \times 15/60 = 0.225$. (2d) and (2e) constrain the admissible charging power for single EV, where the battery maximum SoC is $\bar{x}_i$ and maximum charging power is $\bar{u}_i$.

Charging station constraints: (2f) limits the total available charging power $\bar{P}$ at the charging station. We note that in this paper, we assume there is a known $\bar{P}$ decided by the charging station's transformer.

Temporal logic of charging sessions: (2g) describes the temporal logic that the charging session has to satisfy, as no charging power shall be delivered whenever EV i has not arrived ($t < t_{i,arrival}$) or has departed ($t \geq t_{i,depart}$).

Other Factors to generalize: The optimization problem that we introduced in (2a) is a general model that can be further generalized. For example, battery health is another important factor to be considered in EV charging. A more complex version of this optimization problem could potentially include penalty terms in the objective to discourage sudden change in charging power, which prevents battery health from deteriorating.

IV. REAL-TIME SCHEDULING

In this section, we propose a forecasting module to inform charging station operators day-ahead carbon intensity, and extend the framework described in Sec III to real-time settings.

Offline algorithm assumes we have access to all the information including future arrival statistics. While in the real-time algorithm, we assume we can only decide charging rate based on the predicted carbon intensity and information of arrived electric vehicles. An online algorithm is crucial for the real life applicability of our charging strategy because there are two main factors that changes daily in our optimization problem: daily average carbon intensity and daily EV arrival pattern. From figure 5 and 6, we can see that daily EV arrival varies from 5 vehicles to more than 40 vehicles per day. An online scheduling algorithm will be able to adjust scheduling the given arrival of EV's. On top of that, our discussion of carbon intensity also demonstrated seasonal volatility. With EV arrival uncertainty combined with carbon intensity uncertainty, and online algorithm allows the charging station to adjust its charging strategy given different scenario.

A. Carbon Intensity Day-Ahead Forecasting

We use the 2021 CAISO data (shown in Fig 2) and CAISO day-ahead load forecast data $Load(t)$ to fit and evaluate the carbon predictor model. The grid's day-ahead carbon intensity $\hat{C}(t)$ is closely related to the future electricity demand, calendar features, and past carbon intensity patterns. We randomly choose 80% data to fit the linear regression model and rest data for testing the model performance. The resulting forecasting model

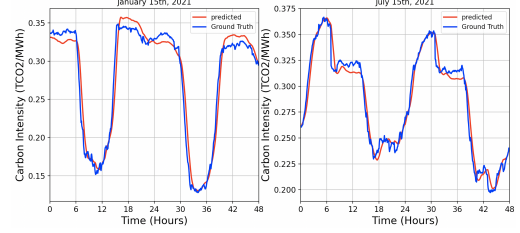


Fig. 4. The ground truth (blue) and predicted carbon intensity (red) for two-day sample in January and July respectively.

is as follows:

$$\begin{aligned} \hat{C}(t) = & \beta_0 + \beta_1 \times Minute + \beta_2 \times Hour + \beta_3 \times Day \\ & + \beta_4 \times Month + \beta_5 \times Load(t) + \beta_6 \times MA(C(t))_{24} \\ & + \beta_7 \times MA(C(t))_{12} + \beta_8 \times MA(C(t))_1; \end{aligned} \quad (3)$$

where $MA(\cdot)_a$ denotes the moving average of past a hours. Fig 4 illustrates the ground truth carbon intensity and forecasted carbon intensity using the fitted model (3). We note that other statistical and machine learning models can also be used for carbon intensity prediction. We use the linear regression model since it is simple and provides accurate prediction result. For the CAISO case study, the overall testing mean squared absolute error is 0.0085. We also demonstrate such forecasts can be readily integrated into the proposed charging scheduler without affecting the algorithm performance. Power grid CO2 emission intensity forecast is also investigated in [21] by considering additional features and an autoregressive model. We calculate and forecast the average carbon intensity rather than marginal carbon intensity, as in this study, we focus on the operating strategies for single EV charging station. Such single-station operation scheme shall pose minimal effects to transmission grid carbon profiles. We refer to [22] for a more complete review and open-source data on electricity grid carbon intensities.

B. Online Algorithm

With the predicted carbon intensity $\hat{C}(t)$, we are ready to implement proposed carbon-aware charging scheduler in an online fashion. Denote the optimization problem (2) as **SCH** with input information coming from the charging infrastructure and EVs' peak power constraints $\bar{P}$ and $\bar{u}$, predicted carbon intensity, the active charging vehicles $\mathcal{V}_k$ along with their states $x_{\mathcal{V}_k}$ where $k \in \mathcal{K}, \mathcal{K} := \{1, 2, 3, \dots, K\}$, the algorithm is summarized in Algorithm 1.

At each time step $k = 1, 2, \dots, K$, we take the forecasted carbon intensity for the next T steps, $\mathcal{C}_k := \{\hat{C}(k), \dots, \hat{C}(k+T-1)\}$, the current available EVs $\mathcal{V}_k$ and their SoC levels $x_{\mathcal{V}_k}$, and solves the optimization (2). With the optimized charging schedule $(u_i^*(1), \dots, u_i^*(T), i \in \mathcal{V}_k)$, we assign $u_i^*(1)$ to the charging power of EV i at time step k to $u_i(k) = u_i^*(1)$ for the active vehicles and update state of charge accordingly for every vehicle $x_i(k+1) = x_i(k) + \frac{\delta u_i(k)}{E_i}, \forall i \in \mathcal{V}$.

7.14 Lin 等 (2021): 分时电价充放电模型

6

we assume a constant energy consumption rate g and a constant charging speed $\frac{1}{\alpha}$. The energy consumption for edge (i, j) is thus given by $c_{ij} = ga_{ij}$, the time required to recharge the energy consumed by traveling along edge (i, j) is given by $f_{ij} = \alpha c_{ij}$.

We assume a commercial EV fleet consisting of K homogeneous EVs, each with a load capacity Q and a battery capacity C . The time required to fully recharge the battery from 0 is defined as $B = \alpha C$. At the beginning of the planning horizon, all the EVs are at the depot (node 0) with a full battery. All EVs should return to the depot (node $N + S + 1$) before the end of the planning horizon. During each visit to a station or to the depot, an EV can either pay to charge its battery or make profits by injecting energy back to the grid from its battery. The charging cost and discharging reward vary according to the time period (time-of-use energy pricing). We assume that EVs are allowed to perform either charging or discharging during their station visits, but are not allowed to perform both during the same time period.

The planning horizon is formed of $|T|$ consecutive discrete periods, each of length δ . Each time period t refers to a time interval $[\delta(t - 1), \delta t)$ and is associated with a buying (from the grid) energy price p_b^t and a selling (to the grid) energy price p_s^t . We assume that if an EV is to charge or discharge during a time period, then it has to do so for the full time period. Due to the constant linear charging/discharging assumption, a fixed amount of energy $\frac{\delta}{B}C$ would be charged/discharged during each period, we therefore can calculate a charging cost $P_{re}^t = \frac{\delta}{B}Cp_b^t$ and a discharging reward $P_{dis}^t = \frac{\delta}{B}Cp_s^t$ for each period. EVs are recharged back to full battery capacity during the night using at a lower energy price p_{night} . For the ease of presentation, we define $P_{night} = \frac{p_{night}}{\alpha}$ as the cost of charging per unit of time during the night. The notations are summarized in Table 2.

The problem formulation that is proposed next jointly optimizes the routes and charging/discharging schedule of the K vehicles by minimizing the net electricity cost given that all customer demands are satisfied.

3.2. Problem Formulation

To formulate EVRPTW-TP, the following binary variables are introduced:

$$\begin{aligned} x_{ijk} &= \begin{cases} 1 & \text{if edge } (i, j) \text{ is traveled by vehicle } k, \\ 0 & \text{otherwise,} \end{cases} \\ r_{itk} &= \begin{cases} 1 & \text{if vehicle } k \text{ charges its battery at node } i \text{ at time period } t, \\ 0 & \text{otherwise,} \end{cases} \\ d_{itk} &= \begin{cases} 1 & \text{if vehicle } k \text{ discharges its battery at node } i \text{ at time period } t, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

The continuous variables are

τ_{ik} : arrival time of vehicle k at node i ,

Table 2 Summary of the Notation

	Definition
V_c	Set of customer nodes $V_c = \{1, 2, \dots, N\}$
V_s	Set of station nodes $V_s = \{N + 1, \dots, N + S\}$
V_{od}	Set of depot nodes $V_{od} = \{0, N + S + 1\}$
$V_{c,s}$	$V_{c,s} = V_c \cup V_s$
$V_{c,s,o}$	$V_{c,s,o} = V_c \cup V_s \cup \{0\}$
$V_{c,s,d}$	$V_{c,s,d} = V_c \cup V_s \cup \{N + S + 1\}$
$V_{s,d}$	$V_{s,d} = V_s \cup \{N + S + 1\}$
$V_{s,o}$	$V_{s,o} = V_s \cup \{0\}$
$V_{s,od}$	$V_{s,od} = V_s \cup V_{od}$
$V_{c,od}$	$V_{c,od} = V_c \cup V_{od}$
$V_{c,s,od}$	$V_{c,s,od} = V_c \cup V_s \cup V_{od}$
p	Depot node $p = N + S + 1$
E	Set of all edges
T	Set of charging/discharging periods
N	Number of customers
S	Number of stations
K	Number of EVs
δ	Length of each charging/discharging period
Q	Cargo capacity
C	Battery capacity
B	Amount of time required to fully charge the EV battery from empty
α	The reciprocal of the constant charging speed
g	Energy consumption rate with respect to distance traveled
a_{ij}	Travel distance of edge (i, j)
t_{ij}	Travel time of edge (i, j)
c_{ij}	Energy consumption along edge (i, j)
f_{ij}	The amount of time required to charge the energy consumed along edge (i, j)
e_i	Earliest service start time at node i
l_i	Latest service start time at node i
s_i	Required service time at node i
q_i	The demand at node i
p_{night}	Unit electricity buying (from the grid) price at night (¢/kWh)
p_b^t	Unit electricity buying (from the grid) price during period t (¢/kWh)
p_s^t	Unit electricity selling (to the grid) price during period t (¢/kWh)
P_{night}	The cost of charging per unit of time at night
P_{re}^t	Cost of charging during period t
P_{dis}^t	Reward for discharging during period t

b_{ik} : remaining energy (in terms of charging time) in vehicle k upon arrival to node i ,

u_{ik} : remaining cargo in vehicle k upon arrival to node i .

The decision variables r_{itk} and d_{itk} are only associated with the station and depot nodes, while the remaining variables are associated with all the nodes. EVRPTW-TP is formulated as

8

$$\min \sum_{k=1}^K \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{itk} P_{re}^t - d_{itk} P_{dis}^t] + \sum_{k=1}^K P_{night} [B - b_{pk} - \sum_{t \in T} \delta(r_{ptk} - d_{ptk})], \quad (1)$$

$$\text{s.t.} \quad \sum_{k=1}^K \sum_{j \in V_{c,s,d}} x_{ijk} = 1, \quad \forall i \in V_c, \quad (2)$$

$$\sum_{i \in V_{c,s,o}} x_{ipk} = 1, \quad \forall k \in \{1, 2, \dots, K\}, \quad (3)$$

$$\sum_{j \in V_{c,s,o}} x_{jik} - \sum_{j \in V_{c,s,d}} x_{ijk} = 0, \quad \forall i \in V_{c,s}, k \in \{1, 2, \dots, K\}, \quad (4)$$

$$\tau_{ik} + (t_{ij} + s_i)x_{ijk} - \delta|T|(1 - x_{ijk}) \leq \tau_{jk}, \quad \forall i \in V_{c,s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (5)$$

$$t\delta(r_{itk} + d_{itk}) + t_{ij}x_{ijk} - \delta|T|(1 - x_{ijk}) \leq \tau_{jk}, \quad \forall i \in V_{s,o}, j \in V_{c,s,d}, t \in T, k \in \{1, 2, \dots, K\}, \quad (6)$$

$$e_i \leq \tau_{ik} \leq l_i, \quad \forall i \in V_{c,s,od}, k \in \{1, 2, \dots, K\}, \quad (7)$$

$$b_{0k} = B, \quad \forall k \in \{1, 2, \dots, K\}, \quad (8)$$

$$b_{jk} \leq b_{ik} - f_{ij}x_{ijk} + B(1 - x_{ijk}), \quad \forall i \in V_c, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (9)$$

$$b_{jk} \leq b_{ik} + \sum_{t \in T} \delta r_{itk} - \sum_{t \in T} \delta d_{itk} - f_{ij}x_{ijk} + B(1 - x_{ijk}), \quad \forall i \in V_{s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (10)$$

$$\sum_{t \in T} \delta r_{itk} \leq B - b_{ik}, \quad \forall i \in V_{s,od}, k \in \{1, 2, \dots, K\}, \quad (11)$$

$$\sum_{t \in T} \delta d_{itk} \leq b_{ik}, \quad \forall i \in V_{s,od}, k \in \{1, 2, \dots, K\}, \quad (12)$$

$$r_{itk} + d_{itk} \leq 1, \quad \forall i \in V_{s,od}, t \in T, k \in \{1, 2, \dots, K\}, \quad (13)$$

$$\tau_{ik} - (t-1)\delta \leq \delta|T|(1 - d_{itk} - r_{itk}), \quad \forall i \in V_{s,d}, t \in T, k \in \{1, 2, \dots, K\}, \quad (14)$$

$$u_{jk} \leq u_{ik} - q_{ij}x_{ijk} + Q(1 - x_{ijk}), \quad \forall i \in V_{c,s,o}, j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (15)$$

$$u_{0k} = Q, \quad k \in \{1, 2, \dots, K\}, \quad (16)$$

$$0 \leq b_{jk} \leq B \sum_{i \in V_{c,s,o}} x_{ijk}, \quad \forall j \in V_s, k \in \{1, 2, \dots, K\}, \quad (17)$$

$$u_{ik}, \tau_{ik} \geq 0, \quad \forall i \in V_{c,s,od}, k \in \{1, 2, \dots, K\}, \quad (18)$$

$$x_{ijk} \in \{0, 1\}, \quad \forall i \in V_{c,s,o}, \forall j \in V_{c,s,d}, k \in \{1, 2, \dots, K\}, \quad (19)$$

$$r_{itk}, d_{itk} \in \{0, 1\}, \quad \forall i \in V_{c,od}, t \in T, k \in \{1, 2, \dots, K\}. \quad (20)$$

The objective function (1) minimizes the net cost, i.e. the total cost of charging the batteries minus the total reward earned from discharging the batteries. The first part of the objective function corresponds to the net cost during the planning horizon, while the second part is the cost of fully recharging back all EVs at night. Constraints (2) ensure that every customer is served by exactly one EV. Constraints (3) force all the EVs to return to the depot by the end of the

planning horizon. Constraints (4) guarantee that no route ends at a customer or a station node. Constraints (5) ensure time feasibility of the edges leaving the customer and the depot nodes, while constraints (6) deal with the edges originating from charging stations. Constraints (7) ensure that the time windows of all the nodes are not violated. Constraints (8) indicate that every EV is fully charged before leaving the depot. Constraints (9)–(10) track the battery capacity along the route. Constraints (11) indicate that an EV battery cannot be recharged to a level that exceeds its capacity, while Constraints (12) state that an EV battery cannot be discharged to a level below 0. Constraints (13) indicate that an EV is allowed to discharge or recharge its battery at a station or the depot nodes but is not allowed to discharge and recharge during the same time period. Constraints (14) ensure that an EV cannot start charging/discharging at a station/depot before arrival and before the start of the time period. Constraints (15) ensure that the demands along a route are all satisfied, whereas Constraints (16) state that all EVs have a full cargo at the start of the planning horizon. Constraints (17)–(20) indicate the variables types and limits.

4. Lagrangian Relaxation for the EVRPTW-TP

As shown in Section 6.3, solving EVRPTW-TP to optimality is computationally very challenging. Lagrangian relaxation is a well known algorithm that has been used to address many complex optimization problems. Particularly, in the context of vehicle routing, Lagrangian relaxation has been used to address several variants of VRP (Fisher et al. 1997, Kallehauge et al. 2006).

For the EVRPTW-TP, all the constraints other than Constraints (2) are associated with a particular vehicle k . Given this special structure which is common in vehicle routing problems, Constraints (2) are relaxed and the violation is penalized in the objective function using the Lagrangian multipliers λ_i . The resulting relaxed problem is

$$\begin{aligned}
 Z_{LR}(\lambda) = \min \quad & \sum_{k=1}^K \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{itk} P_{re}^t - d_{itk} P_{dis}^t] \\
 & + \sum_{k=1}^K P_{night} [B - b_{pk} - \sum_{t \in T} \delta(r_{ptk} - d_{ptk})] \\
 & + \sum_{i \in V_c} \lambda_i (1 - \sum_{k=1}^K \sum_{j \in V_{c,s,d}} x_{ijk}), \\
 \text{s.t.} \quad & (3) - (20).
 \end{aligned}$$

Since the EV fleet is homogeneous, problem $Z_{LR}(\lambda)$ decomposes into K identical sub-problems

$$\begin{aligned}
 Z_{SP}(\lambda) = \min \quad & \sum_{i \in V_{s,od}} \sum_{t \in T} [r_{it} P_{re}^t - d_{it} P_{dis}^t] + P_{night} [B - b_p - \sum_{t \in T} \delta(r_{pt} - d_{pt})] \\
 & - \sum_{i \in V_c} \lambda_i \sum_{j \in V_{c,s,d}} x_{ij}, \\
 \text{s.t.} \quad & (3) - (20).
 \end{aligned}$$